# UNIVERSITY OF CAPE COAST
## COLLEGE OF DISTANCE EDUCATION

## COURSE TITLE:

## BUSINESS FINANCE

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**COLLEGE OF DISTANCE EDUCATION, UNIVERSITY OF CAPE COAST**

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# BUSINESS FINANCE MODULES

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## UNIT 1: OVERVIEW OF BUSINESS FINANCE

### Unit Outline

Session 1: Introduction to Business Finance
Session 2: Meaning of Business Finance
Session 3: Role of Finance Manager and Finance Function
Session 4: Objectives of the Firm
Session 5: Decision areas in Business Finance
Session 6: Agency Problem

### Overview

Finance may be defined as the art and science of managing money or funds. Finance can also be defined as the science of money management. It is the provision of money at the time when it is needed and includes activities such as investing, borrowing, lending, budgeting and forecasting. Finance function is the procurement of funds and their effective utilization in business concerns. The concept of finance includes capital, funds, money, and amount. Studying and understanding the concept of finance has become an important part of every business concern and other stakeholders as well. This session looks at the meaning of business finance, role of finance managers, objectives of the firm, decision areas in finance and agency problem.

### Objectives

By the end of this unit, you should be able to:

(a) Understand the business finance
(b) Explain the various types of financial markets
(c) Describe the roles of finance managers
(d) Explain the objectives of the firm
(e) Recognize the decision areas in finance
(f) Explain the agency problem and how to resolve it

## SESSION 1: INTRODUCTION TO BUSINESS FINANCE

### Introduction

We welcome you to this session. This is the beginning of your understanding to finance. You don't have to worry yourself so much if you don't have any understanding of finance. We shall take you through the course step by step so that you will be able to manage your personal finances in a much better way and manage the funds of your organization well too. In this session we look at finance and the field of finance, the financial system, and the different types of financial markets.

### Objectives

By the end of this session, you should be able to:

(a) explain what finance is
(b) outline the interrelated areas of finance
(c) understand the financial system and how it operates
(d) distinguish between money market and capital markets
(e) appreciate the role of financial intermediaries

Now read on …

### 1.1 Introduction to Finance

Finance is the science and art of managing money. At the individual level, it is concerned with how much of their earnings to spend, save and how to invest the savings. At the corporate level, it involves how the firm decides whether to reinvest the profit or distribute the profit back to investors. Business concern needs finance to meet their requirements in the economic world. Any kind of business activity depends on the finance. Hence, it is called as lifeblood of business organization. Whether the business concerns are big or small, they need finance to fulfil their business activities.

### 1.2 The Field of Finance

Finance, as a discipline, is a body of facts, principles, and theories relating to raising and using money by individuals, businesses, and governments. Finance consists of three interrelated areas namely:

- Money and capital markets – which deals with financial markets and financial institutions
- Investment - which focuses on decisions made by individual and institutional investors as they choose securities for their investment portfolios; and
- Financial management – which basically involves decisions within firms.

### 1.3 Finance as a discipline

Finance as a discipline is related to a number of disciplines, notably economics, accounting, law, managerial science and taxation. It is therefore imperative for finance scholars and professionals to understand the relationships between finance and these disciplines.

Economics give a framework for understanding risk analysis, comparative return and pricing theory. Businesses operate in economic environment and therefore must contend with basic economic variables such as interest rates, inflation, taxes, disposable income and gross domestic product. Financial managers in particular must appreciate the impact of economic variables and employ them correctly to achieve financial goals.

Accounting usually referred to as the language of finance, communicates financial information through financial statements and reports from which financial decisions are made. The financial manager's duty is to interpret and use these statements in allocating financial resources judiciously to reap the best return possible for shareholders in the long-term.

Law governs almost every aspect of actions taken by the financial manager. Rules and regulations and laws govern business operations so as to protect shareholders interest. In fact, companies are created by laws and also operate under laws. Hence the financial manager must honour their legal obligation to avoid unnecessary legal action against the business.

Managerial science and statistics provide tools to aid the financial managers' decision making. Time series and trend analysis can be used in forecasting some aspects of the capital budget. In addition, statistical measure of variance can be used for risk analysis while regression analysis and linear programming can help in capital rationing and understanding the behaviour of costs respectively.

### 1.4 What is a financial system?

Financial system is a set of implemented procedures that track financial activities. The financial system is complex, encompassing many different types of private-sector financial institutions, banks, insurance companies, mutual funds, finance companies and investment banks. All of which are heavily regulated by the government. Financial system can be defined at the global, regional or firm-specific level. On a regional scale, the financial system is the system that enables lenders and borrowers to exchange funds. A financial system is the system that covers financial transactions and the exchange of money between investors, lender and borrowers. These are made of intricate and complex models that portray financial services, institutions and markets that link depositors with investors. Finance plays an important role in the economy. As banks, credit unions, and other financial institutions provide credit, they help expand the economy by directing funds from savers to borrowers.

Efficient financial management requires the existence of some objective or goal, because judgment as to whether or not a financial decision is efficient must be made in light of some standard. Although various objectives are possible, we assume in this course that the goal of the firm in respect of shareholders and that of other stakeholders.

### 1.5 Financial intermediation

A financial intermediary (such as a bank) simultaneously interacts with savers (or lenders) and borrowers and produces a set of services which facilitate the transformation of its liabilities (such as deposits) into assets (such as loans). A financial intermediary is an entity that facilitates a financial transaction between two parties. The function of facilitating liabilities (or assets) into assets (or liabilities) is called intermediation. Through intermediation financial intermediaries allow indirect lending (and borrowing) between savers and borrowers. Such an intermediary or a middleman could be a firm or an institution. Some examples of financial intermediaries are banks, insurance companies, pension funds and investment banks.

#### 1.5.1 Functions of financial intermediaries

- The primary function of financial intermediaries is to convert savings into investments.

- Financial intermediaries move funds from parties with excess capital to parties needing funds
- Intermediaries like commercial banks provide storage facilities for cash and other liquid assets, like precious metals.
- They assist clients to grow their money via investment. Intermediaries like mutual funds and investment banks use their experience to offer investment products to help their clients maximize returns and reduce risks.

### 1.5.2 Advantages of financial intermediaries

Financial intermediation can improve economic efficiency in at least five ways, by:

(i) Facilitating transactions: They help in saving time and cost. Since these intermediaries deal with a large number of customers, they enjoy economies of scale.

(ii) Facilitating portfolio creation: Through a financial intermediary, savers can pool their funds, enabling them to make large investments, which in turn benefits the entity in which they are investing.

(iii) Easing household liquidity constraints: FIs provide liquidity when they convert an asset into cash easily and quickly without loss of value in terms of money. When FIs issue claims against themselves and supply funds they, especially banks, always try to maintain their liquidity

(iv) Spreading risks over time: They help in lowering the risk of an individual with surplus cash by spreading the risk via lending to several people. Also, they thoroughly screen the borrower, thus, lowering the default risk.

(v) Reducing the problem of asymmetric information: They accumulate and process information, thus lowering the problem of asymmetric information.

## 1.6 Money markets

The money market is the market for short-term (one year or less) credit. At the wholesale level, it involves large-volume trades between institutions and traders. At the retail level, it includes money market mutual funds bought by individual investors and money market accounts opened by bank customers.

The money market is the mechanism through which holders of temporary cash surpluses meet holders of temporary cash deficits. The money market is characterized by a high degree of safety and a relatively low return in interest. The money market arises because for most individuals and institutions, cash inflows and outflows are rarely in perfect harmony with each other, and the holding of idle surplus cash is expensive.

### 1.6.1 Money Market instruments

The key money market instruments include:

- Treasury bills and short-term government notes
- savings accounts,
- money market deposit accounts
- Mutual funds: A mutual fund is a type of financial vehicle made up of a pool of money collected from many investors to invest in securities such as stocks, bonds, money market instruments, and other assets.
- Certificates of deposit: A certificate of deposit (CD) is a short-term security with a fixed interest rate and maturity date issued by a bank that seeks to raise funds from the secondary money market.

- Commercial paper: commercial paper is an unsecured, short-term debt instrument issued by a corporation, typically for the financing of account payables and inventories and meeting short-term liabilities.
- Call money: Call money is a very short-term bank loan that does not contain regular principal and interest payments. It is often used by brokerage firms to finance margin accounts.
- Interbank market
- Repurchase agreement /repo: A repurchase agreement (repo) is a form of short-term borrowing for dealers in government securities. In **repo**, a dealer sells government securities to investors, usually on an overnight basis, and buys them back the following day
- Time deposit: A time deposit is an interest-bearing bank deposit account that has a specified date of maturity.

### 1.6.2 Characteristics of the Money Market

- Money market investors seek mainly *safety* and *liquidity*, plus the opportunity to earn some *interest income*.
- Because funds invested in the money market represent only temporary cash surpluses and are usually needed in the near future, money market investors are especially sensitive to *risk*.
- *Original maturities* on money market instruments range from as short as one day on many loans to banks and security dealers to a full year on some bank deposits and T-bills.
- But because there are so many money market securities outstanding, some of which reach maturity each day, investors have a wide menu of *actual maturities* from which to make their selections.
- It is not a single market but a collection of markets for several instruments
- It is wholesale market of short-term debt instruments

## 1.6 Capital Market

The market where investment instruments like bonds, equities and mortgages are traded is known as the capital market. The primal role of this market is to make investment from investors who have surplus funds to the ones who are running a deficit. The capital market offers both long term and overnight funds. The different types of financial instruments that are traded in the capital markets are: Equity instruments such as shares, Credit market instruments, insurance instruments, foreign exchange instruments, hybrid instruments and derivative instruments. There are two types of capital market: Primary market and Secondary market.

### 1.6.1 Primary Market

It is that market in which shares, debentures and other securities are sold for the first time for collecting long-term capital. This market is concerned with new issues. Therefore, the primary market is also called NEW ISSUE MARKET. In this market, the flow of funds is from savers to borrowers (industries), hence, it helps directly in the capital formation of the country. The money collected from this market is generally used by the companies to modernize the plant, machinery and buildings, for extending business, and for setting up new business unit. The features of the primary market include: it relates to new issues of securities such as shares, for example, the first time MTN issued shares in Ghana, you or some friends and family members bought some shares using mobile money or from some selected banks and financial institutions in Ghana. The market in which you bought the shares for the first

time is called the primary market. Another feature is that the primary market has no particular place. This means that the transaction can be done anywhere using the telephone, in offices and even at homes. The primary market comes before secondary market. Before the securities of a company or government can be traded on the secondary market, it must first be sold in the primary market. If this is not done, then trading in the secondary market becomes impossible. There are various methods of raising capital in the primary market such as public issue, offer for sale, placing, right issue etc. these methods you have done in Company and Partnership Law. You remember right. Good student. So, the legal issues you learned are also applicable in Business Finance too.

### 1.6.2 Secondary Market

The secondary market is that market in which the buying and selling of the previously issued securities is done. The securities that were sold in the primary market are traded in the secondary market. The transactions of the secondary market are generally done through the medium of stock exchange. The chief purpose of the secondary market is to create liquidity in securities. If an individual has bought some security and he now want to sell it, he or she can do so through the medium of stock exchange to sell or purchase through the medium of stock. Trading on the stock exchange requires the services of the broker. In Ghana, we have the Ghana Stock Exchange, in USA we have the New York Stock Exchange, Nigeria have the Nigerian Stock Exchange. In any particular country we have one stock exchange for capital formation of a country. The secondary market has the following features: It creates liquidity, comes after primary market, has a particular place and encourage new investments in financial securities.

**Summary**

In this session, we explained the meaning of finance and finance as a discipline. We also looked at the financial system and the role of financial intermediaries. Financial markets such as money market and capital markets and instruments traded in these markets were all explained. We explained that, capital markets can be primary market or secondary markets. Why are you smiling? You understanding the issues, right? Don't give up.

**Self-Assessment Questions**

**Exercise 1.1**

## SESSION 2 Meaning of Business Finance

### Introduction

After looking at the introduction to finance, let us explain the meaning of finance in this session. We shall also look at the types of finance available to businesses and the role of finance to businesses as well. It is important that we take the course in finance serious because it is need in every business organization.

### Objectives

By the end of this session, you should be able to:

(a) explain the meaning of finance
(b) describe business finance
(c) identify the types of finance
(d) discuss the role of business finance

### 2.1 Meaning of Finance

Finance may be defined as the art and science of managing money or funds. Finance can also be defined as the science of money management. It is the provision of money at the time when it is needed and includes activities such as investing, borrowing, lending, budgeting and forecasting. Finance function is the procurement of funds and their effective utilization in business concerns. The concept of finance includes capital, funds, money, and amount. Studying and understanding the concept of finance has become an important part of every business concern and other stakeholders as well.

### 2.2 Definition of Business Finance

According to B.O Wheeler, 'business finance is that part of the business activity concerned with the acquisition and conservation of capital funds in meeting the financial needs and overall objectives of business enterprise' In the words of Parhter and Wert, 'Business finance deals primarily with raising, administering and disbursing funds by privately owned business units operating in nonfinancial fields of industry'. In the context of business, we shall also refer to business finance as financial management. What then is financial management?

Financial management is an integral part of overall management. It is mainly concerned with the effective funds' management in the business. According to J.F Bradely, financial

management is the area of business management devoted to a judicious use of capital and careful selection of sources of capital in order to enable the spending unit move in the direction of reaching its goals. Thus, its focus on ensuring that business activities are not affected due to lack of available funds as and when required.

In the words of J.L Massie, financial management is the operational activity of a business that is responsible for obtaining and effectively utilising the funds necessary for efficient operations of the entity. It is thus concerned with the efficient utilization of an important economic resources. Weston and Brigham also defined financial management in simple terms as an area of financial decision-making that harmonizes individual motives and enterprise goals.

Thus, Financial Management is mainly concerned with the effective funds' management in the business. In simple words, Financial Management as practiced by business firms can be called as Corporate Finance or Business Finance

Corporate finance is concerned with budgeting, financial forecasting, cash management, credit administration, investment analysis and procurement of funds for business activities and other business needs necessary for the business to adopt modern technology and application suitable to the global competitive environment.

According to the Encyclopaedia of Social Sciences, "Corporation finance deals with the financial problems of corporate enterprises. These problems include the financial aspects of the promotion of new enterprises and their administration during early development, the accounting problems connected with the distinction between capital and income, the administrative questions created by growth and expansion, and finally, the financial adjustments required for the bolstering up or rehabilitation of a corporation which has come into financial difficulties".

In this regard, financial managers must see to the effect that funds are procured in a manner that risk, cost and control considerations are properly balanced in any given situation and that there is optimum utilization of funds for sole benefit of the organization.

**2.3 Types of Finance**

Finance plays a major role in all aspects of business activities, hence, making it an important and integral part of every business concern in the contemporary world. Finance can be broadly classified into two major parts: Private finance and Public finance.

Private finance – This includes funds provided by individual investors, firms, corporate financial institutions to finance business activates as and when required.

Public finance – This includes revenue and disbursement of governments such as the central government, state and municipal assemblies as well as state-owned enterprises.

**2.4 The Role of Business Finance**

Businesses are, in effect, investment agencies or intermediaries. This is to say that their role is to raise money from members of the public, and from other investors, and to invest it. Usually, money will be obtained from the owners of the business (the shareholders) and from long-term lenders, with some short-term finance being provided by banks (perhaps in the

form of overdrafts), by other financial institutions and by other businesses being prepared to supply goods or services on credit (trade payables (or trade creditors)).

Businesses typically invest in real assets such as land, buildings, plant and inventories (or stock), though they may also invest in financial assets, including making loans to, and buying shares in, other businesses. People are employed to manage the investments, that is, to do all those things necessary to create and sell the goods and services in the provision of which the business is engaged. Surpluses remaining after meeting the costs of operating the business – wages, raw material costs, and so forth – accrue to the investors. Of crucial importance to the business will be decisions about the types and quantity of finance to raise, and the choice of investments to be made. Business finance is the study of how these financing and investment decisions should be made in theory, and how they are made in practice.

**2.5 A Practical Subject**

Business finance is a relatively new subject. Until the 1960s it consisted mostly of narrative accounts of decisions that had been made and how, if identifiable, those decisions had been reached. More recently, theories of business finance have emerged and been tested so that the subject now has a firmly based theoretical framework – a framework that stands up pretty well to testing with real-life events. In other words, the accepted theories that attempt to explain and predict actual outcomes in business finance broadly succeed in their aim.

Business finance draws from many disciplines. Financing and investment decision making relate closely to certain aspects of economics, accounting, law, quantitative methods and the behavioural sciences. Despite the fact that business finance draws what it finds most useful from other disciplines, it is nonetheless a subject in its own right. Business finance is vital to the business. Decisions on financing and investment go right to the heart of the business and its success or failure. This is because:

- such decisions often involve financial amounts that are very significant to the business concerned;
- once made, such decisions are not easy to reverse, so the business is typically committed in the long term to a particular type of finance or to a particular investment.

Although modern business finance practice relies heavily on sound theory, we must be very clear that business finance is an intensely practical subject, which is concerned with real-world decision making.

**Summary**

Waaw!!! You have been able to complete this session so fast. Maybe this is because you are understanding the concepts and they are becoming very interesting to you. In this session, we defined finance and business finance. We then looked at the types of finance and the role of business finance. We ended the session with the practical aspect of business finance. You did well. Keep it up.

**Self-Assessment Questions**
**Exercise 1.2**

## SESSION 3: ROLE OF FINANCE MANAGER AND FINANCE FUNCTION

### Introduction

The main objective of business finance is to arrange sufficient finance for meeting short-term and long term-needs. A financial manager will have to concentrate on the following areas of finance function. This session looks at the role of finance manager and the finance function in any business organization.

### Objectives

By the end of this session, you should be able to:

(a) discuss the roles of finance manager
(b) explain the finance function
(c) describe the contemporary roles of the finance function

Now read on …

### 3.1 Role of Finance Manager

The finance manager performs different roles in different business organizations. He or she is needed in almost all organization. The following are the major roles played by the finance manager.

#### 3.1.1 Estimating financial requirements

The first task of a financial manager is to estimate short-term and long-term financial requirements of his business. For that, he will prepare a financial plan for present as well as

for future. The amount required for purchasing tangible non-current assets as well as needs for working capital will have to be ascertained.

### 3.1.2 Deciding capital structure

Capital structure refers to kind and proportion of different securities for raising funds. After deciding the quantum of funds required it should be decided which type of securities should be raised. It may be wise to finance tangible non-current assets through long-term debts. Even here if gestation period is longer than share capital may be the most suitable. Long-term funds should be employed to finance working capital also, if not wholly then partially. Entirely depending on overdrafts and cash credits for meeting working capital needs may not be suitable. A decision about various sources for funds should be linked to the cost of raising funds.

### 3.1.3 Selecting a source of finance

An appropriate source of finance is selected after preparing a capital structure which includes share capital, debentures, financial institutions, public deposits etc. If finance is needed for short-term periods then banks, public deposits and financial institutions may be the appropriate. On the other hand, if long-term finance is required then share capital and debentures may be the useful.

### 3.1.4 Selecting a pattern of investment

When funds have been procured then a decision about investment pattern is to be taken. The selection of an investment pattern is related to the use of funds. A decision will have to be taken as to which assets are to be purchased? The funds will have to be spent first on tangible non-current assets and then an appropriate portion will be retained for working capital and for other requirements.

### 3.1.5 Proper cash management

Cash management is an important task of finance manager. He has to assess various cash needs at different times and then make arrangements for arranging cash. Cash may be required to purchase of raw materials, make payments to creditors, meet wage bills and meet day to day expenses. The idle cash with the business will mean that it is not properly used.

### 3.1.6 Implementing financial controls

An efficient system of financial management necessitates the use of various control devices. They are Return on Investment (ROI), cost-volume-profit (CVP) analysis, cost control, ratio analysis, cost and internal audit. ROI is the best control device in order to evaluate the performance of various financial policies.

### 3.1.7 Proper use of surpluses

The utilization of profits or surpluses is also an important factor in business finance. A judicious use of surpluses is essential for expansion and diversification plans and also in protecting the interests of shareholders. The ploughing back of profits is the best policy of further financing but it clashes with the interests of shareholders. A balance should be struck in using funds for paying dividend and retaining earnings for financing expansion plans.

## 3.2 Finance Function

The objective of finance function is to arrange as much funds for the business as are required from time to time. This function has the following objectives.

### 3.2.1 Assessing the financial requirements

The main objective of finance function is to assess the financial needs of an organization and then finding out suitable sources for raising them. The sources should be commensurate with the needs of the business. If funds are needed for longer periods then long-term sources like share capital, debentures, term loans may be explored.

### 3.2.2 Proper utilization of funds

Though raising of funds is important but their effective utilisation is more important. The funds should be used in such a way that maximum benefit is derived from them. The returns from their use should be more than their cost. It should be ensured that funds do not remain idle at any point of time. The funds committed to various operations should be effectively utilised. Those projects should be preferred which are beneficial to the business.

### 3.2.3 Increasing profitability

The planning and control of finance function aims at increasing profitability of the concern. It is true that money generates money. To increase profitability, sufficient funds will have to be invested. Finance Function should be so planned that the concern neither suffers from inadequacy of funds nor wastes more funds than required. A proper control should also be exercised so that scarce resources are not frittered away on uneconomical operations. The cost of acquiring funds also influences profitability of the business.

### 3.2.4. Maximizing value of firm

Finance function also aims at maximizing the value of the firm. It is generally said that a concern's value is linked with its profitability

## 3.3 Contemporary role of business finance

Due to recent trends in business environment, financial managers are identifying new ways through which finance function can generate great value to their organization.

### 3.3.1 Current Business Environment

The progress in financial analytics is because of development of new business models, trends in role of traditional finance department, alternations in business processes and progress in technology. Finance function in this vital environment emerged with enormous opportunities and challenges.

### 3.3.2 New Business Model

At the time when internet was introduced, three new e-business models have evolved. They are business-to-business (B2B), business-to-customer (B2C) and business-to-employees (B2E) future of financial analytics can be improved with the help of this new model of business.

Traditionally, financial analytical is emphasizing on utilization of tangible assets like cash, machinery etc., whereas some companies are mainly focused on intangible assets which are not easy to evaluate and control. Hence financial analytics solved this problem by:

(i) Recognizing the complete performance of organization.
(ii) Determining the source through which value of intangible assets can be evaluated and increased.
(iii) Predict the trends in market.
(iv) The abilities of information system are encouraged.
(v) Minimizes the operating costs and enterprise-wide investments are effectively controlled and upgrade the business processes.

### 3.3.3 Changing Role of the Finance Department

The role of finance function has been changing simultaneously with the changes in economy. These changes are mainly due to Enterprise Resource Planning (ERP), shared services and alternations in its reporting role. In the field of transaction processing, the role of financial staff has been widened up because of automated financial transactions. Now financial executives are not just processing and balancing transactions but they are focusing on decision-making processes. International organizations are facilitating their customers by providing financial information and facility to update both finance and non-finance functions from any place around the world. It resulted in the department of decision support in the organization. Finance professionals are held responsible for supplying suitable analytical tools and methods to decision makers.

(i) Business Processes: With the evolution of business processes, queries regarding business are becoming more complicated. In order to solve, it requires analytics with high level of data integration and organizational collaboration. In the last few decades, organizations are replacing function based legacy systems with new methods like ERP, BRP etc., in order to get accurate and consistent financial and non-financial information. In 1990's organizations were applying some modern systems like supply chain management, Customer Relationship Management (CRM) and many others to encourage their transactions. Overall organizations were building strong relations with customers.
(ii) Technology: With the developments in technology, ERP, internet, data warehousing have also improved. Internet helps in increasing the sources of acquiring financial data, whereas ERP vendors are building their own financial analytics which helps in evaluating the performance, planning and estimating, management and statutory reporting and financial consolidation. Till now, data warehousing solutions used to emphasize on developing elements of analytical infrastructure such as data stores, data marts and reporting applications but in future these data ware housings provide advances analytical abilities to data stores.
(iii) Integrated Analytics: To survive in this competitive environment, organizations must have advanced level of integrated financial analytics. Integrated financial analytics are useful for organizations to evaluate, combine and share information inside and outside the organization. Hence, with the progress in role of finance function, the financial analytics are used in organizations effectively.

**Summary**

Wonderful work you have done. You were able to read and understand this session in record time. You remembered that we looked at the role of finance managers and the finance function in the business organization. You further learned the contemporary roles of business finance. This implies that the roles change from time to time with the passage of time. Finance, therefore, is a living subject that we cannot just throw away.

**Self-Assessment Questions**
**Exercise 1.3**

## SESSION 4: OBJECTIVES OF THE FIRM

### Introduction

A firm is a commercial enterprise, a company or a business entity that buys and sells products and/or services to consumers with the aim of making a profit. It can also be defined as a business organization that buys or hires factors of production in order to produce goods and services to be sold at a profit. What is the objective of a firm? In whose interests is it governed? Is the primary objective or purpose of the firm's existence subordinated only to the shareholders or to all stakeholders, among which shareholders are just one of the interest groups without special status? There are two main theories on the primary firm's objective. These are shareholder theory and stakeholder theory that explain the objectives of the firm. We shall also look at other theories as well.

### Objectives

By the end of this session, you should be able to:

(a) distinguish between profit maximization and wealth maximization
(b) explain the arguments for and against wealth maximization
(c) discuss the satisficing as an objective of the firm
(d) state why corporate social responsibility is an objective of the firm

Now read on …

### 4.1 Shareholder theory or Wealth Maximization

According to the shareholder theory, the primary purpose of a firm is usually defined as value maximization (for shareholders). By this we refer to maximization of a firm's equity, which is in fact the present value of expected benefits (cash flows) that shareholders can expect from the firm. According to this definition, a firm's value can be maximized only when expected benefits are maximized in the long-run. If a firm is governed (predominantly or exclusively) in the shareholders' interests, then its governance aims at the **shareholder value maximization**. Wealth maximization is also known as value maximization or net present worth maximization. In this vein, the shareholder theory explains the rationale of the value maximization for shareholders.

#### 4.1.1 Favourable Arguments for Wealth Maximization

Several academics question why shareholders, as one of the firm's interest groups, should be treated as privileged. According to Easterbrook and Fischel (1996), this is because the shareholders take the majority of firm's risk. Namely, they are entitled to the residual after all other firm's obligations are paid. Their "claims" are subordinated to all other claims and are therefore entitled to the net cash flow of the firm. Consequently, a firm has to be managed in accordance with their interests. Some arguments in favour of this objective are as follows;

(i) Wealth maximization is superior to the profit maximization because the main aim of the business concern under this concept is to improve the value or wealth of the shareholders.
(ii) It considers both time and risk of the business concern.
(iii) Wealth maximization considers the comparison of the value to cost associated with the business concern. Total value detected from the total cost incurred for the business operation. It provides exact value of the business concern.
(iv) It ensures the economic interest of the society.
(v) Wealth maximization provides efficient allocation of resources.

#### 4.1.2 Unfavourable Arguments for Wealth Maximization

Shareholder wealth maximization being the ultimate goal of a business may not true in practice, mostly because of different ways in which the firm is conceived, its purpose, legal foundation, and political environment. Like parliamentary sovereignty, shareholder value is a doctrine that arises in a specific institutional context and sharply contrasts with the operating doctrines in other systems of corporate governance, such as in Germany and Japan.
Therefore, one cannot argue that that the shareholder value is the most appropriate objective a modern firm should follow. The arguments unfavourable to this are

(i) Wealth maximization leads to prescriptive idea of the business concern but it may not be suitable to present day business activities.
(ii) Wealth maximization creates ownership-management controversy.
(iii) Management alone enjoy certain benefits.

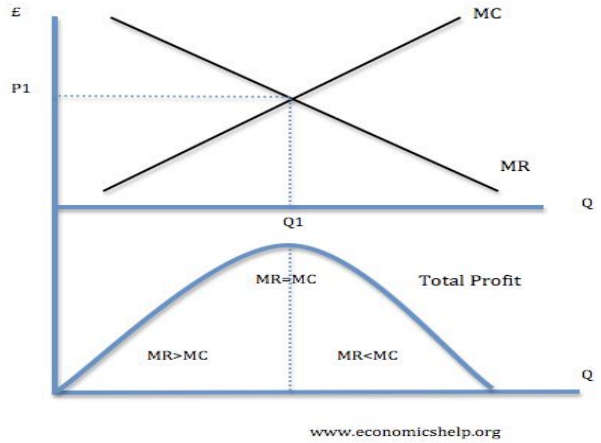
(iv) It can be activated only with the help of the profitable position of the business concern.

### 4.2 Stakeholder Theory

There is no universal view on the definition of the stakeholder theory, although the term is widely recognized as the obligation of the management to take into account the interests of anyone who has a significant "stake" in a firm. Proponents of the stakeholder approach argue that not only shareholders should be entitled to a firm's residual. This is important (at least) in a situation when workers engage in firm-specific investments in human capital, for example, in technology intensive production. In such firms, most of the added value comes from innovations, specialized products and specialized services. Such activities often demand investments in human capital, which make workers involved in the risk of a firm. In countries with strong trade unions, as for example in Denmark, workers are protected from dismissal at short notice and are in case of dismissal entitled to appropriate compensation.

### 4.3 Profit Maximization

Profit maximization aim of any kind of economic activity is earning profit. A business concern is also functioning mainly for the purpose of earning and maximizing the profit. Profit is the measuring techniques to understand the business efficiency of the concern. Profit is achieved when the Total Revenue (TR) is greater than the Total Costs (TC). Therefore, a firm can maximize profits if it produces at an output where marginal revenue (MR) = marginal cost (MC). This is illustrated in the diagram below




#### 4.3.1 Favourable Arguments for Profit Maximization

The following points below are in favour of the profit maximization objectives of the business concern:

(i) Profit is the parameter of the business operation.
(ii) Profit reduces risk of the business concern.
(iii) Profit is the main source of finance.
(iv) Profitability meets the social needs also.

#### 4.3.2 Unfavourable Arguments for Profit Maximization

The following are against the objectives of profit maximization:

(i) Profit maximization leads to exploiting workers and consumers.
(ii) Profit maximization creates immoral practices such as corrupt practice, unfair trade practice, etc.
(iii) Profit maximization objectives leads to inequalities among the stakeholders such as customers, suppliers, public shareholders, etc.

(iv) It does not consider risk of the business concern. Risks may be internal or external which will affect the overall operation of the business concern.

(v) It ignores the time value of money: Profit maximization does not consider the time value of money or the net present value of the cash inflow. It leads certain differences between the actual cash inflow and net present cash flow during a particular period.

**4.4 Satisficing**

Satisficing means the firm is producing satisfactorily but not maximum profit. Firms are more likely to satisfice than maximise, that is, they will make what is acceptable and satisfactory rather than achieve the optimal solution. This means that the firm will make sufficient profits in order to keep shareholders happy but will not waste time and resources seeking the optimal solution. A firm that is satisficing may produce a range of outputs that are within its target level of profits rather than the specific profit-maximising output.

**4.5 Sales Maximisation Theory**

Suggests that managers want the firms they work for to be as large as possible. As working as a high-level manager for a very large corporation is an extremely prestigious position. Managers may be receiving sales-related bonuses. In this case, the manager might increase sales up to the point where **MR is zero** and **TR is maximised.** A further argument is that sales are the key to **market share** and possibly **market power**, and that growth is the key to future profits and managerial security. Managers may be reluctant to undertake short-term risky ventures, even if the profits are large as failure to achieve success could terminate their careers.

Firms often seek to increase their market share even if it means less profit. This could be for the following reasons:

i. Increasing market share may force rivals out of business. E.g. the growth of supermarkets have led to the demise of many local shops. Some firms may actually engage in predatory pricing which involves making a loss to force a rival out of business.
ii. Increased market share increases monopoly power and may enable the firm to put up prices and make more profit in the long run.
iii. Managers prefer to work for bigger companies as it leads to greater prestige and higher salaries.

**4.6 Social Responsibility**

Some large firms now include 'social responsibility' among their objectives, which means they pay due regard to the needs of the stakeholders of the business. A firm may incur extra expense to choose products which don't harm the environment or products not tested on animals. Alternatively, firms may be concerned about local community / charitable concerns. These so-called activities of corporate citizenship are likely to increase the firm's costs and so reduce its level of profits. However, some stakeholders argue that such caring activities do not always conflict with profit maximisation. For example, good staff are easier to attract and retain by firms that are seen as socially responsible, and this reduces the costs of staff turnover

**Summary**
In this session, we have been able to look at the various objectives of the firm from profit maximization, wealth maximization, satisficing, sales maximization and social responsibility. So, you understand that the firm is not just established but for a particular objective.

**Self-Assessment Questions**
**Exercise 1.4**

## SESSION 5: DECISION AREAS IN BUSINESS FINANCE

**Introduction**
Financial management is concerned with the acquisition, financing, and management of assets with some overall goal in mind. Thus, the decision function of financial management can be broken down into three major areas: Investment decisions, Financing decisions; and Asset management decisions.

**Objectives**

By the end of this session, you should be able to:

(a) explain investment decision of a firm
(b) discuss the financing decisions of the firm
(c) explain the asset management decisions of the firm
(d) outline the working capital management decisions of a firm

**5.1 Investment Decision (capital budgeting)**

The investment decision is the most important of the firm's three major decisions when it comes to value creation. Decisions on investment, which take time to mature, have to be based on the returns which that investment will make. Unless the project is for social reasons only, if the investment is unprofitable in the long run, it is unwise to invest in it now. More often than not, it would be good to know what the present value of the future investment is, or how long it will take to mature (give returns). It could be much more profitable putting the planned investment money in the bank and earning interest, or investing in an alternative project. For example, how much of the firm's total assets should be devoted to cash or to inventory? Also, the flip side of investment – disinvestment – must not be ignored. Assets that can no longer be economically justified may need to be reduced, eliminated, or replaced.

**5.2 Financing Decision (capital structure)**

The second major decision of the firm is the financing decision. Here the financial manager is concerned with the makeup of the liabilities and the equity side of the statement of financial position. If you look at the mix of financing for firms across industries, you will see marked differences. Some firms have relatively large amounts of debt, whereas others are almost debt free. Does the type of financing employed make a difference? If so, why? And, in some sense, can a certain mix of financing be thought of as best?

In addition, dividend policy must be viewed as an integral part of the firm's financing decision. The dividend-pay-out ratio determines the amount of earnings that can be retained in the firm. Retaining a greater amount of current earnings in the firm means that fewer Ghana cedis will be available for current dividend payments. The value of the dividends paid to shareholders must therefore be balanced against the opportunity cost of retained earnings lost as a means of equity financing.

Once the mix of financing has been decided, the financial manager must still determine how best to physically acquire the needed funds. The mechanics of getting a short-term loan, entering into a long-term lease arrangement, or negotiating a sale of bonds or shares must be understood.

**5.3 Asset Management Decision**

The third important decision of the firm is the asset management decision. Once assets have been acquired and appropriate financing provided, these assets must still be managed efficiently. The financial manager is charged with varying degrees of operating responsibility over existing assets. These responsibilities require that the financial manager be more concerned with the management of current assets than with that of non-current assets. A large share of the responsibility for the management of non-current assets would reside with the operating managers who employ these assets.

**5.4 Dividend Decision**

The financial manager must decide whether profit should be distributed or retained, or distribute a proportion and retain the balance. The policy (dividend policy) decision must be within the context of the business' objectives.

**5.5 Working Capital Management**

It refers to a business' short-term assets, such as inventory, and its short-term liabilities, such as money owed to suppliers. Managing a business' working capital is a day-to-day activity that ensures the business has sufficient resources to continue its operations and avoid costly interruptions. In view of this, the following questions must be answered:

- How much cash and inventory should be kept on hand?
- Should the business sell on credit?
- How will the business obtain short-term credit?
- If we borrow short-term, how and when should we do it?

**Summary**

At the end of this session, studied the decision areas in finance. Investment decision, financing decision, asset management decision, dividend decision and working capital management decisions were explained in this session. You see why we said finance is a living subject that we cannot throw away?

**Self-Assessment Questions**

**Exercise 1.5**

**SESSION 6: AGENCY PROBLEM**

**Introduction**

You are welcome to the final session of unit 1. In this session, we shall discuss the agency problem and how it can be resolved. The causes of agency problems will also be explained in this session. At the end of this session, we expect that you are geared for the course in business finance at the University of Competitive Choice. UCC, we are the loviest of them all.

**Objectives**

By the end of this session, you should be able to:

(a) explain agency theory
(b) describe the agency problem
(c) identify the causes of agency problem
(d) suggest ways of solving agency problem in organizations

Now read on …

**6.1 Agency theory**

One area of continuing financial research has been agency theory. This theory examines the relationship between the owners and the managers of the business. Agency relationship arises whenever one or more individuals, called principal(s);

- Hires another individual or organization, called an agent, to perform some service and
- Then delegates decision-making authority to that agent.

In business finance context, the primary agency relationships are those between shareholders and managers, and managers and debt-holders.

**6.2 Meaning of agency problem**

In privately owned businesses, management and owners are usually the same people. Management operates the business to satisfy its own goals, needs, and financial requirements. However, as a company moves from private to public ownership, management now represents all the owners. This places management in the agency position of making decisions that will be in the best interest of all shareholders. Because of diversified ownership interests, conflict between managers and shareholders can arise that will impact financial decisions of the business. Such conflicts are called agency problem.

**6.3 Forces that give rise to agency problem**

The forces that give rise to agency problem between management and shareholders include:

(i) Separation of ownership from control: Those who own the company do not manage it.
(ii) Information symmetry: A situation where management has a clear advantage of hiding financial and non-financial reports from owners. Management therefore know it all while shareholders know only what the managers wish them to know.
(iii) Goals of managers may differ from those of owners. For instance, management may
    1. seek to entrench managerial power by creating job security for themselves
    2. increase managerial rewards and enhancements
    3. pursue their own social objectives, such as traveling ostentatiously with their family and undertaking prestigious projects.

**6.4 How agency problem arises**

Agency problem can also exist between management and debt-holders/creditors when management takes decisions which adversely affect the safety of creditors' debts such as:

(i) Approving excessive dividends, the payment of which may pose danger to future payments of interest and principal.

(ii) Going for more debt without considering the implications on existing debt-holders.
(iii)Undertaking highly risky business ventures.

Suppose you hire someone to sell your car and you agreed to compensate him a flat fee when the car is bought. The agent's incentive in this case is to make the sale, not necessarily to get you the best price. If you paid a commission of say 5% of the sales price instead of a flat fee, then this problem might not exist. In much the same way, just as managers are not the owners of the business, they might be tempted to act in ways that are not in the best interest of the owners. For example, they might buy luxury corporate cars for their travel, or overindulge in expense-account dinners. They might shy away from attractive but risky projects because they are worried about the safety of their jobs than the potential for superior profits. They might be engaged in empire building, adding unnecessary or hire employees. Such problems can arise because the managers of the firm may have their own axes to grind.

**6.5 Agency costs**

To reduce agency problems and associated moral hazard problems created by management actions, stockholders must incur agency costs. The three (3) major categories of agency costs include:

(i) expenditure to monitor managerial actions such as audit costs
(ii) expenditures to structure the firm in a way that will limit undesirable managerial behaviour such as appointing outside investors to the board of directors, and
(iii)opportunity costs which are incurred when shareholders impose restrictions.

**6.6 Ways of resolving agency problems**

Agency problem between shareholders and managers can be resolved as follows:

(i) Tying managerial compensation to financial performance of the firm in general, or to the performance of the share value on the stock market in particular. Managerial compensation may take the form of remuneration increases, promotion, etc.
(ii) Direct intervention by shareholders to influence the decisions of management through:
    - Talking with the firm's management and making suggestions regarding how the business should be run, i.e. lobbying
    - Proposal made by majority shareholder(s) to be voted on at the AGM, even if it is opposed by management.
(iii)Threat of firing management if they are not acting in stockholders' interest.
(iv)Threat of takeovers, particularly hostile takeovers (a takeover usually opposed by management) which occurs when the firm's stocks have been undervalued relatively due to poor management.
(v) In the same vein, the agency problem between management and creditors can be resolved by debt-holders as follows;
    - Debt-holders can protect their interest by securing debt against the company's assets.
    - Use of restrictive covenants to restrict the firm from engaging in highly risky ventures, from paying excessive dividends and from engaging in high levels of gearing.

**Summary**

Finally, we have ended this session and the unit. We learned agency theory and the agency problem. We examined the causes of agency problem and ways of resolving them. We also looked at the costs associated with agency problem.

**Self-Assessment Questions**
**Exercise 1.6**

## UNIT 2: SOURCES OF FINANCE

**Unit Outline**

Session 1: Trade Credit, Instalment Credit and Bank Credit
Session 2: Customers' Advances, Factoring, Bill of Exchange
Session 3: Merits and demerits of short-term finance
Session 4: Shares, Debentures, Bonds and Retained Earnings
Session 5: Venture Capital, Angel Investors and Leasing
Session 6: Capitalization

**Overview**

Growing a business from the first seed of an idea is not a smooth linear journey and it's not as simple as going from A to B. The destination is seldom decided as the business idea takes form, becomes a reality and then grows into a successful enterprise. The finance journey is continuous; there may never be an arrival point. For any business to travel on a journey it needs at all points of that journey to be appropriately financed. Businesses need to make sure there is the finance to back their growth plans. Businesses are often started on overdrafts or credit cards, or with help from friends or family or by using the family property as collateral. But soon after that the business will need to be financed so it can stand on its own two feet if it is to be a sustainably growing proposition.

Financing is needed to start a business and ramp it up to profitability. There are several sources
to consider when looking for start-up financing. But first you need to consider how much money you need and when you will need it. The financial needs of a business will vary according to the type and size of the business. For example, processing businesses are usually capital intensive, requiring large amounts of capital. Retail businesses usually require less capital. Debt and equity are the two major sources of financing. Government grants to finance certain aspects of a business may be an option. Also, incentives may be available to locate in certain communities and/or encourage activities in particular industries. In this unit, we shall discuss both short-term and long-term sources of finance and also examine their relative merits and demerits.

**Objectives**

By the end of this Unit, you should be able to:

(a) discuss the sources of short-term funds for a business
(b) explain the merits and demerits of short-term funds
(c) demonstrate understanding of customers' advances, factoring, bill of exchange
(d) discuss bonds, shares and retained earnings as long-term sources of funds
(e) distinguish between angel investors and venture capital firms
(f) describe capitalizations and its various types

## SESSION 1: TRADE CREDIT, INSTALMENT CREDIT AND BANK CREDIT

### Introduction

Sources of finance are the most explored area especially for the entrepreneurs about to start a new business. It is perhaps the toughest part of all the efforts. There are various sources of finance classified based on time period, ownership and control, and source of generation of finance. On the basis of a time period, sources are classified into long term, medium term, and short term. Ownership and control classify sources of finance into owned capital and borrowed capital. Internal sources and external sources are the two sources of generation of capital. All the sources of capital have different characteristics to suit different types of requirements.

### Objectives

By the end of this session, you should be able to:

(a) state and explain the sources of short-term funds
(b) discuss the types of bank credit
(c) outline the merits and demerits of instalment credit
(d) distinguish between bank loan and cash credit

Now read on …

### 1.1 Short-term sources of funds

Short-term financing means financing for a period of less than one year. Need for short-term finance arises to finance the current assets of a business like an inventory of raw material and finished goods, debtors, minimum cash and bank balance etc. Short term financing is also named as working capital financing. The following are examples of short-term sources: Trade credit, Bank credit such as Loans and advances, Cash credit, Overdraft, Discounting of bills; and the rest include Customers' advances, Factoring services, Instalment credit and Loans from credit unions. We shall take you through these in the next discussion.

### 1.2 Trade Credit

Trade credit refers to credit granted to manufactures and traders by the suppliers of raw material, finished goods, components, etc. Usually business enterprises buy supplies on a 30 to 90 days credit. This means that the goods are delivered but payments are not made until the expiry of period of credit. This type of credit does not make the funds available in cash but it facilitates purchases without making immediate payment.

Trade credit a short-term credit facility extended by the creditors to the debtors. Normally, it is common for the traders to buy the material and other supplies from the suppliers on credit basis. After selling the stocks, the traders pay the cash and buy fresh stocks (also known as inventories) again on credit. Sometimes, the suppliers may insist on the buyer to sign a bill (bill of exchange). This bill is called bills payable.

### 1.3 Instalment credit

Instalment credit is another source of short-term financing, in which the borrowed amount is paid in equal installments with interest. It is also called instalment plan or hire-purchase plan. Instalment credit is granted to the enterprise by the suppliers on the assurance that the repayment would be done in fixed installment at regular intervals of time. It is mostly used to acquire long-term assets used in production processes.

Instalment credit is now-a-days a popular source of finance for consumer goods like television, refrigerators as well as for industrial goods. Only a small amount of money is paid at the time of delivery of such articles. The balance is paid in a number of instalments. The supplier charges interest for extending credit. The amount of interest is included while deciding on the amount of instalment. Another comparable system is the hire purchase system under which the purchaser becomes owner of the goods after the payment of last instalment. Sometimes commercial banks also grant instalment credit if they have suitable arrangements with the suppliers.

#### 1.3.1 Advantages of instalment credit

The advantages of instalment credit are as follows:

(i) Convenient Mode of Payment – It implies that instalment credit is an easy mode of payment as it divides the burden of payment in easy instalments paid at regular intervals.

(ii) Protecting Blockage of Funds – It refers to the fact that instalment credit helps the enterprise in saving capital, which can be used for other productive activities. In helps the enterprise in purchasing goods and services by making a part of payment.

(iii) Facilitating Modernization – It implies that instalment credit helps the enterprise in acquiring new machines and technology even in the absence of sufficient funds for the time being.

(iv) Quick Possession of Assets – It requires very little paperwork to transfer the ownership of assets from one party to another.

#### 1.3.2 Disadvantages of instalment credit

The disadvantages of instalment credit are as follows:

(i) Influence on Liquidity Position – It refers to the impact of instalments on the liquidity position of the enterprise. The payment of instalment is considered as an additional burden on the short-term capital of the enterprise.

(ii) Extra Cost – It refers to the extra amount to be paid by the enterprise in the procurement of goods through instalment credit. If an enterprise buys goods on instalment credit then it needs to pay higher amount as compared to one-time payment. This happens because the instalments include the amount of borrowing and the interest.

(iii) Extra Liability – It refers to the extra burden imposed on the enterprise in case of default. If the enterprise fails to pay the instalment amount in the allotted time, it may seriously affect the image of the enterprise. Therefore, it becomes the liability of the enterprise to pay instalment on time.

### 1.5 Bank credit

Commercial banks grant short-term finance to business firms which is known as bank credit. When bank credit is granted, the borrower gets a right to draw the amount of credit at one

time or in instalments as and when needed. Bank credit may be granted by way of loans, cash credit, overdraft and discounted bills. Banks sometimes require securities on credits granted.

### 1.5.1 Bank loans

When a certain amount is advanced by a bank repayable after a specified period, it is known as bank loan. Such advance is credited to a separate loan account and the borrower has to pay interest on the whole amount of loan irrespective of the amount of loan actually drawn. Usually loans are granted against security of assets. These loans are usually given for a period of one year or less. The loans can be given by other financial institutions such as savings and loans companies and microfinance institutions.

### 1.5.2 Cash credit

It is an arrangement whereby banks allow the borrower to withdraw money up to a specified limit. This limit is known as cash credit limit. Initially this limit is granted for one year. This limit can be extended after review for another year. However, if the borrower still desires to continue the limit, it must be renewed after three years. Rate of interest varies depending upon the amount of limit. Banks ask for collateral security for the grant of cash credit. In this arrangement, the borrower can draw, repay and again draw the amount within the sanctioned limit. Interest is charged only on the amount actually withdrawn and not on the amount of entire limit. It operates in the form of a credit card. Example Barclays Bank Credit Card.

### 1.5.3 Bank overdraft

When a bank allows its depositors or account holders to withdraw money in excess of the balance in his account up to a specified limit, it is known as overdraft facility. This limit is granted purely on the basis of credit-worthiness of the borrower. In this system, the borrower has to show a positive balance in his account. Interest is charged only on the overdrawn money. Rate of interest in case of overdraft is less than the rate charged under cash credit. The granting of the overdraft facility is based to the application submitted by the client of the bank. This makes available short-term funds for the business.

### 1.5.4 Discounting bills

Banks also advance money by discounting bills of exchange, promissory notes etc. When these documents are presented before the bank for discounting, banks credit the amount to customer's account after deducting discount. The amount of discount is equal to the amount of interest for the period of bill. Sometimes it could be in the form of invoice discounting. Taking a credit on the face value of outstanding credit sales.

## 1.6 Securities for bank credit

Loans and advances are granted by bank on personal security of the borrower as well as on the security of some tangible assets, besides the standing of the firm. Securities against credit may be of two types: personal security and security of tangible assets. Personal security means the credit-worthiness of the borrower. Banks judge the credit-worthiness of the borrower on the basis of his or her financial soundness and past dealings with the bank. Securities for intangible assets include: Moveable goods, shares certificates, documents of title to goods e.g. bills of lading; fixed deposit receipts, Life insurance policies and Precious metals like gold, diamond etc.

**Summary**

In this session, we found out that there are different sources of short-term funds a business can fall on at any given time. We examined trade credit, instalment payments and bank

credit. We discussed the different types of bank credit and securities that are needed for a bank credit to be granted.

**Self-Assessment Questions**
**Exercise 2.1**

## SESSION 2: CUSTOMERS' ADVANCES, FACTORING, BILL OF EXCHANGE

**Introduction**

Dear students, welcome to the continuation of the short-term sources of funds for a business. Here, we shall discuss customers' advances, factoring and bill of exchange. These will give us the idea that, it is not always that we need physical cash to start business.

**Objectives**

By the end of this session, you should be able to:

(a) explain what is meant by customers' advance
(b) examine factoring as a source of funds for a business
(c) outline the advantages and disadvantages of customers advances
(d) describe a bill of exchange as a source of short-term funds
(e) explain the importance features of a bill of exchange

Now read on …

### 2.1 Meaning of customers' advance

Sometimes businessmen insist on their customers to make some advance payment. It is generally asked when the value of order is quite large or things ordered are very costly. Customers' advance represents a part of the payment towards price on the product(s) which will be delivered at a later date. Customers generally agree to make advances when such goods are not easily available in the market or there is an urgent need of goods. A firm can meet its short-term requirements with the help of customers' advances.

Customer advances may be defined as the part of payment made in advance by the customer to the enterprise for the procurement of goods and services in the future. It is also called Cash before Delivery (CBD). The customers pay the amount of advance, when they place the order of goods and services required by them. The method of procuring goods and services depends on the characteristic and value of the product. Customer advances allow customers to defer their payment for some time and fulfil their other obligation on priority.

#### 2.1.1 Advantages of customers' advance

The advantages of customer advances are as follows:

(i) Free from Interest Burden – It implies that the enterprise does not require paying any interest on customer advances
(ii) No Security Required – It implies that the enterprise does not need to keep any security to raise customer advances.
(iii) No Repayment Obligation – It refers to the fact that the enterprise is free to decide whether to refund money, if the order is cancelled by customers.

### 2.1.2 Disadvantages of customers' advance

The disadvantages of customer advances are as follows:

(i) Limited to Selected Enterprise – It refers to the fact that the benefits of customer advances can be availed only by those enterprises, which have goodwill in the market.
(i) Limited Period Offer – Limited period offer implies that the customer advances can be availed only for limited period of time. The allotted time for the advances and delivery of goods and services is fixed and cannot be extended.
(ii) Limited Amount of Advance – It refers to the fact that customers made only a part of payment, which may not fulfil the fund requirement of the enterprise. In addition, the amount of advance is proportional to the value of the product; therefore, it may vary from product to product.

## 2.2 Factoring

Factoring companies offer a range of services in the area of sales administration and the collection of amounts due from trade receivables. A factor can take over the administration of sales invoicing and accounting for a client company, together with collecting amounts due from trade receivables and chasing up any slow payers.

A factor can offer a cash advance against the security of trade receivables, allowing a company ready access to cash as soon as credit sales are made. For an additional fee, a factor can take on any bad debts that may arise through non-payment. This is called non-recourse factoring, since here the factor does not have recourse to the company for compensation in the event of non-payment.

Factoring comprises complementary financial services, which is provided to the borrowers. The borrower has freedom to select the set of services provided by the factoring enterprise. Factoring is a transaction whereby a business sells its statement of receivable to a factor at a discount rate, to raise fund for financing short-term projects.

### 2.2.1 Advantages of factoring

The advantages of factoring are as follows:

(i) Flexibility – Flexibility helps in ensuring that an enterprise can relax certain procedures for extending better services to clients.

(ii) Cope up with Uncertainties – It refers to the fact that factoring enables the enterprise to meet seasonal demands for cash whenever required.

(iii) Better Fund Usage – It refers to the efficient utilization of funds because the enterprise takes decision with the help of advice provided by the factor.

### 2.2.2 Disadvantages of factoring

The disadvantages of factoring are as follows:

(i) Discounted Value – Discounting value refers to the fact that factoring enterprise does not pay the face value of the account receivable to the client enterprise. The factoring enterprise pays discounted value of the account receivable to the client enterprise.

(ii) Failure in maintaining Customer Relation – It refers to the fact that if the factor is not able to satisfy the client enterprise, he/she would not get business from the enterprise in future.

## 2.3 Bill of Exchange

A bill of exchange is a document in which an individual asks the recipient to make payment for goods and services received to a third party at a future date. The individual who writes the bill is known as drawer and the individual who receives the bill is known as drawee. The individual who pays the bill is known as payee. One of the popular forms of bills of exchange is cheque, which is widely used in the market. The cheques used are usually post-dated cheques.

**The important features of bills of exchange are as follows:**

(i) Produces in written format, such as draft or cheque.

(ii) It includes an order made by one party to second party to pay the amount to a third party.

(iii) The bills of exchange make it mandatory for the drawer and drawee to sign the bill. If the bill does not contain the signs of drawer and drawee, it cannot be considered as a legal document.

(iv) It mentions the time at which the payment is to be made.

**Summary**

Holala!!! Again, you have done it. You just learnt about customers' advance, factoring and bill of exchange as sources of funds for a business within a period of one year or less. We looked at the merits and demerits of customers' advances and critical features of a bill of exchange. We hope you will exploit these sources when establishing your business in the future.

**Self-Assessment Questions**

**Exercise 2.2**

## SESSION 3: MERITS AND DEMERITS OF SHORT-TERM FINANCE

### Introduction

The short-term financing is an ultimate source to manage the working capital requirements of an enterprise. The need of short-term funds varies from enterprise to enterprise and depends on the nature of business and production cycle. Short-term loans help business concerns to meet their temporary requirements of money. They do not create a heavy burden of interest on the organisation. But sometimes organisations keep away from such loans because of uncertainty and other reasons. Let us examine the purpose, merits and demerits of short-term finance.

### Objectives

By the end of this session, you should be able to:

(a) state and explain the purpose of short-term finance
(b) discuss the merits of short-term finance
(c) examined the demerits of short-term finance

Now read on …

### 3.1 Purpose of Short-term Finance

After establishment of a business, funds are required to meet its day to day expenses. For example, raw materials must be purchased at regular intervals, workers must be paid wages regularly, water and power charges have to be paid regularly. Thus, there is a continuous necessity of liquid cash to be available for meeting these expenses. For financing such requirements short-term funds are needed. The availability of short-term funds is essential.

Inadequacy of short-term funds may even lead to closure of business. Short-term finance serves following purposes:

(i) Smooth Running of Business – It refers to the continuity in the operations of the enterprise. The short-term fund helps in fulfilling day-to-day financial needs of the enterprise. It facilitates the smooth running of business operations by meeting day to day financial requirements.

(ii) It enables firms to hold stock of raw materials and finished product.

(iii) With the availability of short-term finance goods can be sold on credit. Sales are for a certain period and collection of money from debtors takes time. During this time gap, production continues and money will be needed to finance various operations of the business.

(iv) Increase in Productivity – It refers to the use of short-term fund to enhance the production level of the enterprise. The proper utilization of fund for the procurement of labour, material, and machine can increase the productivity of the enterprise. Short-term finance becomes more essential when it is necessary to increase the volume of production at a short notice.

(v) Fulfilment of Emergency Needs – It implies that short-term finance is needed to meet any urgent or emergency requirement of funds. Short-term funds are also required to allow flow of cash during the operating cycle. Operating cycle refers to the time gap between commencement of production and realisation of sales.

(vi) Business Cycle – Business cycle brings fluctuations in the demand and supply of products and services. If there is boom in the market, the demand of the product increases. In such a situation, the enterprise needs to raise the short-term funds to increase the supply of the product. Therefore, the enterprise requires short-term funds to cope up with the temporary rise in the demand of the product.

(vii) Liquidity Position – Liquidity position refers to the capability of an enterprise to meet the immediate cash need for managing the current liabilities. The short-term financing helps in maintaining the liquidity position of an enterprise by providing excess cash to an enterprise.

(viii) Operational Efficiency – It requires the replacement of old machines and equipment by new ones and employment of skilled labour and application of latest technology to improve the efficiency of the enterprise. The short-term financing helps in fulfilling these requirements of an enterprise.

(ix) Nature of Business – It refers to the different types of business operations carried out by enterprises. The requirement of short-term financing varies as per the nature of business of the enterprise. For example, a manufacturing enterprise requires more short-term capital because it needs to procure raw material on a regular basis.

(x) Selling goods on Credit – Selling goods on credit reduces the cash balance of the enterprise. Therefore, if an enterprise decides to sell goods on credit, it needs to retain sufficient amount of short-term fund to carry out its day-to-day operations smoothly.

### 3.2 Merits of short-term finance

(i) Economical

Finance for short-term purposes can be arranged at a short notice and does not involve any cost of raising. The amount of interest payable is also affordable. It is, thus, relatively more economical to raise short-term finance.

(ii) Flexibility

Loans to meet short-term financial need can be raised as and when required. These can be paid back if not required. This provides flexibility.

(iii) No interference in management

The lenders of short-term finance cannot interfere with the management of the borrowing concern. The management retain their freedom in decision making.

(iv) May also serve long-term purposes

Generally business firms keep on renewing short-term credit, e.g., cash credit is granted for one year but it can be extended up to 3 years with annual review. After three years it can be renewed. Thus, sources of short-term finance may sometimes provide funds for long-term purposes.

**3.3 Demerits of short-term finance**

Short-term finance suffers from a few demerits which are listed below:

(i) Fixed Burden

Like all borrowings interest has to be paid on short-term loans irrespective of profit or loss earned by the organisation. That is why business firms use short-term finance only for temporary purposes.

(ii) Charge on assets

Generally short-term finance is raised on the basis of security of moveable assets. In such a case the borrowing concern cannot raise further loans against the security of these assets nor can these be sold until the loan is cleared (repaid).

(iii) Difficulty of raising finance

When business firms suffer intermittent losses of huge amount or market demand is declining or industry is in recession, it loses its creditworthiness. In such circumstances they find it difficult to borrow from banks or other sources of short-term finance. This can be likened to a situation when coronavirus pandemic affected Ghana in 2020. Businesses find it difficult during this period to raise short-term funds for their operations.

(iv) Uncertainty

In cases of crisis, business firms always face the uncertainty of securing funds from sources of short-term finance. If the amount of finance required is large, it is also more uncertain to get the finance. This was the case in Ghana from 2017 to 2019 when the country was hit by the banking crisis that resulted in some banks been merged into one consolidated bank, some banks been closed down and a number of microfinance companies losing their licenses to operation in the country.

(v) Legal formalities

Sometimes certain legal formalities are to be complied with for raising finance from short-term sources. If shares are to be deposited as security, then transfer deed must be prepared. Such formalities take lot of time and create lot of complications.

**Summary**

In this session, we studied the purpose of short-term finance and examined the merits and demerits of short-term finance. These will guide us anytime we are going for short-term financing for a business. These factors discussed must be thoroughly analysed and tactically personalized before going for the money. What might be advantageous to one business might be disadvantageous to the other.

**Self-Assessment Questions**
**Exercise 2.3**

## SESSION 4: SHARES, DEBENTURES, BONDS AND RETAINED EARNINGS

### Introduction

Long-term financing means capital requirements for a period of more than 5 years to 10, 15, 20 years or maybe more depending on other factors. Capital expenditures in tangible non-current assets like plant and machinery, land and building etc. of a business are funded using long-term sources of finance. Part of working capital which permanently stays with the business is also financed with long-term sources of finance. In this session we shall examine the different types of long-term financing such as shares, bonds and debentures.

### Objectives

By the end of this session, you should be able to:

(a) explain security finance
(b) discuss shares as a source of finance for a business
(c) discuss debentures as source of long-term funds
(d) explain bonds as a source of finance for a business
(e) describe the relevance of retained earnings as a source of business funds

Now read on …

### 4.1 Sources of long-term finance

The sources of long-term finance refer to the institutions or agencies from which, or through which finance for a long period can be procured. In the case of sole traders and partnership firms, long-term funds are generally provided by the owners themselves and by retained profits. But in companies whose financial requirements are rather large, the following are the sources from, or through which long-term funds are raised: shares, bonds, debentures, leasing,

### 4.2 Long-term security finance

If the finance is mobilized through issue of securities such as shares, bonds and debenture, it is
called security finance. It is also called as corporate securities. This type of finance plays a major role in the field of deciding the capital structure of the company. Security finance consists of the following important characteristics:

(i) Long-term sources of finance.
(ii) It is also called as corporate securities.
(iii) Security finance includes both shares and debentures.
(iv) It plays a major role in deciding the capital structure of the company.
(v) Repayment of finance is very limited.
(vi) It is a major part of the company's total capitalization.

Types of Security Finance Security finance may be divided into two major types:

- Ownership securities or capital stock or equity finance; example shares
- Creditorship securities or debt capital; example bonds and debentures

### 4.3 Shares

The ownership securities also called capital stock or equity financing is commonly called shares. Shares are the most universal method of raising finance for the business concern. Ownership capital consists of the following types of securities: Equity Shares and Preference Shares in Ghana. These shares are usually issued at no par value.

#### 4.3.1 Equity Shares

Equity Shares also known as ordinary shares. Equity shareholders are the real owners of the company. They have a control over the management of the company. Equity shareholders are eligible to get dividend if the company earns profit. Equity share capital cannot be redeemed during the lifetime of the company. The liability of the equity shareholders is the value of unpaid value of shares. Equity shares have the following important features:

(i) Maturity of the shares: Equity shares have permanent nature of capital, which means it has no maturity period. It cannot be redeemed during the lifetime of the company.
(ii) Residual claim on income: Equity shareholders have the right to get income left after paying fixed rate of dividend to preference shareholder. The earnings or the income available to the shareholders is equal to the profit after tax minus preference dividend.
(iii) Residual claims on assets: If the company wound up, the ordinary or equity shareholders have the right to get the claims on assets. These rights are only available to the equity shareholders.
(iv) Right to control: Equity shareholders are the real owners of the company. Hence, they have power to control the management of the company and they have power to take any decision regarding the business operation.

(v) Voting rights: Equity shareholders have voting rights in the meeting of the company with the help of voting right power; they can change or remove any decision of the business concern. Equity shareholders only have voting rights in the company meeting and also, they can nominate proxy to participate and vote in the meeting instead of the shareholder.

(vi) Pre-emptive right: Equity shareholder pre-emptive rights. The pre-emptive right is the legal right of the existing shareholders. It is attested by the company in the first opportunity to purchase additional equity shares in proportion to their current holding capacity.

(vii) Limited liability: Equity shareholders are having only limited liability to the value of shares they have purchased. If the shareholders are having fully paid up shares, they have no liability. We are fully aware you discussed this under Company and Partnership Law.

### 4.3.2 Preference Shares

Preference shares are part of the corporate securities of a firm. It is the shares, which have preferential right to get dividend and get back the initial investment at the time of winding up of the company. Preference shareholders are eligible to get fixed rate of dividend and they have voting rights in accordance with the Companies Act of 2019, Act 992. It means a preference shareholder enjoys two rights over equity shareholders :(a) right to receive fixed rate of dividend and (b) right to return of capital. After settling the claims of outsiders, preference shareholders are the first to get their dividend and then the balance will go to the equity shareholders. However, the preference shareholders do not have any voting rights in the annual general meetings of the company. Preference shares may be classified into the following major types in Ghana:

(i) Cumulative preference shares: Cumulative preference shares have right to claim dividends for those years which have no profits. If the company is unable to earn profit in any one or more years, C.P. Shares are unable to get any dividend but they have right to get the comparative dividend for the previous years if the company earned profit

(ii) Non-cumulative preference shares: Non-cumulative preference shares have no right to enjoy the above benefits. They are eligible to get only dividend if the company earns profit during the years. Otherwise, they cannot claim any dividend.

(iii) Redeemable preference shares: When, the preference shares have a fixed maturity period it becomes redeemable preference shares. It can be redeemable during the lifetime of the company. The Companies Act of 2019, S.62 has provided certain restrictions on the return of the redeemable preference shares.

(iv) Irredeemable Preference Shares: Irredeemable preference shares can be redeemed only when the company goes for liquidator. There is no fixed maturity period for such kind of preference shares.

(v) Convertible Preference Shares: Convertible preference shareholders have right to convert their holding into equity shares after a specific period. The constitution of the company must authorize the right of conversion.

(vi) Non-convertible Preference Shares: These shares, cannot be converted into equity shares.

### 4.4 Debentures

Creditorship Securities also known as debt finance which means the finance is mobilized from the creditors. Debenture and Bonds are the two major parts of the Creditorship Securities. Debenture is a document issued by the company. It is a certificate issued by the

company under its seal acknowledging a debt. Debentures are the loans taken by the company. It is a certificate or letter issued by the company under its common seal acknowledging the receipt of loan. A debenture holder is the creditor of the company. A debenture holder is entitled to a fixed rate of interest on the debenture amount. Payment of interest on debenture is the first charge against profits. Apart from the loans from financial institutions, a company may raise loans through debentures. This is an additional source of long-term finance. The payment of interest and principal amounts on these debentures is subject to the terms and conditions of issue of debentures. Debentures may be divided into the following major types:

(i) Unsecured debentures: Unsecured debentures are not given any security on assets of the company. It is also called simple or naked debentures. This type of debenture is traded as unsecured creditors at the time of winding up of the company.
(ii) Secured debentures: Secured debentures are given security on assets of the company. It is also called as mortgaged debentures because these debentures are given against any mortgage of the assets of the company.
(iii) Redeemable debentures: These debentures are to be redeemed on the expiry of a certain period. The interest is paid periodically and the initial investment is returned after the fixed maturity period.
(iv) Irredeemable debentures: These kinds of debentures cannot be redeemable during the life time of the business concern. The Companies Act of 2019 called it perpetual debentures.
(v) Convertible debentures: Convertible debentures are the debentures whose holders have the option to get them converted wholly or partly into shares. These debentures are usually converted into equity shares. Conversion of the debentures may be: Non-convertible debentures; Fully convertible debentures and Partly convertible debentures.

**4.5 Bonds**

Bonds may be used to raise financing for a specific activity. They are a special type of debt financing because the debt instrument is issued by the company. Bonds are different from other debt financing instruments because the company specifies the interest rate and when the company will pay back the principal (maturity date). Also, the company does not have to make any payments on the principal (and may not make any interest payments) until the specified maturity date. The price paid for the bond at the time it is issued is called its face value. When a company issues a bond, it guarantees to pay back the principal (face value) plus interest. From a financing perspective, issuing a bond offers the company the opportunity to access financing without having to pay it back until it has successfully applied the funds. The risk for the investor is that the company will default or go bankrupt before the maturity date. However, because bonds are a debt instrument, they are ahead of equity holders for company assets.

**4.6 Retained Earnings**

Retained earnings are another method of internal sources of finance. Actually, is not a method of raising finance, but it is called as accumulation of profits by a company for its expansion and diversification activities. Retained earnings are called under different names such as; self-finance, inter finance, and plugging back of profits. According to the Companies

Act of 2019, certain amount of the net profits after tax of a financial year have to be compulsorily transferred to reserve by a company before declaring dividends for the year. Under the retained earnings sources of finance, a part of the total profits is transferred to various reserves such as general reserve, replacement fund, reserve for repairs and renewals, reserve funds and secrete reserves, etc.

**Summary**

This session was centred on shares, bonds, debentures and retained earnings as sources of finance for a business for a long period of time. We explained the different types of shares business can issue. We want you to remember that under the topic valuation of financial securities, we shall revisit the topic on shares and bonds as the major financial securities used in Ghana and how they are valued. So, when going for funding, there is the need to do the valuation and see how it will be beneficial for your business or company.

**Self-Assessment Questions**

**Exercise 2.4**

## SESSION 5: VENTURE CAPITAL, ANGEL INVESTORS AND LEASING

**Introduction**

Venture capitalist, business angels and leasing companies are investors in usually small unquoted companies with high growth potentials. But why are the opportunities there at all? If the rewards were attractive enough, surely established sources of money would be competing to invest instead? The answer lies in part in the cost of doing relatively small investments. There are good reasons that larger institutions can't afford to take the risk, while individuals can. In this session, we examine other sources of long-term sources of finance such as business angels, venture capitals and leasing and how they are potentially good for small and medium size companies.

**Objectives**

By the end of this session, you should be able to:

(a) identify venture capital firms that provide funding for businesses
(b) describe angel investors and how they provide funding for businesses
(c) explain leasing as a source of long-term finance
(d) discuss the nature of finance lease

Now read on …

## 5.1 Venture capital

Venture capitalists invest in businesses with the potential for high returns – those with products or services with a unique selling point, or competitive advantage. They invest in a portfolio where a significant number of businesses may fail, so those that succeed have to compensate for those losses. They also want proven track records, and so rarely invest at the start-up stage. Like angel investors, venture capitalists bring a wealth of experience to the business. They are unlikely to get involved in the day-to-day running of the business but will often help focus the business strategy.

Securing VC investment can be a complex, costly and time-consuming process. A detailed business plan is a must. And legal fees will be incurred through the deal negotiation, regardless of whether investment is ultimately secured. Corporate venture capital (CVC) is another growing source of funding for small businesses. It describes a wide variety of equity investment undertaken by a corporation, or its investment entity, into a high-growth and high-potential, privately-held business. CVC performs the same economic role that private venture capital plays – the identification and nurturing of the innovative businesses of the future. This formal and direct relationship is usually three-fold:

- by making a financial investment in return for an equity stake in the business;
- by offering debt finance to fund growth activities for an agreed return;
- by offering non-financial support for an agreed return, such as providing access to established marketing or distribution channels, or knowledge transfer. It is important that the company's aims are aligned with those of the business.

Venture capital firms are usually focused on creating an investment portfolio of businesses with high growth potential resulting in high rates of returns. These businesses are often high-risk investments. They may look for annual returns of 25 to 30 percent on their overall investment portfolio. Because these are usually high-risk business investments, they want investments with expected returns of 50 percent or more. Assuming that some business investments will return 50 percent or more while others will fail, it is hoped that the overall portfolio will return 25 to 30 percent.

More specifically, many venture capitalists subscribe to the 2-6-2 rule of thumb. This means that typically two investments will yield high returns, six will yield moderate returns (or just return their original investment), and two will fail.

**5.2 Angel investors or Business Angels**

Angel Investors or Business angels are individuals who make equity investments in businesses with growth potential, businesses in the early stages of development, or in established businesses looking for expansion capital. Angels back high-risk opportunities, with the potential for high returns. Some invest on their own, others through an angel syndicate or club. However, the most significant trend is for angels to invest in syndicates or groups, generally with a lead angel. At seed stage, lower amounts of funding may be available. Businesses in the growth stage may be able to attract higher amounts from angel syndicates. Angels can offer multiple rounds of finance and frequently co-invest with other sources of equity and co-investment funds as further growth finance is required.

When taking on angel investment, a business should look beyond the capital they put in. Most angels can bring valuable first-hand experience of growing businesses, often early-stage businesses. Their skills and experience will be shared with the business, as well as their network of contacts. Most focus on investments within a small geographic area, and so have local knowledge and local networks. Angels often make investment decisions quickly, without complex assessments. However, tracking down the right angel can take time.

Angel investors are individuals and businesses that are interested in helping small businesses survive and grow. So, their objective may be more than just focusing on economic returns. Although angel investors often have somewhat of a mission focus, they are still interested in profitability and security for their investment. So, they may still make many of the same demands as a venture capitalist. Angel investors may be interested in the economic development of a specific geographic area in which they are located. Angel investors may focus on earlier stage financing and smaller financing amounts than venture capitalists.

**5.3 Leasing**

This method has become quite common among the manufacturing companies. Leasing facility is usually provided through the mediation of leasing companies who buy the required plant and machinery from its manufacturer and lease it to the company that needs it for a specified period on payment of an annual rent. For this purpose, a proper lease agreement is made between the lessor (leasing company) and lessee (the company hiring the asset). Such agreement usually provides for the purchase of the machinery by the lessee at the end of the lease period at a mutually agreed and specified price. It may be noted that the ownership remains with the leasing company during the lease period.

Sometimes, a company, to meet its financial requirements, may sell its own existing fixed asset (machinery or building) to a leasing company at the current market price on the condition that the leasing company shall lease the asset back to selling company for a specified period. Such an arrangement is known as ‘Sell and Lease Back’. The company in such arrangement gets the funds without having to part with the possession of the asset involved which it continues to use on payment of annual rent for the lease.

A lease is a method of obtaining the use of assets for the business without using debt or equity financing. It is a legal agreement between two parties that specifies the terms and conditions for the rental use of a tangible resource such as a building and equipment. Lease

payments are often due annually. The agreement is usually between the company and a leasing or financing organization and not directly between the company and the organization providing the assets. When the lease ends, the asset is returned to the owner, the lease is renewed, or the asset is purchased. A lease may have an advantage because it does not tie up funds from purchasing an asset. It is often compared to purchasing an asset with debt financing where the debt repayment is spread over a period of years. However, lease payments often come at the beginning of the year where debt payments come at the end of the year. So, the business may have more time to generate funds for debt payments, although a down payment is usually required at the beginning of the loan period.

It may seem strange to see leasing referred to as a source of long-term finance, but in fact it is very similar to secured lending. Leases may be divided into two types:

(i) Operating leases
It is often possible to hire an asset, say an item of plant, that is perhaps required only occasionally, rather than purchasing it. Usually, the owner carries out any maintenance necessary. The decision whether to buy the asset or to lease it will perhaps be affected by financing considerations. Basically, though, it is an operating decision, which would be made according to which approach would be cheaper.

(ii) Finance leases
Here the potential user identifies an asset in which it wishes to invest, negotiates price, delivery and so on, and then seeks a supplier of finance to buy it. Having arranged for the asset to be purchased, the user leases it from the purchaser. Naturally, the lease payments will need to be sufficient to justify the owner's expenditure, in terms both of capital repayment and of interest.

**5.4 The nature of finance leasing**

It is finance leases that concern us here since they are effectively term loans with capital repayable by instalments. This is an important source of finance, which has been estimated to provide as much as 20 per cent of the total finance for new capital expenditure by businesses over recent years (Drury and Braund 1990).

In the past, finance leasing has been believed to be popular with users partly because, while it is tantamount to borrowing, neither the asset nor the obligation to the owner appeared on the balance sheet of the user business. Those trying to assess the business's financial position could overlook such a source of off-balance sheet finance. However, accounting regulations have now put leasing on the balance sheet. Businesses are now required to show both the leased assets and the capital value of the obligation to the owner on the face of the balance sheet.

Another feature of finance leasing, which was apparently a major reason was its considerable tax efficiency in some circumstances. Lease payments are fully deductible for corporation tax purposes by the borrowing business. This includes the capital portion of the payment. Until 1984, the capital cost of items of plant attracted 100 per cent first-year capital allowance in their year of acquisition. As far as the borrowing business was concerned, leasing rather than buying a non-current asset would deny it the opportunity to claim the first-year allowance. On the other hand, leasing would still enable the business to claim 100 per cent of the cost of the asset, but over its life, rather than in the first year. In many cases, however, even where businesses were to buy the asset by raising finance from, say, a term loan, profits were

insufficient for the full benefit to be gained from the large first-year allowance. This was particularly true where recession restricted profitability.

With leasing, the right to claim capital allowance passed to the 'lender'. If the profits from leasing and other activities were such as to put the 'lender' in a position to take full advantage of them, some of this advantage could be passed on to the 'borrower' in lower lease payments. Thus the 'borrower' could give up some of the advantage of the first-year allowance in exchange for a lower lease charge.

**Summary**

In this session, we looked at other sources of financing a business on long-term basis. We examined venture capital firms, angel investors and leasing. Under leasing, we focused on finance lease by looking at the nature of finance lease. All these sources are real and available for businesses to use.

**Self-Assessment Questions**

**Exercise 2.5**

## SESSION 6: CAPITALIZATION

### Introduction

Financial planning and decision play a major role in the field of business finance which consists of the major area of business finance such as, capitalization, financial structure, capital structure, leverage and financial forecasting. Financial planning includes the following important parts: Estimating the amount of capital to be raised; determining the form and proportionate amount of securities and formulating policies to manage the financial plan. Having looked at the various sources of finance for a business, we shall discuss whether such a business is over capitalized, undercapitalized and watered capitalized.

### Objectives

By the end of this session, you should be able to:

(a) define capital
(b) explain the types of capital
(c) understand business capitalization

Now read on …

### 6.1 Meaning of capital

The term capital refers to the total investment of the company in terms of money, and assets. It is also called as total wealth of the company. When the company is going to invest large amount of finance into the business, it is called as capital. Capital is the initial and integral part of new and existing business concern. The capital requirements of the business concern may be classified into two categories: fixed capital and working capital.

### 6.1.1 Fixed Capital

Fixed capital is the capital, which is needed for meeting the permanent or long-term purpose of the business concern. Fixed capital is required mainly for the purpose of meeting capital expenditure of the business concern and it is used over a long period. It is the amount invested in various fixed or permanent assets, which are necessary for a business concern. According to the definition of Hoagland, "Fixed capital is comparatively easily defined to include land, building, machinery and other assets having a relatively permanent existence".

### Character of Fixed Capital

- Fixed capital is used to acquire the tangible non-current assets of the business concern.
- Fixed capital meets the capital expenditure of the business concern.
- Fixed capital normally consists of long period.
- Fixed capital expenditure is of nonrecurring nature.
- Fixed capital can be raised only with the help of long-term sources of finance.

### 6.1.2 Working Capital

Working capital is the capital which is needed to meet the day-to-day transaction of the business concern. It may cross working capital and net working capital. Normally working capital consists of various compositions of current assets such as inventories, bills, receivable, debtors, cash, and bank balance and prepaid expenses. We know you remember this in your first year when you took Foundation of Accounting course. You now understand the relevant of Accounting right. According to the definition of Bonneville, "any acquisition of funds which increases the current assets increase the Working Capital also for they are one and the same". Working capital is needed to meet the following purpose:

- Purchase of raw material
- Payment of wages to workers
- Payment of day-to-day expenses
- Maintenance expenditure etc.

## 6.2 Meaning of Capitalization

Capitalization is one of the most important parts of financial decision, which is related to the total amount of capital employed in the business concern. Understanding the concept of capitalization leads to solve many problems in the field of business finance. Because there is a confusion among the capital, capitalization and capital structure.

Capitalization refers to the process of determining the quantum of funds that a firm needs to run its business. Capitalization is only the no-par value of share capital and debenture and it does not include reserve and surplus. According to Guthman and Dougall, "capitalization is the sum of the no-par value of shares and bonds outstanding". Capitalization is the balance sheet value of shares and bonds outstands". From the meaning so far, we can say capitalization relates to long-term funds of the business. This long-term finance relates to equity finance and debt finance of the business. Capitalization may be classified into the following three important types based on its nature:

- Over Capitalization
- Under Capitalization
- Water Capitalization

### 6.2.1 Over Capitalization

Over capitalization refers to the company which possesses an excess of capital in relation to its activity level and requirements. In simple means, over capitalization is more capital than actually required and the funds are not properly used. According to Bonneville, Dewey and Kelly, over capitalization means, "when a business is unable to earn fair rate on its outstanding securities".

Example

A company is earning a sum of GHS50,000 and the rate of return expected is 10%. This company will be said to be properly capitalized. Suppose the capital investment of the company is GHS60,000, it will be over capitalization to the extent of GHS100,000. The new rate of earning would be: 50,000/60,000×100=8.33%. When the company has over capitalization, the rate of earnings will be reduced from 10% to 8.33%.

#### 6.2.1.1 Causes of Over Capitalization

Over capitalization arise due to the following important causes:

- Over issue of capital by the company.
- Borrowing large amount of capital at a higher rate of interest.
- Providing inadequate depreciation to the non-current assets.
- Excessive payment for acquisition of goodwill.

- High rate of taxation.
- Under estimation of capitalization rate.

#### 6.2.1.2 Effects of Over Capitalization

Over capitalization leads to the following important effects:

- Reduce the rate of earning capacity of the shares.
- Difficulties in obtaining necessary capital to the business concern.
- It leads to fall in the market price of the shares.
- It creates problems on re-organization.
- It leads under or misutilisation of available resources.

#### 6.2.1.3 Remedies for Over Capitalization

Over capitalization can be reduced with the help of effective management and systematic design of the capital structure. The following are the major steps to reduce over capitalization.

- Efficient management can reduce over capitalization.
- Redemption of preference share capital which consists of high rate of dividend.
- Reorganization of equity share capital.
- Reduction of debt capital.

### 6.2.2 Under Capitalization

Under capitalization is the opposite concept of over capitalization and it will occur when the company's actual capitalization is lower than the capitalization as warranted by its earning capacity. Under capitalization is not the so-called inadequate capital. Under capitalization was defined by Gerstenberg, as the rate at which a company's profit is exceptionally high in the same industry. Hoagland defined under capitalization as "an excess of true assets value over the aggregate of stocks/shares and bonds outstanding".

#### 6.2.2.1 Causes of Under Capitalization

Under capitalization arises due to the following important causes:

- Under estimation of capital requirements.
- Under estimation of initial and future earnings.
- Maintaining high standards of efficiency.
- Conservative dividend policy.
- Desire of control and trading on equity.

#### 6.2.2.2 Effects of Under Capitalization

Under Capitalization leads certain effects in the company and its shareholders.

- It leads to manipulate the market value of shares.
- It increases the marketability of the shares.
- It may lead to more government control and higher taxation.
- Consumers feel that they are exploited by the company.
- It leads to high competition.

#### 6.2.2.3 Remedies of Under Capitalization

Under Capitalization may be corrected by taking the following remedial measures:

- Under capitalization can be compensated with the help of fresh issue of shares.
- Increasing the no-par value of share may help to reduce under capitalization.
- Under capitalization may be corrected by the issue of bonus shares to the existing shareholders.

- Reducing the dividend per share by way of splitting up of shares.

### 6.2.3 Watered Capitalization

If the capital of the company is not mentioned by assets of equivalent value, it is called as watered capital. In simple words, watered capital means that the realizable value of assets of the company is less than its book value. According to Hoagland's definition, "A capital is said to be watered when its true value is less than its book value."

#### 6.2.3.1 Causes of Watered Capital

Generally watered capital arises at the time of incorporation of a company but it also arises during the life time of the business. The following are the main causes of watered capital:

- Acquiring the assets of the company at high price.
- Adopting ineffective depreciation policy.
- Worthless intangible assets are purchased at higher price.

**Summary**

In this session, we defined capitalization as the process of determining the quantum of funds that a firm needs to run its business. This may lead to a firm been over capitalized, undercapitalized or watered. The causes and effects of these types of capitalization have been explained in this session. When going for funding, it is imperative to understand the needs of your business and ensure that you get the funding that is appropriate for the smooth operations of the business.

**Self-Assessment Questions**

**Exercise 2.6**

## UNIT 3: FINANCIAL STATEMENT ANALYSIS

**Unit Outline**

Session 1: Meaning and Significance of Analysis of Financial Statement
Session 2: Pre-Requisites of Financial Analysis
Session 3: Horizontal Analysis and Common Size Statement
Session 4: Trend Analysis
Session 5: Ratio Analysis
Session 6: Introduction to Distress Analysis

**Overview**

You have learnt about the financial statements (Income Statement, Statement of Financial Position and Cash Flow Statement) of companies. Basically, these are summarised financial reports which provide the operating results and financial position of companies, and the detailed information contained therein is useful for assessing the operational efficiency and financial soundness of a company. This requires proper analysis and interpretation of such information for which a number of techniques (tools) have been developed by financial experts. In this Unit, we will have an overview of these techniques.

**Objectives**

By the end of this unit, you should be able to:

(a) define financial analysis
(b) differentiate between analysis and interpretation
(c) discuss the significance of financial statement analysis
(d) understand the pre-requisites of financial analysis
(e) discuss the tools to be used for analysis of financial statements
(f) perform analysis of a company using vertical and horizontal analysis
(g) understand the different types of ratios used in analysing financial statements
(h) evaluated distress levels of public companies

## SESSION 1: MEANING AND SIGNIFICANCE OF ANALYSIS OF FINANCIAL STATEMENT

### Introduction

Dear students, we welcome you to this session. This is the introduction to financial statement analysis. This session is crucial in that we cannot just accept whatever that is in the financial statements without subjecting them to critical review. In this session, we look at the meaning of financial analysis, significance of the analysis and objectives of financial analysis.

### Objectives

By the end of this session, you should be able to:

(i) define financial analysis
(j) differentiate between analysis and interpretation
(k) discuss the significance of financial statement analysis
(l) explain the objectives of financial statement analysis

Now read on …

### 1.1 Meaning of Analysis of Financial Statements

The process of critical evaluation of the financial information contained in the financial statements in order to understand and make decisions regarding the operations of the firm is called 'Financial Statement Analysis'. It is basically a study of relationship among various financial facts and figures as given in a set of financial statements, and the interpretation thereof to gain an insight into the profitability and operational efficiency of the firm to assess its financial health and future prospects.

Financial statement analysis is a judgemental process which aims to estimate current and past financial positions and the results of the operation of an enterprise, with primary objective of determining the best possible estimates and predictions about the future conditions. It essentially involves regrouping and analysis of information provided by financial statements to establish relationships and throw light on the points of strengths and weaknesses of a business enterprise, which can be useful in decision-making involving comparison with other

firms (cross sectional analysis) and with firms' own performance, over a time period (time series analysis).

The term 'financial analysis' includes both 'analysis and interpretation'. The term analysis means simplification of financial data by methodical classification given in the financial statements. Interpretation means explaining the meaning and significance of the data. These two are complimentary to each other. Analysis is useless without interpretation, and interpretation without analysis is difficult or even impossible.

**1.2 Significance of Analysis of Financial Statement**

Financial analysis is the process of identifying the financial strengths and weaknesses of the firm by properly establishing relationships between the various items of the statement of financial position and the income statement. Financial analysis can be undertaken by management of the firm, or by parties outside the firm, viz., owners, trade creditors, lenders, investors, labour unions, analysts and others. The nature of analysis will differ depending on the purpose of the analyst. A technique frequently used by an analyst need not necessarily serve the purpose of other analysts because of the difference in the interests of the analysts. Financial analysis is useful and significant to different users in the following ways:

(i) Finance manager

Financial analysis focusses on the facts and relationships related to managerial performance, corporate efficiency, financial strengths and weaknesses and creditworthiness of the company. A finance manager must be well-equipped with the different tools of analysis to make rational decisions for the firm. The tools for analysis help in studying accounting data so as to determine the continuity of the operating policies, investment value of the business, credit ratings and testing the efficiency of operations. The techniques are equally important in the area of financial control, enabling the finance manager to make constant reviews of the actual financial operations of the firm to analyse the causes of major deviations, which may help in corrective action wherever indicated.

(ii) Top management

The importance of financial analysis is not limited to the finance manager alone. It has a broad scope which includes top management in general and other functional managers. Management of the firm would be interested in every aspect of the financial analysis. It is their overall responsibility to see that the resources of the firm are used most efficiently and that the firm's financial condition is sound. Financial analysis helps the management in measuring the success of the company's operations, appraising the individual's performance and evaluating the system of internal control.

(iii) Trade payables

Trade payables, through an analysis of financial statements, appraises not only the ability of the company to meet its short-term obligations, but also judges the probability of its continued ability to meet all its financial obligations in future. Trade payables are particularly interested in the firm's ability to meet their claims over a very short period of time. Their analysis will, therefore, evaluate the firm's liquidity position.

(iv) Lenders

Suppliers of long-term debt are concerned with the firm's long-term solvency and survival. They analyse the firm's profitability over a period of time, its ability to generate cash, to be able to pay interest and repay the principal and the relationship between various sources of funds (capital structure relationships). Long-term lenders analyse the historical financial statements to assess its future solvency and profitability.

(v) Investors
Investors, who have invested their money in the firm's shares, are interested about the firm's earnings. As such, they concentrate on the analysis of the firm's present and future profitability. They are also interested in the firm's capital structure to ascertain its influences on firm's earning and risk. They also evaluate the efficiency of the management and determine whether a change is needed or not. However, in some large companies, the shareholders' interest is limited to decide whether to buy, sell or hold the shares.

(vi) Labour unions
Labour unions analyse the financial statements to assess whether it can presently afford a wage increase and whether it can absorb a wage increase through increased productivity or by raising the prices.

(vii) Others: The economists, researchers, students, etc., analyse the financial statements to study the present business and economic conditions. The government agencies need it for price regulations, taxation and other similar purposes.

**1.3 Objectives of Analysis of Financial Statements**

Analysis of financial statements reveals important facts concerning managerial performance and the efficiency of the firm. Broadly speaking, the objectives of the analysis are to apprehend the information contained in financial statements with a view to know the weaknesses and strengths of the firm and to make a forecast about the future prospects of the firm thereby, enabling the analysts to take decisions regarding the operation of, and further investment in the firm. To be more specific, the analysis is undertaken to serve the following purposes (objectives):

- to assess the current profitability and operational efficiency of the firm as a whole as well as its different departments so as to judge the financial health of the firm.
- to ascertain the relative importance of different components of the financial position of the firm.
- to identify the reasons for change in the profitability/financial position of the firm.
- to judge the ability of the firm to repay its debt and assessing the short-term as well as the long-term liquidity position of the firm.

Through the analysis of financial statements of various firms, an economist can judge the extent of concentration of economic power and pitfalls in the financial policies pursued. The analysis also provides the basis for many governmental actions relating to licensing, controls, fixing of prices, ceiling on profits, dividend freeze, tax subsidy and other concessions to the corporate sector.

**Summary**
In this session we defined financial analysis and financial statement analysis. We discuss the significance and objectives of financial statement analysis and stated that looking at the different users of financial statements, the importance of analysis might be different to all these user groups. Read the rest and get the full knowledge you so desire.

**Self-Assessment Questions**
**Exercise 3.1**

## SESSION 2: PRE-REQUISITES OF FINANCIAL ANALYSIS

**Introduction**
Before performing a company's analysis, the analyst must first perform an analysis of the economy and the industry in which the company operates. The analyst must also know the tools available for the analysis. These are discussed in this session.

**Objectives**
By the end of this session, you should be able to:
(a) understand the pre-requisites of financial analysis
(b) explain economic analysis
(c) discuss important aspects of industry analysis
(d) undertake analysis of a company
(e) discuss the tools to be used for analysis of financial statements

Now read on …

### 2.1 Economic Analysis
Economic analysis can be looked at from different perspectives such as mathematics that uses the Lagrange theory and the Kuhn-Tucker theory; the accounting perspective and the finance perspective. For the purpose of business finance, we shall be looking at economic analysis from the perspective of accounting and finance. With respect to information on the economy, the analyst must evaluate how the company performs in different economic environments. Armed with this information and economic forecasts, the analyst will be able to develop

better forecasts of how the particular company will perform in the future. Economic data that is needed includes:

- Production and income;
- Employment;
- Consumption;
- Investment activity;
- Interest rates;
- Exchange rates;
- Share prices; and
- Inflation.

**2.2 Industry analysis**

The analyst should take a close look at the industry or industries in which the company operates. In Ghana, there are different industries such the hospitality, printing, media and entertainment, manufacturing, transport, banking and finance, mining, education, trading and the petroleum. The important aspects of the industry analysis are:

- Nature of competition;
- Market share for each company in the industry;
- Labour conditions;
- Regulatory conditions;
- Price elasticity of demand and supply;
- Sensitivity of demand to economic conditions.

In the analysis of an industry and a company, it is important to understand the sources of value added. In basic economics, we have learned that a firm creates value when it has a comparative or competitive advantage. When analysing an industry and a company, the financial analyst must identify the comparative or competitive advantage that provides economic pro t and sustainable growth in the future. The economic pro t (i.e. the economic value added) is the income of the company in excess of the firm's cost of capital. A company's sustainable growth is the growth rate that it can keep up without having to resort to additional financial leverage.

One way to analyse these advantages is to use the five factors that are outlined by Michael Porter:

a. Threat of new entrants – the industry/company may have an advantage if there are barriers to new entrants.
b. Bargaining power of buyers – the stronger the bargaining power of buyers, the less advantage that a/an industry/ company has.
c. Bargaining powers of suppliers – the greater the bargaining power of suppliers, the lower the economic profit.
d. Rivalry among competitors – the more competitive the industry, the lower the economic profits for any member of the industry
e. Threat of substitutes – the greater the ability of competitors to imitate the products or services of a company, the lower the potential economic profit. Patents, trademarks and copyrights will lower the threat of substitutes.

These are often referred to as Porter's Five Forces. As the analyst examines the industry and the company – past, present and future – these questions must be addressed:

(i) What are the sources of the industry's and company's economic profit?
(ii) What are the sources of the company's sustainable growth?

**2.3 Company analysis**

The analysis of a company requires looking closely at the company's financial history and recent events, with a goal of assembling the future prospects of the company. The types of information that an analyst must gather include:

- Financial statement data and related disclosures;
- Major news items in recent years;
- Position and market share in the industry;
- International investment;
- Where the company is in its life cycle (i.e. high growth/ development, maturing, declining);
- Contributions of major product, divisions or subsidiaries to the company's performance;
- Research and development e orts;
- Sensitivity of company to commodity prices (e.g. oil); and
- Major litigation.

**2.4 Tools of Analysis of Financial Statements**

The most commonly used techniques of financial analysis are explained in this session.

**2.4.1 Comparative Statements**

These are the statements showing the profitability and financial position of a firm for different periods of time in a comparative form to give an idea about the position of two or more periods. It usually applies to the two important financial statements, namely, statement of financial position (balance sheet) and income statement (statement of profit and loss) prepared in a comparative form. The financial data will be comparative only when same accounting principles are used in preparing these statements. If this is not the case, the deviation in the use of accounting principles should be mentioned as a footnote. Comparative figures indicate the trend and direction of financial position and operating results. This analysis is also known as 'horizontal analysis'.

**2.4.2 Common Size Statements**

These are the statements which indicate the relationship of different items of a financial statement with a common item by expressing each item as a percentage of that common item. The percentage thus calculated can be easily compared with the results of corresponding percentages of the previous year or of some other firms, as the numbers are brought to common base. Such statements also allow an analyst to compare the operating and financing characteristics of two companies of different sizes in the same industry. Thus, common size statements are useful, both, in intra-firm comparisons over different years and also in making inter-firm comparisons for the same year or for several years. This analysis is also known as 'Vertical analysis'.

**2.4.3 Trend Analysis**

It is a technique of studying the operational results and financial position over a series of years. Using the previous years' data of a business enterprise, trend analysis can be done to observe the percentage changes over time in the selected data. The trend percentage is the percentage relationship, in which each item of different years bears to the same item in the base year. Trend analysis is important because, with its long run view, it may point to basic changes in the nature of the business. By looking at a trend in a particular ratio, one may find whether the ratio is falling, rising or remaining relatively constant. From this observation, a problem is detected or the sign of good or poor management is detected.

**2.4.4 Ratio Analysis**

It describes the significant relationship which exists between various items of a balance sheet and a statement of profit and loss of a firm. As a technique of financial analysis, accounting ratios measure the comparative significance of the individual items of the income and position statements. It is possible to assess the profitability, solvency and efficiency of an enterprise through the technique of ratio analysis.

### 2.4.5 Cash Flow Analysis

It refers to the analysis of actual movement of cash into and out of an organisation. The flow of cash into the business is called as cash inflow or positive cash flow and the flow of cash out of the firm is called as cash outflow or a negative cash flow. The difference between the inflow and outflow of cash is the net cash flow. Cash flow statement is prepared to project the manner in which the cash has been received and has been utilised during an accounting year as it shows the sources of cash receipts and also the purposes for which payments are made. Thus, it summarises the causes for the changes in cash position of a business enterprise between dates of two balance sheets.

**Summary**

In this session, we took you through the pre-requites of financial statement analysis. We discuss economic analysis, industry analysis and company analysis. We also explained the various tools used for analysing financial statements. In the subsequent session, we shall see the use of these tools. We are most confident you will understand these tools and know how to use them appropriately.

**Self-Assessment Questions**

**Exercise 3.2**

## SESSION 3: HORIZONTAL ANALYSIS AND COMMON SIZE STATEMENT

**Introduction**

Among the tools for analysing financial statements is vertical and horizontal analysis. We shall take you through the two tools and give you merits and demerits of each and when each is supposed to be used. Wish you a successful study.

**Objectives**

By the end of this session, you should be able to:

(a) explain vertical analysis
(b) discuss horizontal analysis
(c) discuss the advantages and disadvantages of vertical and horizontal analysis
(d) perform analysis of a company using vertical and horizontal analysis

Now read on …

### 3.1 Meaning of Horizontal Analysis

Horizontal analysis (otherwise known as year-on-year analysis or comparative statements) involves comparing particular year's results with that of the previous year. These are the statements showing the profitability and financial position of a firm for different periods of time in a comparative form to give an idea about the position of two or more periods. It usually applies to the two important financial statements, namely, statement of financial position and income statement prepared in a comparative form.

The financial data will be comparative only when same accounting principles are used in preparing these statements. If this is not the case, the deviation in the use of accounting principles should be mentioned as a footnote. Comparative figures indicate the trend and direction of financial position and operating results.

#### 3.1.1 Advantages of Comparative Analysis

(i) Figures for two or more periods are placed side by side to facilitate inter-firm comparison and in horizontal analysis.
(ii) It brings out more clearly the nature and trend of current changes that affect the enterprise.
(iii) It helps in estimating weakness in the operating cycle, financial health and future position of the business.

#### 3.1.2 Disadvantages of Comparative Analysis

(i) It may be misleading, if frequent changes have been allowed in principles and procedures over the periods.
(ii) Without the knowledge of internal analysis, it may be extremely misleading.
(iii) It does not show the relation of any items to total assets or to total liabilities or to total net sales in a year.
(iv) Proper comparison between two or more firms cannot be possible by this statement because there is no common base of comparison.

### 3.2 Steps in Horizontal Analysis

If the financial statement is re-casted for comparing all the elements of financial conditions from year to year in absolute term as well as in percentage then the re-casted statement is called comparative financial statement. This statement is designed to provide time view of various elements of financial statement. This in chapter two of this book is what we referred to as time series analysis. This statement is made by:

(i) Showing absolute money values of each elements of income statement and statement of financial position of different periods.
(ii) Showing increase or decrease in absolute money value of each elements by deducting elements of current period from past period.
(iii) Showing increase or decreasing value in terms of percentage
$$= \frac{increasing\ /decreasing\ amount}{amount\ in\ past\ period} \times 100\%.$$
(iv) Keeping in mind that the principles and procedures followed in the collection, and presentation of data should not materially differ over the periods.

#### 3.2.1 Illustration

The following are the income statements of Kankam Company Limited for the years ended 31/12/2016 and 31/12/2017.

| | **31/12/16 (GH₵'000)** | **31/12/17 (GH₵'000)** |
|---|---|---|
| Net Sales | 170,000 | 190,400 |
| Less: Cost of goods sold | 105,000 | 120,000 |
| Gross Profit (P) | 65,000 | 70,400 |
| Administrative expenses (A) | 13,200 | 14,960 |
| Selling expenses: | | |
| Advertisement expenses | 3,000 | 4,000 |
| Other selling expenses | 40,800 | 41,800 |
| Total selling expenses (B) | 43,800 | 45,800 |
| Operating expenses (A + B) | 57,000 | 60,760 |
| Operating Profit (D) [D = P – (A + B)] | 8,000 | 9,640 |
| Other Incomes (E) | 6,400 | 9,200 |
| Other expenses (F) | 6,800 | 4,800 |
| Profit before tax (PBT) [PBT = D + E – F] | 7,600 | 14,040 |
| Income tax (T) | 3,800 | 6,200 |
| Profit after tax (PAT) [PAT = PBT – T] | 3,800 | 7,840 |

**REQUIRED**
Prepare a comparative income statement and comment on the performance of the company.

**3.2.2 Solution**
**Comparative Income Statement of Kankam Company Ltd. for the years ended 31 DECEMBER, 2016 and 2017**

| **Particulars** | **31/12/16 (GH₵'000)** | **31/12/17 (GH₵'000)** | **Amount of increase (+) or decrease (-) (GH₵)** | **Percentage increase (+) or decrease (-) (%)** |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Net Sales | 170,000 | 190,400 | (+) 20,400 | (+) 12.0 |
| Less: Cost of goods sold | 105,000 | 120,000 | (+) 15,000 | (+) 14.3 |
| Gross Profit (P) | 65,000 | 70,400 | (+) 5,400 | (+) 8.3 |
| Administrative expenses (A) | 13,200 | 14,960 | (+) 1,760 | (+) 13.3 |
| Selling expenses: | | | | |
| Advertisement expenses | 3,000 | 4,000 | (+) 1,000 | (+) 33.3 |
| Other selling expenses | 40,800 | 41,800 | (+) 1,000 | (+) 2.5 |
| Total selling expenses (B) | 43,800 | 45,800 | (+) 2,000 | (+) 4.6 |
| Operating expenses (A + B) | 57,000 | 60,760 | (+) 3,760 | (+) 6.6 |
| Operating Profit (D) [D = P – (A + B)] | 8,000 | 9,640 | (+) 1,640 | (+) 20.5 |
| Other Incomes (E) | 6,400 | 9,200 | (+) 2,800 | (+) 43.8 |
| Other expenses (F) | 6,800 | 4,800 | (–) 2,000 | (–) 29.4 |
| Profit before tax (PBT) [PBT = D + E – F] | 7,600 | 14,040 | (+) 6,440 | (+) 84.7 |
| Income tax (T) | 3,800 | 6,200 | (+) 2,400 | (+) 63.2 |
| Profit after tax (PAT) [PAT = PBT – T] | 3,800 | 7,840 | (+) 4,040 | (+) 106.3 |

**Notes:** Calculation for percentage increase (+) or decrease (-). This is to show how the percentages were calculated. We undertook this for net sales and cost of goods sold. You can however, continue with the calculations.

(i) $\frac{20,400}{170,000} \times 100\% = 12\%$

(ii) $\frac{15,000}{105,000} \times 100\% = 14.3\%$

Comparative income statement shows the income and expenses of two periods of the same company, absolute changes of each items for the year ended 31/12/2017 over 31/12/2016 and also shows percentage change. The following comments can be made on the performance of Kankam Company Ltd.:

(i) Sales of Kankam Ltd. has increased by GH₵20,400,000 during the year 2017 over 2016. And the cost of goods sold has also increased by GH₵15,000,000 in the same period. i.e., sales have improved by 12% and cost of goods sold has increased by 14.3%. So, gross profit has not improved markedly. Cost of goods usually increased due to higher quantity of sales or due to higher input cost. In the instant question the sale value has increased, therefore, it is clear cost of goods sold has increased due to higher quantity of sales. If such quantity had been sold at previous price, then sales value has been increased with higher amount. But here sales value has not increased significantly.

(ii) It indicates that the addition in sales has been due to lowering of sale price. It is also clear from advertisement expenses. The increase in advertisement expenses (33.3%) has been much higher than the percentage increase in net sales (12%). This indicates there was unfavourable market conditions that necessitates mass advertisement and reduction of sale price to motivate higher quantity of sales. Such situation may also arise due to new

product launching where huge advertisement is necessary and reduction of sale price is necessary.

(iii) There has been a substantial improvement in other incomes, both in relative term (43.8%) and in absolute term (GH₵2,800,000). Similarly, there has been a considerable reduction in other expenses in relative term (29.4%) as well as in absolute term (GH₵2,000,000). These items have been responsible for the increase in profit before tax (PBT) for the period under study by 84.7%. It implies that more emphasis has been given by the management of the company on earning non-operating profits as compared to the operating profits.

### 3.3 Common Size Statement

Common size financial statement is a restated financial statement showing percentage of total items with common base for comparison. It is also known as the vertical analysis. This statement is made in the following procedure:

(i) Assets side is classified into non-current assets (NCA) and current assets (CA), showing individually and in total. Then total assets (TA) are taken as common base 100%. Then $\frac{NCA}{TA} \times 100\%$, $\frac{CA}{TA} \times 100\%$, etc. are calculated.

(ii) Liabilities side is classified into non-current liabilities (NCL), and current liabilities (CL) showing individually and in total. Then percentage of each liability to total liability is calculated.

(iii) Income statement is classified into sales, cost of goods sold, operating expenses, net profit, interest, tax, EAT (earning after tax) etc. Then percentage of each element to sales is calculated.

#### 3.3.1 Advantages of Common Size Statements

(i) It shows the changes over the years in relation to total assets, total liabilities, sales.
(ii) It helps in inter-firm comparison with common base.
(iii) It helps in vertical analysis of figures.
(iv) It helps in understanding financial strategy of the firms in comparison.
(v) It shows the relative efficiency of each cost items of two firms.

#### 3.3.2 Disadvantages of Common Size Statement

(i) It shows the percentage of each item to the total in each period but not changes of each item from period to period.
(ii) Its observations are not very useful because there are more definite norms for the proportion of each item to total.
(iii) If there is no uniform costing system, no same accounting practice among all firms of an industry, then it is meaningless for studying the comparative financial position of two firms.

### 3.4 Illustration

From the following income statement, prepare a common size statement and also interprets the result.

**Income Statement for the year ended 31 DECEMBER**

| | 2016 | 2017 |
|---|---|---|
| | **(GH₵)** | **(GH₵)** |

| | | |
|---|---|---|
| Net Sales | 1,050,000 | 1,350,000 |
| Less: Cost of goods sold | 570,000 | 645,000 |
| Gross Profit | 480,000 | 705,000 |
| Less: Other operating expenses | 150,000 | 216,000 |
| Operating profit | 330,000 | 489,000 |
| Less: Interest on long term debt | 60,000 | 51,000 |
| Profit before tax | 270,000 | 438,000 |

**3.4.1 Solution**

**Comparative Common Size Statement for the year ended 31st DECEMBER**

| | **2016** | **2017** |
|---|---|---|
| Net Sales | 100% | 100% |
| **Less:** Cost of goods sold = $\frac{Cost\ of\ sales}{Net\ Sales} \times 100\%$ | 54.3% | 47.8% |
| Gross Profit= $\frac{Gross\ Profit}{Net\ Sales} \times 100\%$ | 45.7% | 52.2% |
| **Less:** Other operating expenses = $\frac{Other\ Operating\ Expenses}{Net\ Sales} \times 100\%$ | 14.3% | 16% |
| Operating profit $\frac{Operating\ profit}{Net\ Sales} \times 100\%$ | 31.4% | 36.2% |
| **Less:** Interest on long term debt= $\frac{Interest}{Net\ Sales} \times 100\%$ | 5.7% | 3.8% |
| Profit before tax (PBT) $\frac{Profit\ before\ Tax}{Net\ Sales} \times 100\%$ | 25.7% | 32.4% |

**Comment**

(i) The PBT to net sales has increased from 25.7% in the year 2016 to 32.4% in the year 2017. It indicates that the profit earning capacity of the company has improved during the study period. This improvement in the profitability of the company has been mainly due to significant reduction in the cost of goods sold of the company. It may occur due to a reduction in input prices or may occur due to improvement in the efficiency of the company (i.e. reduction in waste or thievery). As other operating expenses has higher in 2017 so, it is clear that company has been operated with tight supervision, tight inventory control for reduction of COGS (Cost of Goods Sold).

(ii) The interest on long term debt to net sales has declined from 5.7% in the 2016 to 3.8% in 2017. It implies that the financial burden of the company has reduced significantly during the study period. Higher operating profit or fund from operation has been utilised for repayment of long-term debt, so that the financial risk associated with the company has declined significantly during the study period.

### 3.4.2 Illustration

From the following figures prepare a common size comparative statement and comment on the result.

| | 2013 | 2014 | 2015 | 2016 | 2017 |
|---|---|---|---|---|---|
| **Particulars** | **(GH₵)** | **(GH₵)** | **(GH₵)** | **(GH₵)** | **(GH₵)** |
| Cost of Materials | 150,000 | 220,000 | 250,000 | 200,000 | 200,000 |
| Labour cost | 200,000 | 140,000 | 150,000 | 150,000 | 175,000 |
| Conversion cost | 150,000 | 150,000 | 140,000 | 200,000 | 175,000 |
| Total Manufacturing cost | 500,000 | 510,000 | 540,000 | 550,000 | 550,000 |
| Sales Revenue | 1,200,000 | 1,100,000 | 1,000,000 | 1,000,000 | 950,000 |
| Gross Profit | 700,000 | 590,000 | 460,000 | 450,000 | 400,000 |
| Other operating expenses | 300,000 | 220,000 | 200,000 | 200,000 | 180,000 |
| Operating profit | 400,000 | 370,000 | 260,000 | 250,000 | 220,000 |

### 3.4.3 Solution:

**Comparative Common Size Statement**

| **Particulars** | **2010** | **2011** | **2012** | **2013** | **2014** |
|---|---|---|---|---|---|
| Cost of Materials= $\frac{Cost\ of\ material}{Sales} \times 100\%$ | 12.5% | 20% | 25% | 20% | 21.05% |
| Labour cost= $\frac{Labour\ Cost}{Sales} \times 100\%$ | 16.67% | 12.73% | 15% | 15% | 18.42% |
| Conversion cost= $\frac{Conversion\ cost}{Sales} \times 100\%$ | 12.50% | 13.64% | 14% | 20% | 18.42% |
| Total Manufacturing cost = $\frac{Total\ Mnufacturing\ cost}{Sales} \times 100$ | 41.67% | 46.36% | 54% | 55% | 57.89% |
| Sales Revenue | 100% | 100% | 100% | 100% | 100% |
| Gross Profit $\frac{Gross\ Profit}{Sales} \times 100$ | 58.33% | 53.64% | 46% | 45% | 42.11% |

| | | | | | |
|---|---|---|---|---|---|
| Other operating expenses $\frac{Other\ Operating\ Expenses}{Sales} \times 100$ | 25% | 20% | 20% | 20% | 18.95% |
| Operating profit $\frac{Operating\ Profit}{Sales} \times 100$ | 33.33% | 33.64% | 26% | 25% | 23.16% |

**Comments**

From the above analysis it can be concluded that there was a clear upward rising trend in the manufacturing cost of goods sold during the study period. As a result, gross profit to sales had decreased very significantly during the same period. It was 58.3% in the year 2013 which ultimately reduced to 42.11% in the ultimate year of the study period i.e. 2017. It implies that operational efficiency of the company had reduced very significantly during the study period. It might also be a result of a change in management where the initial controls have been relaxed paving the way for inefficiency.

**Summary**

In this session, we took you through horizontal analysis and vertical analysis. These tools gave us a clear picture of what the company was doing. You are not supposed to use the two at the same time. As an analyst you decide which one is more suitable for the purpose of your analysis.

**Self-Assessment Questions**

**Exercise 3.3**

## SESSION 4: TREND ANALYSIS

### Introduction

Having looked at vertical and horizontal analysis in the previous session, we take a close look at trend analysis in this session. We define trend analysis and show you how to conduct trend analysis for a company. We also provide advantages and disadvantages of trend analysis. We wish you well as you read.

### Objectives

By the end of this session, you should be able to:

(a) explain the meaning of trend analysis
(b) understand how trend analysis is done
(c) outline the advantages and disadvantages of trend analysis

Now read on …

### 4.1 Meaning of Trend Analysis

The financial statements may be analysed by computing trends of series of information. Trend analysis determines the direction upwards or downwards and involves the computation of the percentage relationship that each item bears to the same item in the base year. In case of comparative statement (which we have already looked at), an item is compared with itself in the previous year to know whether it has increased or decreased or remained constant. Common size analysis (also discussed earlier in the chapter) is to ascertain whether the proportion of an item (say cost of revenue from operations) is increasing or decreasing in the common base (say sales operations).

But in case of trend analysis, we learn about the behaviour of the same item over a given period, say, during the last 5 years. Take for example, administrative expenses, whether they are exhibiting increasing tendency or decreasing tendency or remaining constant over the period of comparison. Generally, trend analysis is done for a reasonably long period. Many companies present their financial data for a period of 5 or 10 years in various forms in their annual reports.

### 4.2 Method of Calculating Trend Percentage

The method of calculating trend percentages involves the calculation of percentage of necessary items with the same items of a base year. i.e., trend percentages are not calculated for all items of financial statements. They are calculated only for major items since the purpose is to highlight important changes. On the other hand, any year may be taken as base year – generally the earliest year is taken as base year. Any intervening year may also be taken as the base year. This might depend on various factors such as macroeconomic and microeconomic indicators, socio-cultural factors and political situation in a particular country.

Thus, trend percentage or trend ratio:

$$= \frac{Value\ of\ each\ item\ in\ the\ financial\ statement\ of\ any\ period}{Value\ of\ same\ item\ in\ the\ financial\ statement\ of\ the\ base\ period} \times 100\%$$

While calculating trend percentages care should be taken regarding the following matters:

(i) The uniform costing system should be followed from year to year for

horizontal analysis.

(ii) The accounting principles and practices should follow consistency convention throughout the period of analysis. Without such consistency, the comparison will be adversely affected.

(iii) The base year should be that normal year which is representative of the items shown in the statement.

(iv) Trend percentages should be calculated only for items having logical relationship with one another.

(v) Trend percentages should be studied after considering the absolute figures on which they are based, otherwise, they may give misleading results. For example, one expense may increase from GH₵100 to GH₵200 while other expenses may increase from GH₵10,000 to GH₵15,000. In the first case, trend percentage will show 100% increase while in second case it will show 50% increase. This is misleading because in the first case, the change though 100% is not significant in real terms as compared to second case.

(vi) The figures for the current year should also be adjusted in the light of price level changes as compared to the base year before calculating the trend percentage, otherwise comparison will be meaningless.

**4.3 Advantages of trend analysis**

(i) It shows the trend of items with passage of time.

(ii) It shows the nature and rate of movement of various financial factors.

(iii) It shows horizontal and vertical analysis to reflect the behaviour of various financial items with passage of time.

(iv) It helps in estimating the financial factor in future.

**4.4 Disadvantages of trend analysis**

(i) If there is no uniform accounting system year after year, then trend ratios give misleading result.

(ii) It does not take into consideration the inflation accounting system. So, figures of the base period are incomparable with the figures of current period in case of inflation.

(iii) Trend ratios must always be read with absolute data on which they are based, otherwise the conclusion drawn may be misleading. It may be said that a 10% change in trend ratio may represent an absolute change of GH₵1,000 only in one item, while a 10% change in another item may represent an absolute change of GH₵10,000.

**4.5 Illustration**

Compute the trend ratios from the following data and comment.

| | **Balances as on 31st December** | | | |
|---|---|---|---|---|
| | **2014 (GH₵)** | **2015 (GH₵)** | **2016 (GH₵)** | **2017 (GH₵)** |
| Cost of material consumed | 200,000 | 250,000 | 200,000 | 180,000 |
| Labour cost | 150,000 | 150,000 | 200,000 | 125,000 |
| Other expense | 150,000 | 200,000 | 100,000 | 150,000 |
| Cost of sales | 500,000 | 600,000 | 500,000 | 455,000 |
| Profit | 300,000 | 300,000 | 250,000 | 345,000 |

| | | | | |
|---|---|---|---|---|
| Sales | 800,000 | 900,000 | 750,000 | 800,000 |

**4.2.1 Solution**

Computation of Trend Ratio (%). Assuming a base year to be 2014.

| | **2014** | **2015** $\frac{2015}{2014} \times 100\%$ | **2016** $\frac{2016}{2014} \times 100\%$ | **2017** $\frac{2017}{2014} \times 100\%$ |
|---|---|---|---|---|
| Cost of material consumed | 100 | 125 | 100 | 90 |
| Labour cost | 100 | 100 | 133.3 | 83.3 |
| Other expense | 100 | 133.3 | 66.7 | 100 |
| Cost of sales | 100 | 120 | 100 | 91 |
| Profit | 100 | 100 | 83.3 | 115 |
| Sales | 100 | 112.5 | 93.8 | 100 |

**Comment**

The reduction in cost of sales in the year 2016 and 2017 is mainly due to reduction in cost of material consumed. Except that there is fluctuating trend in all the items disclosed in the financial statement during the period under study. So, no definite conclusion can be drawn from the above analysis. Though we cannot draw any conclusion from the analysis, it should be noted that the percentage calculation above 100% represents an increment and those below 100% indicate a decrease. Any percentage that is the same as 100% as in the base year implies that there are no changes during the period under review.

**Summary**

In this session, we took you through trend analysis as a tool for analysing the performance of a company. We explained the meaning, advantages and disadvantages of trend analysis. You noticed the slight difference between trend analysis and horizontal analysis. Good student, keep it up.

**Self-Assessment Questions**

**Exercise 3.4**

## SESSION 5: RATIO ANALYSIS

### Introduction

Dear students, we introduce you to ratio analysis in business finance. Financial statements are like fine perfumes to be sniffed and not swallowed and hence the figures must be properly looked at. In this session, we shall discuss five different sets of ratios used for measuring the performance of a company.

### Objectives

By the end of this session, you should be able to:

(a) understand the different types of ratios used in analysing financial statements
(b) measure the profitability of a firm
(c) understand the liquidity position of a firm
(d) monitor the efficiency and productivity of a firm
(e) evaluate the investment returns of the firm
(f) estimate the leverage position of the firm

Now read on …

### 5.1 Types of Financial Ratios

To evaluate the financial condition and performance of a firm, the financial analyst needs certain yardsticks. The yardstick generally used is a ratio or index, relating two pieces of financial data to each other. The relevant ratios can be grouped under the following headings:

(i) Profitability ratios show the firms return on its investments.
(ii) Liquidity ratios measure how easily the firm can lay its hands-on cash.
(iii) Efficiency ratios or turnover ratios measure how productively the firm is using its assets.
(iv) Market (Investment) ratios.
(v) Leverage ratios show how heavily the company is in debt

### 5.2 Profitability Ratios

Users of accounts will want to know how much profit a business has made and compare with previous periods or with other entities. Profit needs to be related to the size of the entity and the capital invested in it. There are four main profitability ratios:

#### 5.2.1 Return on Capital Employed

$$\text{ROCE} = \frac{Profit}{Capital}\ x\ 100$$

The problem with this ratio is that both profit and capital can be defined in several different ways. The main definitions of profit and capital are listed below:
Profit:

- Operating profit
- Net profit before interest and taxation
- Net Profit before taxation
- Net profit after taxation
- Net profit after taxation and preference dividend.

Capital

- Total assets
- Total assets less intangible assets
- Total assets less current liabilities
- Shareholders' funds
- Shareholders' funds less preference shares
- Shareholders' funds plus total liabilities (could be same as total assets)

The method to adopt really depends on your purpose. However, the numerator and denominator must be compatible. You may need to calculate ROCE for different purposes. The main methods are:

1. $\frac{Net\ Profit\ before\ Taxation}{Average\ Shareholder's Funds}\ x\ 100$

2. $\frac{Net\ Profit\ after\ Taxation}{Average\ Shareholder's Fund}\ x\ 100$

3. $\frac{Net\ Profit\ after\ Taxation\ and\ Preference\ Dividend}{Average\ Shareholder's Fund\ less\ Preference\ Shares}\ x\ 100$

4. $\frac{Net\ Profit\ before\ Taxation\ and\ Interest}{Average\ Shareholder's Fund\ plus\ Long-term\ Loans}\ x\ 100$

ROCE helps to avoid making sweeping statements about profit. For example, a profit of $500m may not be high after all if it was generated by a $10b capital employed or a profit of $500 might be acceptable if the capital invested was $2,000. High or low may be viewed in relative sense.

**5.2.2 Gross Profit Ratio**

This ratio enables us to judge how successful the business has been at trading. It tells us how much profit the business has made in relation to the amount of sales that it had made.

$$\text{Gross Profit ratio} = \frac{Gross\ Profit}{Total\ Sales\ Revenue}\ x\ 100$$

**5.2.3 Mark-up ratio**

This ration measures the amount of profit added to the cost of goods sold.

$$\text{Mark-up ratio} = \frac{Gross\ Profit}{Cost\ of\ Goods\ Sold}\ x\ 100$$

### 5.2.4 Net Profit Ratio

This ratio is used to calculate the proportion of revenue that finds its way into profits.

$$\text{Net Profit ratio} = \frac{Net\ Profit\ before\ Taxation}{Total\ Sales\ Revenue}\ x\ 100$$

This ratio is normally used in making internal comparisons. If it is used to make inter-company comparisons, allow for different circumstances.

## 5.3 Liquidity Ratios

These ratios measure the extent to which assets can be quickly turned into cash. It cannot always be assumed that profitability is everything that is desirable. It must be stressed that accounting is needed, not just to calculate profitability, but also to know whether or not the business will be able to pay its creditors, expenses, loans falling due etc at the correct times. Failure to ensure that these payments are covered effectively could mean that the business would have to be closed down. A reasonably sound liquidity position enables an organization to obtain the needed financing to take advantage of investment opportunities as well as respond to operational emergencies. A deficit liquidity position could lead to financial distress and as such the financial manager should examine stock and in particular, the flow of liquid resources as well as timing of cash inflows and outflows. The two common liquidity ratios are:

### 5.3.1 Current (asset) Ratio

This is one way of viewing a company working capital position. It is most commonly used for short-term solvency and the best single indicator of the extent to which claims of short-term creditors are covered by assets that can be converted to cash in a period corresponding to maturity of the debt claims. Hence, current asset shows the number of times current assets will settle current liabilities, and hence a high ratio is required if firms are to remain liquid.

$$\text{Current Ratio} = \frac{Current\ Assets}{Current\ Liabilities}$$

It is usually expressed as a factor e.g. 2:1 or 2 to 1.

### 5.3.2 Quick (Acid Test) Ratio

To get a better picture of a company's ability to meet its short-term obligations, Acid Test (or Quick) ratio is resorted to. This is necessary since it will not be easy to dispose of stocks in the short-term. The Quick ratio therefore compares a firm's current liabilities with its quick assets (cash, short-term marketable securities and current receivables). This stringent test measures the firm's ability to settle short-term debts immediately or within days without relying on the sale of inventories and prepaid expenses.

$$\text{Acid Test Ratio} = \frac{Current\ Assets - Inventories}{Current\ Liabilities}$$

## 5.4 Efficiency Ratios

These ratios report on how efficient management has employed the limited resources in an attempt to increase profit margins and to maximize shareholders worth. We will discuss four main ratios here.

### 5.4.1 Stock/Inventory Turnover Ratio

The stock turnover ratio measures the average rate of speed with which stocks move through the business. The stock turnover ratio gives a measure of the quality as well as the liquidity of the stock component of the current assets. The greater the turnover ratio, the more efficient the entity would appear to be purchasing (or producing) and selling goods. A turnover ratio of say 2 would suggest that the business has about 6 months of sales in stock.

$$\text{Stock/Inventory Turnover Ratio} = \frac{Cost\ of\ Sales}{Average\ Inventories}$$

### 5.4.2 Fixed Asset/ Tangible Non-current Asset Turnover Ratio

Fixed assets (e.g. plant and machinery) enable the business to function more efficiently, so a high level of fixed assets ought to generate more sales. Fixed asset turnover ratio is used to validate this.

$$\text{Fixed Asset Turnover Ratio} = \frac{Total\ Sales\ Revenue}{Tangible\ Non{-}current\ Assets\ at\ net\ book\ value}$$

This ratio is expressed as a factor and could be in percentage too. It is useful only when compared with previous periods or with other entities.

### 5.4.3 Trade Debtor Collect Period (Average Collection Period ((ACP))

This represents the average length of days between the day sales are made (or invoice is sent out) and the day customers pay their bills. An increase in collection days may suggest that customer balances are becoming piled up. The shorter the ACP, the better it is for the business and the more efficient the business is enforcing its credit policy.

$$\text{ACP} = \frac{Average\ Trade\ Debtors\ (Accounts\ Receivables)}{Total\ Credit\ Sales}\ x\ 365\ days$$

### 5.4.4 Trade Creditor Payment Period (Average Payment Period-APP)

This ratio gives the average age of a business accounts payable. It evaluates the probability that payables are made on time. Like the trade debtor collection period, it is not usually expressed as a percentage. An upward trend would suggest that the business is having difficulty in finding the cash to pay its creditors.

$$\text{APP} = \frac{Average\ Trade\ Creditors\ (Accounts\ Payables)}{Total\ Credit\ Purchases}\ x\ 365\ days$$

## 5.5 Market (Investment) Ratios

The ratios examined earlier on are probably of interest to all users of accounts – creditors employees, managers, shareholders etc. There are however some other ratios which are primarily of interest to prospective investors. These ratios are called leverage or investment ratios.

### 5.5.1 Dividend Yield

This ratio usually applies to ordinary shareholders and may be calculated as follows:

$$\text{Dividend Yield} = \frac{Dividend\ per\ Share}{Market\ Price\ per\ Share}\ x\ 100$$

The dividend yield measures the rate of return an investor gets by comparing the cost of his shares with the dividend paid. For example, if an investor buys 100 £1 ordinary shares at a market rate of 200p (£2) per share and the dividend was 10p, his yield would be 5% i.e. 10/200 × 100

**5.5.2 Dividend Cover**

This ratio shows the number of times that ordinary dividend could be paid out of current earnings. Usually, the dividend is described as being X times covered by the earnings. Thus, if dividend is covered twice, the company would be paying out half of its earnings as an ordinary dividend.

$$\text{Dividend Cover} = \frac{Net\ Profit\ after\ Tax\ and\ Preference\ Dividend}{Paid\ and\ Proposed\ Ordinary\ Dividends}$$

**5.5.3 Earnings Per Share**

This ratio enables us to put the profit into context and to avoid looking at it in simple absolute terms. It's usually looked at from the ordinary shareholders perspective.

$$\text{Earnings per share} = \frac{Net\ Profit\ after\ Tax\ and\ Preference\ Dividend}{Number\ of\ Ordinary\ Shares\ in\ issue\ during\ the\ year}$$

**5.5.4 Price/Earnings Ratio**

The P/E ratio enables a comparison to be made between the earnings per share and the market price. It tells us that the market price is X times the earnings. It means that it will take X years before we recover the market price paid for the share out of the earnings.

$$\text{P/E Ratio} = \frac{Market\ Price\ per\ Share}{Earnings\ per\ Share}$$

**5.5.5 Capital Gearing Ratio**

Like ROCE, there are so many ways of calculating it. The two most common forms being as follows:

$$\text{Capital Gearing ratio} = \frac{Borrowings}{Shareholders'Fund + Borrowings}$$

Or

$$\text{Capital Gearing ratio} = \frac{Borrowings}{Shareholders'Fund}$$

The 1st method tells us X% of the company has been financed by borrowings.
Shareholders' funds include:

- Ordinary share capital
- Preference share capital
- Share premium account
- Capital reserves
- Other reserves

- Profit and loss account

Borrowing may include:
- Preference share capital
- Debentures
- Loans
- Overdrafts
- Provisions
- Accruals
- Current liabilities
- Other amounts due for payment

The recommended formula however, is:

$$\text{Capital Gearing ratio} = \frac{Preference\ Shares + Long-term\ Loans}{Shareholders' Fund + Long-term\ Loans}$$

**5.6 Leverage Ratios**

These are ratios used to help in the prediction of the long-term solvency of a firm. They look at the long-term commitment to creditors or accounts payables. These ratios actually look at the capital structure of a company where they examine whether the firm is been financed from within using equity or from outside using debts. Each has advantages and disadvantages. If the firm is financed from debts, it means that there will be maximum control and checks and balances to ensure that the board of directors are doing the right thing. If it is financed from within through equity, it may mean that, the firm has total control over its operations and interest payments will not be made and that money can be reinvested into the business activities of the firm. Failure to meet the claims of outsiders might also mean that legal issues may arise and the assets may be sold or the firm may be liquidated to satisfy the external obligations.

**5.6.1 Debt Ratio**

This ratio measures the percentage of total assets financed from debt. The higher the ratio, the more the firm's assets are provided by creditors relative to the owners. When this happens, we say, the firm is highly geared. The debt ratio is computed by dividing total liabilities by total assets.

$$\text{Debt Ratio} = \frac{Total\ Liabilities}{Total\ Assets}$$

Total liabilities include both current liabilities and long-term liabilities. Creditors prefer low debt ratio, because the lower the ratio, the greater the cushion against creditors' losses in the event of liquidation.

**5.6.2 Debt to Equity Ratio**

The claims against the total assets of the firm are divided into those of the creditors and those of the owners. This total claim provides the measure of solvency of the firm that indicates the margin of safety for the creditors. This ratio is calculated as follows:

$$\text{Debt Ratio} = \frac{Total\ Debts}{Shareholders' Equity\ (Net\ Worth)}$$

This ratio indicates the relative proportions of debt and equity in financing the assets of a firm.
A ratio of 2:1 is considered as ideal, although it depends upon industry.

**5.7 Illustration**

The following Income Statement of Kooko Limited for the year 31-12-2020 is given below:

| | | |
|---|---|---|
| Sales | | 500,000.00 |
| Cost of Sales: | | |
| Opening inventories | 76,250.00 | |
| Add Purchases | 315,250.00 | |
| Carriage and Freight | 2,000.00 | |
| Wages | 5,000.00 | |
| | 398,500.00 | |
| Closing Inventory | (98,500.00) | **(300,000.00)** |
| Gross Profit | | 200,000.00 |
| Add Other Incomes: | | |
| Interest on Securities | | 1,500.00 |
| Dividend on Shares | | 3,750.00 |
| Profit on Sale of Shares | | 750.00 |
| | | **206,000.00** |
| Expenses: | | |
| Administrative Expenses | 101,000.00 | |
| Selling and Distribution Expenses | 12,000.00 | |
| Non-operating Expenses | 2,000.00 | |
| Financial Expenses | 7,000.00 | (122,000.00) |
| Net Profit | | **84,000.00** |

Required: Calculate:
(1) Gross profit margin
(2) Expense ratio
(3) Operating ratio
(4) Net Profit ratio
(5) Operating Profit ratio

**Solution**

1. Gross Profit margin = $\frac{Gross\ Profit}{Sales}\ x\ 100$

$= \frac{200,000}{500,000}\ x\ 100$

= 40%

2. Expense ratio = $\frac{Operating\ Expense}{Sales}\ x\ 100$

$= \frac{113,000}{500,000} x\ 100$

= 22.6%

3. Operating ration = $\frac{Cost\ of\ goods\ Sold + Operating\ Expenses}{Net\ Sales}\ x\ 100$

$= \frac{300,000+113,000}{500,000}\ x\ 100$

= 82.60%

4. Net Profit ratio = $\frac{Net\ Profit}{Net\ Sales}\ x\ 100$

$= \frac{84,000}{500,000}\ x\ 100$
= 16.8%

5. Operating Profit ratio = $\frac{Operating\ Profit}{Net\ Sales}\ x\ 100$
Operating Profit = Sales – Operating Expenses + Administrative Expenses
$= \frac{87,000}{500,000}\ x\ 100$ = 17.40%

**5.8 Illustration**

The Statement of Financial Position of Kaako Limited as at 31-12-2020 was as follows:

| | | | |
|---|---|---|---|
| Non-Current Assets | | | |
| Plant and Machinery | | | 24,000.00 |
| Land and Building | | | 40,000.00 |
| Furniture and Fittings | | | 16,000.00 |
| | | | **80,000.00** |
| Current Assets | | | |
| Inventories | | 12,000.00 | |
| Accounts Receivables | | 12,000.00 | |
| Investments (Short-term) | | 4,000.00 | |
| Cash in Hand | | 12,000.00 | |
| | | **40,000.00** | |
| Current Liabilities | | | |
| Accounts Payables | 16,000.00 | | |
| Bank Overdraft | 4,000.00 | | |
| Current Taxation | 4,000.00 | | |
| Future Taxation | 4,000.00 | **(28,000.00)** | |

| | |
|---|---|
| Net Working Capital | 12,000.00 |
| | **92,000.00** |
| Financed by | |
| Equity Share Capital | 40,000.00 |
| Capital Reserve | 8,000.00 |
| Retained Earnings | 12,000.00 |
| | **60,000.00** |
| Non-Current Liabilities | |
| 8% Loan on Mortgage | 32,000.00 |
| | **92,000.00** |

From the above, calculate the following:

(1) Current ratio
(2) Quick ratio
(3) Debt ratio
(4) Debt to equity ratio
(5) Capital Gearing ratio

Solution

1. Current ratio = $\frac{Current\ Assets}{Current\ Liabilities}$

$= \frac{40,000}{28,000}$

= 1.43:1

2. Quick ratio = $\frac{Current\ Assets - Inventories}{Current\ Liabilities}$

$= \frac{40,000 - 12,000}{28,000}$

= 1:1

3. Debt ratio = $\frac{Total\ Liabilities}{Total\ Assets}$

$= \frac{32,000+28,000}{80,000+ 40,000}$

$= \frac{60,000}{120,000}$

= 0.5

4. Debt to Equity ratio = $\frac{Total\ Debts}{Shareholders' Equity\ (Net\ Worth)}$

$= \frac{60,000}{60,000}$

$= 1$

5. Capital Gearing ratio $= \frac{60,000}{60,000+60,000}$
$= 0.5$

**Summary**
Waaaw!!! This has session had been interesting and tough right? But it is worth the time spent. We took you through five different sets of ratios used for analysing the performance of a company. These are profitability ratios showed the firms return on its investments; liquidity ratios measured how easily the firm can lay its hands-on cash; efficiency ratios or turnover ratios measured how productively the firm is using its assets; market (Investment) ratios and leverage ratios showed how heavily the company was in debt.

**Self-Assessment Questions**
**Exercise 3.5**

## SESSION 6: INTRODUCTION TO DISTRESS ANALYSIS

**Introduction**
Financial analysts and other users of financial statements have long recognized the intricacy that financial statements may be masked or window dressed or manipulated or massaged and doubted the reliability of the financial information. Events of financial crises such as Enron, Merck, Xerox, and WorldCom that occurred recently in U.S. and the takeover of UT Bank and Capital Bank by GCB Bank in Ghana in 2017, and the closure of several microfinance and investment banks in 2018-2019 increased the concerns on the creditworthiness of the disclosed financial data. Under this background, many practitioners, academicians and even market agencies tended to express a negative attitude toward the trustworthiness of accounting information. The situation in Ghana in 2017, where these two financial institutions no longer exist, and UT Bank delisted from the Ghana Stock Exchange brings into focus the need for distress analysis.

**Objectives**
By the end of this session, you should be able to:
(a) explain the meaning of financial distress
(b) outline the causes of financial distress
(c) estimate the distress level of companies

Now read on …

### 6.1 Meaning of Distress
Distress means acute financial hardship or crisis. Corporate distress or sickness means such a situation of a firm when it is unable to meet its debt. In other words, when value of the total

assets of a company is insufficient to discharge its total external liabilities, the said company can be said to be a 'Distress Company'. Corporate Distress also includes the following:

(a) Unfavourable liquidity position;
(b) Adverse long-term solvency position;
(c) Outdated production process;
(d) Deteriorating selling status;
(e) Poor administrative set-up; and
(f) Overall adverse economic condition.

Corporate distress, therefore, is a situation when the financial status of a company moves towards bankruptcy or insolvency. Ross (1996) stated that financial distress is a situation where a firm's operating cash flows are not sufficient to satisfy current obligations and the firm is forced to take corrective action. Dewing (1952) identified four causes of business or corporate failures: excessive competition, unprofitable expansion, cessation of public demand for the firm's products or services, and excessive payment of capital charges. Beaver (1966) defined corporate failure as bankruptcy, overdrawn of bank account, or non-payment of a dividend to shareholders.

**6.2 Edward I. Altman's Multivariate Analysis for public companies**

Edward I. Altman (born 1941) was a Professor of Finance, Emeritus, at New York University's Stern School of Business. He is best known for the development of the Altman Z-score for predicting bankruptcy which he published in 1968. Edward I. Altman developed a Multivariate Model of corporate distress prediction on the basis of Multiple Discriminant Analysis (MDA). In his study, Altman selected 33 failed and 33 non-failed firms, of which 22 accounting and non-accounting ratios, which had been deemed to be the predictors of corporate distress, were taken into consideration. Of the 22 accounting ratios, Prof. Altman selected 5 ratios which had been deemed as the best predictors of corporate distress prediction. The purposes of these five selected ratios are as follows:

1. To measure liquidity position of the firms.
2. To measure reinvestment of earnings of the firms.
3. To measure profitability of the firms.
4. To measure financial leverage condition of the firms.
5. To measure sales-generating ability of firm's Assets.

In 1968, the following Discriminant Function was developed by Altman:

$Z = 1.2\ X_1 + 1.4\ X_2 + 3.3\ X_3 + 0.6\ X_4 + 1.0\ X_5$

Where

Z = Overall Index

The five variables are

$X_1$ = Working Capital to Total Assets (a liquidity measure)

$X_2$ = Retained Earnings to Total Assets (a measure of reinvestment of earnings)

$X_3$ = EBIT to Total Assets (a profitability measure)

$X_4$ = Market Value of Equity & Preference to Book Value of Total Debt (a measure of leverage)

$X_5$ = Sales to Total Assets (a measure of sales-generating ability of the firm's assets)

**6.3 Analysis of Value of Z-score**

i. If the calculated value of Z-score is greater than 2.99, it is predicted that the firm belongs to non-bankrupt class (i.e., non-failed firm or non-distress firm).
ii. If the calculated value of Z-score is smaller than 1.81, it is predicted that the firm belongs to bankrupt class (i.e., failed firm or distress firm).

iii. If the calculated value of Z-score of a firm falls between 1.81 and 2.99 (referred to as Grey Area), it is predicted that the firm consists of both bankrupt and non-bankrupt class (i.e., mixture of failed and non-failed elements or distress and non-distress) and, therefore, requires further investigation to determine its solvency status. As per Altman's Multivariate Model of Distress Prediction:

   a. If Z > 2.99: Non-failed or non-distressed firm.
   b. If Z < 1.81: Failed or distressed firm.
   c. If Z ≥ 1.81 but ≤ 2.99: Mixture of failed and non-failed elements which requires further investigation to determine its solvency status.

**6.4 Illustration**

These are samples from a typical hypothetical company.

$\boldsymbol{X_1}$= Working Capital to Total Assets= $\frac{\text{WORKING CAPITAL}}{TOTAL\ ASSETS}$

$\frac{16501}{126946}$ $\frac{4592}{101464}$

**0.13** **0.05**

$\boldsymbol{X_2}$= Retained Earnings to Total Assets= $\frac{\text{RETAINED EARNINGS}}{TOTAL\ ASSETS}$

$\frac{47408}{126946}$ $\frac{39730}{101464}$

**0.37** **0.39**

$\boldsymbol{X_3}$= EBIT to Total Assets = $\frac{\text{EBIT}}{TOTAL\ ASSETS}$

$\frac{38653}{126946}$ $\frac{23148}{101464}$

**0.30** **0.23**

$\boldsymbol{X_4}$= Market Value of Equity Shares to Book Value of Total Debt= $\frac{\text{MARKET VALUE OF EQUITY}}{BOOKVALUE\ TOTAL\ DEBT}$

$\frac{802500000}{78053000}$ $\frac{531875000}{60124000}$

**10.28** **9.70**

$\boldsymbol{X_5}$= SALES TO TOTAL ASSETS = $\frac{\text{SALES}}{TOTAL\ ASSETS}$

$= \frac{239005}{126946}$ $= \frac{179257}{101464}$

**1.88** **1.77**

$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$

$Z_{2017}$= 1.2(0.1300) +1.4(0.3735) +3.3(0.3045) +0.6(10.2815) +1.0(1.8827)
= 0.1560 + 0.5228 + 1.0048 + 6.1689+ 1.8827
= 9.74 **(Non-distress Firm)**

$Z_{2016}$= 1.2(0.0453) +1.4(0.3916) +3.3(0.2281) +0.6(9.7008) +1.0(1.7667)
= 0.0543 + 0.5482 + 0.7529 + 5.8205+ 1.7667
= 8.94 **(Non-distress Firm)**

**Summary**

We have ended this session on predicting financial distress of a firm. We explained the meaning of distress and the causes of distress. This we believed had prepared you for analysing the distress levels of a company before investing in such a company.

**Self-Assessment Questions**
**Exercise 3.6**

1. What is financial distress?
2. Outline any four causes of financial distress in Ghana.
3. What is the significance of $X_1$ and $X_5$ in distress analysis?

## UNIT 4: TIME VALUE OF MONEY AND VALUING FINANCIAL SECURITIES

**Unit Outline**
Session 1: What is time value of money?
Session 2: Present values and future values
Session 3: Annuities
Session 4: Perpetuities, sinking fund and loan amortization
Session 5: Valuation of bonds
Session 6: Valuation of shares

**Overview**
Companies, big and small, need capital to do their business. The investors provide the capital to a company. A company may need a new factory to manufacture its products, or an airline a few more planes to expand into new territory. The firm acquires the money needed to build the factory or to buy the new planes from investors. The investors, of course, want a return on their investment. In this Unit, we shall discuss time value of money, the concept of compounding and discounting, and the various applications of time value of money such as

annuities, perpetuities, sinking fund and loan amortization. We will also look at the valuation of bonds and shares as major financial securities.

**Objectives**

By the end of this Unit, you should be able to:

(d) explain the concept of time value of money
(e) distinguish between compounding and discounting
(f) appreciate compound interest and how it can be used
(g) determine the present value and future value of money
(h) recognize the application of time value of money in annuities, perpetuities, sinking fund and loan amortization
(i) understand bond price and how it is determined
(j) undertake valuation of different types of shares

**SESSION 1 What is time value of money?**

**Introduction**

Dear lovely students, you are most welcomed to this session. Remember we always say time is money and there is no time to waste? This is the session that actually look at the time value of money. We shall look at compounding and discounting, future value and present value and how they can be related to each other. We shall also discuss the different types of interest in finance. Hey, it's going to be great. Just box on.

**Objectives**

By the end of this session, you should be able to:

(e) explain the concept of time value of money
(f) distinguish between compounding and discounting
(g) relate compounding and future value; discounting and present value
(h) identify the different types of interest rates
(i) understanding compounding periods

Now read on …

### 1.1 Meaning of time value of money

Which would you prefer - GH¢10,000 today or GH¢10,000 in 5 years? Obviously, GH¢10,000 today. You already recognize that there is TIME VALUE TO MONEY!! Why is TIME such an important element in your decision? TIME allows you the opportunity to postpone consumption and earn INTEREST.

The concept, time value of money, is very important in financial decision – making. Conceptually, it means that the value of a unit of money is different in different time periods. Simply put, it means a cedi today is worth more than a cedi tomorrow. The subject-matter of finance is that of determination of value for the present period as well as the future. There are three reasons why a cedi tomorrow is worth less than a cedi today:

- Individuals prefer present consumption to future consumption. To induce people to give up present consumption you have to offer them more in the future.
- When there is monetary inflation, the value of currency decreases over time. The greater the inflation, the greater the difference in value between a cedi or dollar today and a cedi or dollar tomorrow.
- If there is any uncertainty (risk) associated with the cash flow in the future, the less that cash flow will be valued.

Other things remaining equal, the value of cash flows in future time periods will decrease as:

- the preference for current consumption increases;
- expected inflation increases;
- the uncertainty in the cash flow increases.

This is based on the concept that the value of a unit of money is different in different time periods. That is to mean that a cedi in the hand today is worth more than a cedi to be received tomorrow or a cedi paid to you in a day to come from now is less valuable to you than a cedi paid to you today. You would obviously choose a GH¢1,000 today when given the option to choose between GH¢1,000 today and GH¢1,000 in ten years because the immediate receipt of GH¢1,000 provides you the opportunity to put your money to work and earn interest. Secondly, individuals prefer present consumption to future consumption hence to induce people to give up present consumption you have to offer them more in the future. Again, when there is monetary inflation, the value of currency decreases over time. The greater the inflation, the greater the difference in value between a cedi today and a cedi tomorrow. Furthermore, if there is any uncertainty or risk associated with the cash flow in the future, the less that cash flow will be valued.

### 1.3 Compounding and Discounting

The mechanism for factoring in these elements is the discount rate. The discount rate is a rate at which present and future cash flows are traded off. It incorporates: Preference for current consumption (Greater…Higher Discount Rate); Expected inflation (Higher inflation…Higher Discount Rate); and Uncertainty in the future cash flows (Higher Risk...Higher Discount Rate). A higher discount rate will lead to a lower value for cash flows in the future. The discount rate is also an opportunity cost, since it captures the returns that an individual would have made on the next best opportunity.

Discounting future cash flows converts them into cash flows in present value cedis. Just a discounting converts future cash flows into present cash flows. Compounding converts present cash flows into future cash flows.

### 1.4 Future Value

Amount to which an investment will grow after earning interest. Future value, sometimes called compound value, is the amount to which a present amount of money or a series of payments will grow overtime when compounded at a given interest. Future Value is the value at some future time of a present amount of money, or a series of payments, evaluated at a given interest rate. A future value comprises two parts: Future Value = Present Value + Interest

### 1.4.1 Present Value

The current value of a future amount of money, or a series of payments, evaluated at a given interest rate. In most instances, it is the principal amount of money borrowed or lent. The present value is the amount of money invested or borrowed today.

### 1.4.2 Interest

This is the compensation for the use of funds for a specific period. It is the cost of borrowing money or the return from lending money. Interest rates usually are stated as, or understood to be, annual rates. It consists of compensation for the length of time the money is borrowed or invested and compensation for the risk that will not be repaid exactly as promised. That is why in your business law course, it is said that time of payment is not a condition of a contract of sale but rather a warranty. Warranty because, interest can be calculated if the debtor delayed in payment the money. There are two types of interests: simple interest and compound interest.

#### 1.4.2.1 Simple interest

Simple interest is the interest earned only on the original investment or principal, borrowed (lent). The cedi amount of simple interest is a function of three variables: the original amount borrowed (lent), or principal; the interest rate per time period; and the number of time periods for which the principal is borrowed (lent).

The formula for calculating simple interest is: $SI = P_o \times r \times t$
Where
SI = simple interest in cedis
$P_0$ = principal or original amount borrowed (lent) at time period
r = i = interest rate per time period; other represent the 'r' with 'i'.
t = n = number of time periods; others represent the 't' with 'n'.

**Example 1**

Assume that you deposit GH¢100 in a savings paying 8 percent simple interest and keep it there for 10 years. At the end of 10 years, the amount of interest accumulated is determined as follows:

$$SI = P_o \times r \times t$$
$$P_0 = 100$$
$$r = i = 8\% = 0.08$$
$$t = n = 10$$
$$SI = 100 \times 0.08 \times 10 = GH¢80$$

To solve for the future value of the amount at the end of 10 years ($FV_{10}$), we add the interest earned on the principal only to the original amount invested. Therefore

$$FV_{10} = GH¢\ 100 + [100(0.08)(10)] = GH¢\ 180$$

For any simple interest rate, the future value of an account at the end of n periods is

$FV_n = P_0 + SI = P_0 + P_0(r)(t)$ or

Equivalently, $FV_n = P_0[(1+r)(t)]$

**Example 2**

Assume $P_0$ = GH¢ 100,000; r = 10%; t = 1

SI = P x r x t

SI = GH¢ 100,000 x 0.1x 1 = GH¢10,000

$FV_1$ = P+1

= GH¢100,000 + GH¢10,000

= 110,000

If t = 2,

$FV_2$ = GH¢120,000

**Example 3**

Suppose that you deposited GH¢5000 into your account that yields a 25% for five years, what is the value of your money at the end of 5 years assuming the interest is charged under: Simple interest basis?

***Solution***

$FV_5 = P_0(1+r\times t)$

$= 5000(1+0.25\times 5)$

$= 5000(2.25)$

= GH¢ 1,250

**1.4.2.2 Compound interest**

According to Albert Einstein, "the greatest mathematical discovery of all time is compound interest." Compound interest is interest earned on interest. It simply means that the interest from the first period is added to the future present value, and the interest rate the next time around is now being applied to a larger amount. In other words, interest is earned on previously earned interest hence the name compound interest. The general formula for the future (compound) value of a deposit at the end of *n* periods, is: $FV = P_0 (1 + r)^t$

To begin with, consider a person who deposits GH¢100 into a savings account. If the interest rate is 8 percent, compounded annually, how much will the GH¢100 be worth at the end of a year? Setting up the problem, we solve for the future value (which in this case is also referred to as the compound value) of the account at the end of the year.

$FV_1 = P_0 (1+r) = 100(1.08)$ = GH¢108

Interestingly, this first-year value is the same number that we would get if simple interest were employed. But this is where the similarity ends. What if we leave GH¢100 on deposit for two years? The GH¢100 initial deposit will have grown to GH¢108 at the end of the first year at 8 percent compound annual interest. Going to the end of the second year, GH¢108 becomes GH¢116.64, as GH¢8 in interest is earned on the initial GH¢100, and GH¢0.64 is earned on the GH¢8 in interest credited to our account at the end of the first year.

$FV_n = P_0(1+r)^t$

**Example 4**

Suppose that you deposited GH¢5000 into your account that yields a 25% for five years, what is the value of your money at the end of 5 years assuming the interest is charged under: Compound interest basis?

***Solution***

$FV_5 = P_0(1+r)^t$
$= 5000(1.25)^5$
$= 5000(3.05175)$
$=$ GH¢ 15,529

### 1.4.2.3 Compounding Period

A compounding period is the span of time between when interest was last compounded and when it will be compounded again. For example, annual compounding means a full year will pass before interest is compounded again. If interest is compounded semi-annually, it means six months will pass before interest is compounded. This means that in a full year, we shall have 2 compounding periods. This same principle applies when interest is compounded quarterly, monthly and daily. For quarterly, we have 4 compounding periods, monthly we have 12 compounding periods and for daily, we have 365/6 compounding periods. If interest is compounded on weekly basis, we have 52 compounding periods.

General Formula: $FVn = P_o(1 + [i/m])^{mn}$
Where:

- n: Number of Years
- m: Compounding Periods per Year
- i: Annual Interest Rate
- FVn: FV at the end of Year n
- $P_o$: PV of the Cash Flow today

Example:
Julie Miller has GHS1,000 to invest for 2 Years at an annual interest rate of 12%.

| | | |
|---|---|---|
| Annual | $FV_2 = 1{,}000(1+[0.12/1])^{(1)(2)}$ | $= 1{,}254.40$ |
| Semi | $FV_2 = 1{,}000(1+[0.12/2])^{(2)(2)}$ | $= 1{,}262.48$ |
| Quarterly | $FV_2 = GHS1{,}000(1+[0.12/4])^{(4)(2)}$ | $= GHS1{,}266.77$ |
| Monthly | $FV_2 = GHS1{,}000(1+[0.12/12])^{(12)(2)}$ | $= GHS1{,}269.73$ |
| Daily | $FV_2 = GHS1{,}000(1+[0.12/365])^{(365)(2)}$ | $= GHS1{,}271.20$ |

### 1.4.2.4 Continuous Compounding

$FV = PVe^{rt}$
Where e is the base of natural log. The value of e = 2.718
$\lim_{m\to\infty} (I + {}^{r}/_{m})^{m} = e$

**Summary**

In this session, we looked at the meaning of time value of money. We examined how compounding and future value related and discounting and present value are related. We explained that the interest charged on any amount of money could be simple interest or compound interest and in reality, it is the compound interest that is mostly used. We explained that the more the compounding period, the higher the interest to be earned.

**Self-Assessment Questions**
**Exercise 4.1**

## SESSION 2 PRESENT VALUES AND FUTURE VALUES

### Introduction

Remember we looked at discounting and compounding in our previous session? Good!! In this session we shall look at the application of the concept of discounting and compounding.

The value of an amount of money today and the value of the amount in the future. These are issued of present value and future value we shall look at in this session.

**Objectives**
By the end of this session, you should be able to:
(a) calculate the present value of future cash flows
(b) determine the future value of present cash flows
(c) understand the use of Present and Future value Tables
(d) describe the application of the Rule of 72

Now read on …

**2.1 Present Values**
Money can be invested to earn interest. If you are offered the choice between GH¢10,000 now and GH¢10,000 at the time of your graduation, you naturally take the money now to get some interest. Financial managers make the same point when they say that money in hand today has a time value or a cedi today is worth more than a cedi tomorrow. The present value of an amount is the value today of a future cash flow.

Finding present values is called discounting, and it is simply the reverse of compounding. In general, the present value of a cash flow due n years in the future is the amount which, if it were on hand today, would grow to equal the future amount. By solving for PV in the future value equation, the present value, or discounting, equation can be developed and written as

$$\boldsymbol{PV = \frac{FV_n}{(1+i)^n} = FV_t\left(\frac{1}{1+i}\right)^n = FV_n(PVIF_{i,n})}$$

$PVIF_{i,n}$, the Present Value Interest Factor at i% for n periods, is a short-hand way of writing the equation which is equal to $(1+i)^{-n}$ or $\left(\frac{1}{1+i}\right)^n$
In present value problems, the interest rate is also known as the discount rate or capitalization rate. So, the discount rate is the interest rate used to compute present value of future cash flows. The present value is often called the discounted value. To calculate the present value, we discount the future value at the interest rate *i*.

**Example 1**
You need GH¢3,000 next year to buy a new computer. The interest rate is 8% per annum. How much money should you set aside now in order to pay for the purchase of the new computer?
Solution
In doing this we have two main approaches. We can use the formula or use the mathematical table. Let us look at the formula first and the use the table later.

$$\boldsymbol{PV = \frac{FV_n}{(1+i)^n}}$$

$$\boldsymbol{PV = \frac{3{,}000}{(1+0.08)^1} = \frac{3{,}000}{1.08}} = \text{GH¢}\mathbf{2{,}778}$$

Now that GH¢2,777.78 invested for 1 year at 8% will prove just enough for you to buy your computer:

Future value = GH¢**2,778 x 1.08** = GH¢**3,000.00**

This simply shows that the longer time before you make a payment, the less you need to invest today. Can you try if you want to purchase your new computer until the end of 2 years? What was your answer? GH¢**2,572 right?** Present values are always calculated using compound interest.

Now let's consider the use of mathematical table. The use of the present value table. In doing this, we use $\boldsymbol{PV = FV_n(PVIF_{i,n})}$. Where:

PV = Present Value
FVn = Future Value at period n
PVIF = Present Value Interest Factor
i = interest rate
n = time period

We then use the mathematical table, the present value table to find the PVIF.
$\boldsymbol{PV = 3,000(PVIF_{8\%,1})}$. We look for the value of $\boldsymbol{PVIF_{8\%,1}}$ from the table. **This value from the table is also called the discount factor.**

PV = GH¢3,000($PVIF_{i\%,\ n}$)
PV = GH¢3,000($PVIF_{8\%,\ 1}$)

($PVIF_{8\%,\ 1}$) can be read from the Table. From the Table, look for 8% and 1 period, where the two intersect, the value is the Present Value Interest Factor (PVIF). From the Table, the value is 0.9259. Therefore:

PV = GH¢3,000 (0.9259)
PV = GH¢**2,778**

This answer is the same as using the formula. This Present Value Table is available and can be kept with you and used during any of your examinations. You only ensure you do not write anything on it. You are advised to laminate it.

Present Value Interest Factors for One cedi Discounted at i Percent for n Periods: PVIF i,n = $1 / (1 + k)^n$

| Number of Period | Interest Rate per Period | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.0100 | 2% | 3% | 4% | 5% | 6% | 7% | 8% | 9% | 10% | 11% | 12% | 13% | 14% | 15% | 16% |
| 1 | 0.9901 | 0.9804 | 0.9709 | 0.9615 | 0.9524 | 0.9434 | 0.9346 | 0.9259 | 0.9174 | 0.9091 | 0.9009 | 0.8929 | 0.8850 | 0.8772 | 0.8696 | 0.8621 |
| 2 | 0.9803 | 0.9612 | 0.9426 | 0.9246 | 0.9070 | 0.8900 | 0.8734 | 0.8573 | 0.8417 | 0.8264 | 0.8116 | 0.7972 | 0.7831 | 0.7695 | 0.7561 | 0.7432 |
| 3 | 0.9706 | 0.9423 | 0.9151 | 0.8890 | 0.8638 | 0.8396 | 0.8163 | 0.7938 | 0.7722 | 0.7513 | 0.7312 | 0.7118 | 0.6931 | 0.6750 | 0.6575 | 0.6407 |
| 4 | 0.9610 | 0.9238 | 0.8885 | 0.8548 | 0.8227 | 0.7921 | 0.7629 | 0.7350 | 0.7084 | 0.6830 | 0.6587 | 0.6355 | 0.6133 | 0.5921 | 0.5718 | 0.5523 |
| 5 | 0.9515 | 0.9057 | 0.8626 | 0.8219 | 0.7835 | 0.7473 | 0.7130 | 0.6806 | 0.6499 | 0.6209 | 0.5935 | 0.5674 | 0.5428 | 0.5194 | 0.4972 | 0.4761 |
| 6 | 0.9420 | 0.8880 | 0.8375 | 0.7903 | 0.7462 | 0.7050 | 0.6663 | 0.6302 | 0.5963 | 0.5645 | 0.5346 | 0.5066 | 0.4803 | 0.4556 | 0.4323 | 0.4104 |
| 7 | 0.9327 | 0.8706 | 0.8131 | 0.7599 | 0.7107 | 0.6651 | 0.6227 | 0.5835 | 0.5470 | 0.5132 | 0.4817 | 0.4523 | 0.4251 | 0.3996 | 0.3759 | 0.3538 |
| 8 | 0.9235 | 0.8535 | 0.7894 | 0.7307 | 0.6768 | 0.6274 | 0.5820 | 0.5403 | 0.5019 | 0.4665 | 0.4339 | 0.4039 | 0.3762 | 0.3506 | 0.3269 | 0.3050 |
| 9 | 0.9143 | 0.8368 | 0.7664 | 0.7026 | 0.6446 | 0.5919 | 0.5439 | 0.5002 | 0.4604 | 0.4241 | 0.3909 | 0.3606 | 0.3329 | 0.3075 | 0.2843 | 0.2630 |
| 10 | 0.9053 | 0.8203 | 0.7441 | 0.6756 | 0.6139 | 0.5584 | 0.5083 | 0.4632 | 0.4224 | 0.3855 | 0.3522 | 0.3220 | 0.2946 | 0.2697 | 0.2472 | 0.2267 |
| 11 | 0.8963 | 0.8043 | 0.7224 | 0.6496 | 0.5847 | 0.5268 | 0.4751 | 0.4289 | 0.3875 | 0.3505 | 0.3173 | 0.2875 | 0.2607 | 0.2366 | 0.2149 | 0.1954 |
| 12 | 0.8874 | 0.7885 | 0.7014 | 0.6246 | 0.5568 | 0.4970 | 0.4440 | 0.3971 | 0.3555 | 0.3186 | 0.2858 | 0.2567 | 0.2307 | 0.2076 | 0.1869 | 0.1685 |
| 13 | 0.8787 | 0.7730 | 0.6810 | 0.6006 | 0.5303 | 0.4688 | 0.4150 | 0.3677 | 0.3262 | 0.2897 | 0.2575 | 0.2292 | 0.2042 | 0.1821 | 0.1625 | 0.1452 |
| 14 | 0.8700 | 0.7579 | 0.6611 | 0.5775 | 0.5051 | 0.4423 | 0.3878 | 0.3405 | 0.2992 | 0.2633 | 0.2320 | 0.2046 | 0.1807 | 0.1597 | 0.1413 | 0.1252 |
| 15 | 0.8613 | 0.7430 | 0.6419 | 0.5553 | 0.4810 | 0.4173 | 0.3624 | 0.3152 | 0.2745 | 0.2394 | 0.2090 | 0.1827 | 0.1599 | 0.1401 | 0.1229 | 0.1079 |
| 16 | 0.8528 | 0.7284 | 0.6232 | 0.5339 | 0.4581 | 0.3936 | 0.3387 | 0.2919 | 0.2519 | 0.2176 | 0.1883 | 0.1631 | 0.1415 | 0.1229 | 0.1069 | 0.0930 |
| 17 | 0.8444 | 0.7142 | 0.6050 | 0.5134 | 0.4363 | 0.3714 | 0.3166 | 0.2703 | 0.2311 | 0.1978 | 0.1696 | 0.1456 | 0.1252 | 0.1078 | 0.0929 | 0.0802 |
| 18 | 0.8360 | 0.7002 | 0.5874 | 0.4936 | 0.4155 | 0.3503 | 0.2959 | 0.2502 | 0.2120 | 0.1799 | 0.1528 | 0.1300 | 0.1108 | 0.0946 | 0.0808 | 0.0691 |
| 19 | 0.8277 | 0.6864 | 0.5703 | 0.4746 | 0.3957 | 0.3305 | 0.2765 | 0.2317 | 0.1945 | 0.1635 | 0.1377 | 0.1161 | 0.0981 | 0.0829 | 0.0703 | 0.0596 |
| 20 | 0.8195 | 0.6730 | 0.5537 | 0.4564 | 0.3769 | 0.3118 | 0.2584 | 0.2145 | 0.1784 | 0.1486 | 0.1240 | 0.1037 | 0.0868 | 0.0728 | 0.0611 | 0.0514 |
| 21 | 0.8114 | 0.6598 | 0.5375 | 0.4388 | 0.3589 | 0.2942 | 0.2415 | 0.1987 | 0.1637 | 0.1351 | 0.1117 | 0.0926 | 0.0768 | 0.0638 | 0.0531 | 0.0443 |
| 22 | 0.8034 | 0.6468 | 0.5219 | 0.4220 | 0.3418 | 0.2775 | 0.2257 | 0.1839 | 0.1502 | 0.1228 | 0.1007 | 0.0826 | 0.0680 | 0.0560 | 0.0462 | 0.0382 |
| 23 | 0.7954 | 0.6342 | 0.5067 | 0.4057 | 0.3256 | 0.2618 | 0.2109 | 0.1703 | 0.1378 | 0.1117 | 0.0907 | 0.0738 | 0.0601 | 0.0491 | 0.0402 | 0.0329 |
| 24 | 0.7876 | 0.6217 | 0.4919 | 0.3901 | 0.3101 | 0.2470 | 0.1971 | 0.1577 | 0.1264 | 0.1015 | 0.0817 | 0.0659 | 0.0532 | 0.0431 | 0.0349 | 0.0284 |
| 25 | 0.7798 | 0.6095 | 0.4776 | 0.3751 | 0.2953 | 0.2330 | 0.1842 | 0.1460 | 0.1160 | 0.0923 | 0.0736 | 0.0588 | 0.0471 | 0.0378 | 0.0304 | 0.0245 |
| 26 | 0.7720 | 0.5976 | 0.4637 | 0.3607 | 0.2812 | 0.2198 | 0.1722 | 0.1352 | 0.1064 | 0.0839 | 0.0663 | 0.0525 | 0.0417 | 0.0331 | 0.0264 | 0.0211 |
| 27 | 0.7644 | 0.5859 | 0.4502 | 0.3468 | 0.2678 | 0.2074 | 0.1609 | 0.1252 | 0.0976 | 0.0763 | 0.0597 | 0.0469 | 0.0369 | 0.0291 | 0.0230 | 0.0182 |
| 28 | 0.7568 | 0.5744 | 0.4371 | 0.3335 | 0.2551 | 0.1956 | 0.1504 | 0.1159 | 0.0895 | 0.0693 | 0.0538 | 0.0419 | 0.0326 | 0.0255 | 0.0200 | 0.0157 |
| 29 | 0.7493 | 0.5631 | 0.4243 | 0.3207 | 0.2429 | 0.1846 | 0.1406 | 0.1073 | 0.0822 | 0.0630 | 0.0485 | 0.0374 | 0.0289 | 0.0224 | 0.0174 | 0.0135 |
| 30 | 0.7419 | 0.5521 | 0.4120 | 0.3083 | 0.2314 | 0.1741 | 0.1314 | 0.0994 | 0.0754 | 0.0573 | 0.0437 | 0.0334 | 0.0256 | 0.0196 | 0.0151 | 0.0116 |

## 2.2 Future Values

We have already explained the issue of future value. Let us look at its application in business. After using the formula in our previous session, here, we shall look at the use of the Mathematical Table. Using the Future Value Table: $FV_n = P_0(FVIF_{i\%,\ t})$. Where we let $FVIF_{i\%,t}$ (i.e., the future value interest factor at i% for n periods) equal $(1+r)^t$. The expression

$(1+r)^t$ is the *compound factor*. This compound factor can be found using the Future Value Table.

**Example 2**

You have GH¢10,000 invested in a bank account for 6 years. Supposed the bank is currently paying an interest of 6 percent per year on the investment. What is the total value of the investment at the end of the 6 years?

In solving this problem, we can use two approaches. The formula and the Future Value table.

Using the formula approach

$$FV_n = P_0(1+r)^t \text{ Or}$$
$$FV_6 = 10{,}000\ (1+0.06)^6$$
$$FV_6 = 10{,}000\ (1.06)^6$$
$$FV_6 = 10{,}000\ (1.4185)$$
$$FV_6 = 14{,}185$$

Using the Table approach

$$FV_n = P_0(FVIF_{i\%,\ t})$$
$$FV_6 = 10{,}000\ (FVIF_{6\%,\ 6})$$

$(FVIF_{6\%,\ 6})$ can be read from the Table. From the Table, look for 6% and 6 periods, where the two intersect, the value is the Future Value Interest Factor (FVIF). From the Table, the value is 1.4185. Therefore:

$$FV_6 = 10{,}000\ (1.4185)$$
$$FV_6 = 14{,}185$$

Future Value Interest Factors for One Ghana Cedi Compounded at i Percent for n Periods: FVIF $_{i\%,\ n} = (1 + i)^n$

| Number of Period | Interest Rate per Period | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1% | 2% | 3% | 4% | 5% | 6% | 7% | 8% | 9% | 10% | 11% | 12% | 13% | 14% | 15% | 16% |
| 1 | 1.0100 | 1.0200 | 1.0300 | 1.0400 | 1.0500 | 1.0600 | 1.0700 | 1.0800 | 1.0900 | 1.1000 | 1.1100 | 1.1200 | 1.1300 | 1.1400 | 1.1500 | 1.1600 |
| 2 | 1.0201 | 1.0404 | 1.0609 | 1.0816 | 1.1025 | 1.1236 | 1.1449 | 1.1664 | 1.1881 | 1.2100 | 1.2321 | 1.2544 | 1.2769 | 1.2996 | 1.3225 | 1.3456 |
| 3 | 1.0303 | 1.0612 | 1.0927 | 1.1249 | 1.1576 | 1.1910 | 1.2250 | 1.2597 | 1.2950 | 1.3310 | 1.3676 | 1.4049 | 1.4429 | 1.4815 | 1.5209 | 1.5609 |
| 4 | 1.0406 | 1.0824 | 1.1255 | 1.1699 | 1.2155 | 1.2625 | 1.3108 | 1.3605 | 1.4116 | 1.4641 | 1.5181 | 1.5735 | 1.6305 | 1.6890 | 1.7490 | 1.8106 |
| 5 | 1.0510 | 1.1041 | 1.1593 | 1.2167 | 1.2763 | 1.3382 | 1.4026 | 1.4693 | 1.5386 | 1.6105 | 1.6851 | 1.7623 | 1.8424 | 1.9254 | 2.0114 | 2.1003 |
| 6 | 1.0615 | 1.1262 | 1.1941 | 1.2653 | 1.3401 | 1.4185 | 1.5007 | 1.5869 | 1.6771 | 1.7716 | 1.8704 | 1.9738 | 2.0820 | 2.1950 | 2.3131 | 2.4364 |
| 7 | 1.0721 | 1.1487 | 1.2299 | 1.3159 | 1.4071 | 1.5036 | 1.6058 | 1.7138 | 1.8280 | 1.9487 | 2.0762 | 2.2107 | 2.3526 | 2.5023 | 2.6600 | 2.8262 |
| 8 | 1.0829 | 1.1717 | 1.2668 | 1.3686 | 1.4775 | 1.5938 | 1.7182 | 1.8509 | 1.9926 | 2.1436 | 2.3045 | 2.4760 | 2.6584 | 2.8526 | 3.0590 | 3.2784 |
| 9 | 1.0937 | 1.1951 | 1.3048 | 1.4233 | 1.5513 | 1.6895 | 1.8385 | 1.9990 | 2.1719 | 2.3579 | 2.5580 | 2.7731 | 3.0040 | 3.2519 | 3.5179 | 3.8030 |
| 10 | 1.1046 | 1.2190 | 1.3439 | 1.4802 | 1.6289 | 1.7908 | 1.9672 | 2.1589 | 2.3674 | 2.5937 | 2.8394 | 3.1058 | 3.3946 | 3.7072 | 4.0456 | 4.4114 |
| 11 | 1.1157 | 1.2434 | 1.3842 | 1.5395 | 1.7103 | 1.8983 | 2.1049 | 2.3316 | 2.5804 | 2.8531 | 3.1518 | 3.4785 | 3.8359 | 4.2262 | 4.6524 | 5.1173 |
| 12 | 1.1268 | 1.2682 | 1.4258 | 1.6010 | 1.7959 | 2.0122 | 2.2522 | 2.5182 | 2.8127 | 3.1384 | 3.4985 | 3.8960 | 4.3345 | 4.8179 | 5.3503 | 5.9360 |
| 13 | 1.1381 | 1.2936 | 1.4685 | 1.6651 | 1.8856 | 2.1329 | 2.4098 | 2.7196 | 3.0658 | 3.4523 | 3.8833 | 4.3635 | 4.8980 | 5.4924 | 6.1528 | 6.8858 |
| 14 | 1.1495 | 1.3195 | 1.5126 | 1.7317 | 1.9799 | 2.2609 | 2.5785 | 2.9372 | 3.3417 | 3.7975 | 4.3104 | 4.8871 | 5.5348 | 6.2613 | 7.0757 | 7.9875 |
| 15 | 1.1610 | 1.3459 | 1.5580 | 1.8009 | 2.0789 | 2.3966 | 2.7590 | 3.1722 | 3.6425 | 4.1772 | 4.7846 | 5.4736 | 6.2543 | 7.1379 | 8.1371 | 9.2655 |
| 16 | 1.1726 | 1.3728 | 1.6047 | 1.8730 | 2.1829 | 2.5404 | 2.9522 | 3.4259 | 3.9703 | 4.5950 | 5.3109 | 6.1304 | 7.0673 | 8.1372 | 9.3576 | 10.7480 |
| 17 | 1.1843 | 1.4002 | 1.6528 | 1.9479 | 2.2920 | 2.6928 | 3.1588 | 3.7000 | 4.3276 | 5.0545 | 5.8951 | 6.8660 | 7.9861 | 9.2765 | 10.7613 | 12.4677 |
| 18 | 1.1961 | 1.4282 | 1.7024 | 2.0258 | 2.4066 | 2.8543 | 3.3799 | 3.9960 | 4.7171 | 5.5599 | 6.5436 | 7.6900 | 9.0243 | 10.5752 | 12.3755 | 14.4625 |
| 19 | 1.2081 | 1.4568 | 1.7535 | 2.1068 | 2.5270 | 3.0256 | 3.6165 | 4.3157 | 5.1417 | 6.1159 | 7.2633 | 8.6128 | 10.1974 | 12.0557 | 14.2318 | 16.7765 |
| 20 | 1.2202 | 1.4859 | 1.8061 | 2.1911 | 2.6533 | 3.2071 | 3.8697 | 4.6610 | 5.6044 | 6.7275 | 8.0623 | 9.6463 | 11.5231 | 13.7435 | 16.3665 | 19.4608 |
| 21 | 1.2324 | 1.5157 | 1.8603 | 2.2788 | 2.7860 | 3.3996 | 4.1406 | 5.0338 | 6.1088 | 7.4002 | 8.9492 | 10.8038 | 13.0711 | 15.6676 | 18.8215 | 22.5745 |
| 22 | 1.2447 | 1.5460 | 1.9161 | 2.3699 | 2.9253 | 3.6035 | 4.4304 | 5.4365 | 6.6586 | 8.1403 | 9.9336 | 12.1003 | 14.7138 | 17.8610 | 21.6447 | 26.1864 |
| 23 | 1.2572 | 1.5769 | 1.9736 | 2.4647 | 3.0715 | 3.8197 | 4.7405 | 5.8715 | 7.2579 | 8.9543 | 11.0263 | 13.5523 | 16.6266 | 20.3616 | 24.8915 | 30.3762 |
| 24 | 1.2697 | 1.6084 | 2.0328 | 2.5633 | 3.2251 | 4.0489 | 5.0724 | 6.3412 | 7.9111 | 9.8497 | 12.2392 | 15.1786 | 18.7881 | 23.2122 | 28.6252 | 35.2364 |
| 25 | 1.2824 | 1.6406 | 2.0938 | 2.6658 | 3.3864 | 4.2919 | 5.4274 | 6.8485 | 8.6231 | 10.8347 | 13.5855 | 17.0001 | 21.2305 | 26.4619 | 32.9190 | 40.8742 |
| 26 | 1.2953 | 1.6734 | 2.1566 | 2.7725 | 3.5557 | 4.5494 | 5.8074 | 7.3964 | 9.3992 | 11.9182 | 15.0799 | 19.0401 | 23.9905 | 30.1666 | 37.8568 | 47.4141 |
| 27 | 1.3082 | 1.7069 | 2.2213 | 2.8834 | 3.7335 | 4.8223 | 6.2139 | 7.9881 | 10.2451 | 13.1100 | 16.7386 | 21.3249 | 27.1093 | 34.3899 | 43.5353 | 55.0004 |
| 28 | 1.3213 | 1.7410 | 2.2879 | 2.9987 | 3.9201 | 5.1117 | 6.6488 | 8.6271 | 11.1671 | 14.4210 | 18.5799 | 23.8839 | 30.6335 | 39.2045 | 50.0656 | 63.8004 |
| 29 | 1.3345 | 1.7758 | 2.3566 | 3.1187 | 4.1161 | 5.4184 | 7.1143 | 9.3173 | 12.1722 | 15.8631 | 20.6237 | 26.7499 | 34.6158 | 44.6931 | 57.5755 | 74.0085 |
| 30 | 1.3478 | 1.8114 | 2.4273 | 3.2434 | 4.3219 | 5.7435 | 7.6123 | 10.0627 | 13.2677 | 17.4494 | 22.8923 | 29.9599 | 39.1159 | 50.9502 | 66.2118 | 85.8499 |

## 2.3 Nominal and Effective Interest Rate

The nominal interest rate is simply the stated rate. The effective interest rate is the true interest rate. It may differ from the nominal rate depending on the frequency of compounding.

The general formula is:

$\mathbf{re = (1+ {}^{r}/_{m})^{m} - 1}$ for a ***defined compounding period***
$\mathbf{re = e^{r} - 1}$ for a ***continuous compounding period***

**Example 3**
Assume r = 10%, and the compounding period is quarterly.
Then, re $(1 + {}^{0.1}/_{4})^{4} - 1 = 10.4\%$
Therefore, if you took a loan from a bank and the interest is compound quarterly at 10%, you will actually be paying 10.4% p. a.

If interest is compound monthly:
$re = (1+{}^{0.1}/_{12})^{12} - 1$
$= 10.5\%$

**Example 4**
If the stated annual interest rate, 8% is compounded quarterly, what is the effective annual rate of interest?
***Solution***
$\mathbf{re = (1+ {}^{r}/_{m})^{m} - 1}$
$= (1+ {}^{0.08}/_{4})^{4} - 1$
$= 0.0824 = 8.24\%$

**Example 5**
If the nominal interest rate on a loan is 14 percent compounded continuously, what is the effective annual rate?
***Solution***
$\mathbf{r_e = e^{r} - 1} = e^{0.14} - 1$
$r_e = 1.150273799 - 1$
$r_e = 0.150273$
$= 15.03\%$
Therefore, effective interest rate is 15.03%.

**2.4 Nominal and Real Interest Rate**
*Real interest rate* is the compound interest rate that is adjusted for changes in price levels (inflation) or for frequency of compounding. The reverse is true for *nominal interest rate*.

Mathematically,
(1+nominal interest rate) = (1+real interest rate) (1+inflation or %ΔP)
Where %ΔP is the *percentage change in price*
**NB:** if %ΔP is positive, it is called *inflation,* but if %ΔP is negative, it is called *deflation*.
$\Rightarrow (1+i_n) = (1+ i_r)(1+ \%\Delta P)$
$I_n = (1+ i_r)(1+ \%\Delta P) - 1$
Therefore, **nominal interest rate** $(i_n) = (1+ i_r)(1+ \%\Delta P) - 1$

Similarly, for real interest rate $(i_r)$,

$$(1+ i_r) = \frac{(1 + in)}{(1+ \%\Delta P)}$$

$$i_r = \frac{(1 + in)}{(1 + \%\Delta P)} - 1$$

Therefore, **real interest rate**, $i_r = \frac{(in - \%\Delta P)}{(1 + \%\Delta P)} - 1$

Using same approach, $\mathbf{\%\Delta P} = \frac{(in - ir)}{(1 + ir)}$

**Example 6**

Determine the true interest rate of a project if the current interest rate is 40% and inflation is currently 30%.

***Solution***

Current or nominal interest rate = $i_n$ = 40%

Inflation = %ΔP = 30%

True or real interest rate = ir = $\frac{(in - \%\Delta P)}{(1 + \%\Delta P)}$ = (40% - 30%) / (1+30%) = (0.4 – 0.3) /(1+0.3)

= 0.1 / 1.3 = 0.0769 = **7.69%**

**2.5 The Rule of 72**

If you want to double your money, the "Rule of 72" tells you how. A quick way to handle compound interest problems involving doubling your money makes use of the "Rule of 72". This rule states that if the number of years, *t,* for which an investment will be held, is divided into the value 72, and we will get the approximate interest rate, required for the investment to double in value. Mathematically, by applying the "Rule of 72", the approximate rate of interest is given by:

$$r = 72/t$$

Let use the rule to examine this case

Mr. Kazookaw a renowned businessman once made an investment with UCC Investment Bank with a sum of GHS10million and then after five years his investment had amounted to GHS20million. In short, he doubled his money in five (5) years. Therefore, in that view, what compound rate of return did Mr. Kazookaw earn on his Investment? In this regard, the expected rate of return that would allow Mr. Kazookaw to achieve such amount within 5 years' time according to the rule is r = 72/5 = 14.4%.

**Examples 7**

Emmanuella invested GH¢2,500 into a Certificate of Deposit earning 6.5% interest. How long will it take to double Emmanuella's investment?

Solution

72/6.5 = 11 years

**Example 8**

The average Stock Market return since 1926 has been 11%. According to the Rule of 72, how often will an individual investment double?

Solution

72/11 = 6.5 years

**Summary**

You did well by completing the session. We examined future value and present value. We looked at how to calculate the present value and future value using the Tables. We also explained the Rule of 72 and its application. Time value of money is interesting, right? Time is really money, isn't it?

**Self-Assessment Questions**
**Exercise 4.2**

## SESSION 3: ANNUITIES

### Introduction

Having learnt about time value of money, it is important to see its application. In our previous example, we look at the payment of a lump-sum. The payment of a single amount at a go. In this session, we shall examine the payment of the same amount of money for a period of time.

### Objectives

By the end of this session, you should be able to:

(a) understand annuities
(b) explain the different types of annuities
(c) descript the application of time value of money using annuities
(d) calculate present value and future value of annuities

Now read on …

### 3.1 Meaning of annuity

An annuity is an asset that pays its owner a constant stream of cash flows for FIXED period. In other words, it is a series of equal payments made at fixed intervals for a specified number of periods. Annuities are like perpetuities except that annuities do not go on forever.

An annuity is a contract between you and an insurance company in which you make series of payments and, in return, receive regular disbursements, beginning either immediately or at some point in the future.

An annuity is a contract where an insurance company promises to make payments to an annuitant over a specified period of time or for life. One of the purposes for an annuity is to make sure a person does not outlive his income. An annuity is a type of insurance to protect against the risk of financial hardship during retirement.

There are three participants in an annuity contract: the owner, the annuitant and the beneficiary. Most of the time, the annuitant and the owner are the same person, but it is not required. The owner is the purchaser of the annuity, pays the premiums and has the right to surrender the annuity. The owner also is responsible for any taxes due upon surrender or pay-out and is usually the person who names the beneficiary of the contract. The annuitant is the person whose age and life expectancy is going to be used to calculate the benefits of the annuity and who will receive the annuity payments. The beneficiary receives the death benefit upon death of the annuitant or the owner.

### 3.2 Types of Annuities

There are different types of annuities but in Business Finance, we shall look at three types: fixed annuity, variable annuity and immediate income annuity.

### 3.2.1 Fixed annuity

This type of annuity accumulates interest on the funds deposited into the annuity on a fixed rate basis. Every fixed annuity has a current interest rate and a minimum guaranteed interest rate. The current interest rate will always be equal to or higher than the minimum guaranteed interest rate. Although this varies from company to company and contract to contract, the current interest rate is declared on an annual basis, usually after an initial guarantee period. With a fixed annuity, the insurance company assumes the risk of paying at least the minimum guaranteed interest rate.

### 3.2.2 Variable annuity

Different than a fixed annuity, a variable annuity pays varying rates of interest on the funds placed inside the annuity based upon the investment options chosen by the annuity owner. If the investment choices do well, the annuity will do well. If the investment choices do poorly, the annuity will not grow as well or even could lose value. Because the growth of a variable annuity is not guaranteed by the insurance company, the contract holder assumes the risk. Examples of variable annuities in Ghana are: Lifetime needs plus plan and educare plus by Enterprise Insurance Company.

### 3.2.3 Immediate income annuity

This type of annuity begins paying a benefit very soon, usually within 30 days to one year after it is purchased, and usually requires a lump sum payment. This usually is an income protection plan that provides cover of an amount equal to the sum assured, upon the unfortunate event of death, critical illness or permanent disability during the term of the policy. Example of such annuities is the Enhanced family income protection plan by Enterprise Insurance Company and Ultimate protection plan by Star Life Insurance in Ghana.

## 3.3 Categories of Annuities

These three types of annuities can be grouped into two: ordinary annuity and annuity due. If the payments occur at the *end* of each period, as they typically do, the annuity is called an ***ordinary annuity***. The equal-payment house mortgage or instalment credit agreement is example of ordinary annuity. If payments are made at the *beginning* of each period (i.e. payment is due *today*, *immediately*, *now*, *at start*, etc), the annuity is an ***annuity due***. Rental payment for an apartment and life insurance premiums are examples. However, sometimes, payments may be due to commence after some time period, say in 2- or 3-years' time. This is what is referred to as ***deferred annuity***. **NB**: *Deferred annuity* can become an *ordinary annuity* only if payment is due to commence at the end of the period, say, after 12 months or 1 year.

## 3.4 Things Common to all Annuities

There are two distinct phases to an annuity - the accumulation phase and the pay-out or distribution phase. The accumulation phase is the first phase where all the premiums are paid into the annuity and the money grows tax-deferred. The second phase is the pay-out or distribution phase, which is when the annuity actually starts to pay the benefits to the annuitant. There are several pay-out options; a few of these options will be discussed. Make sure your agent goes over each option with you thoroughly so that you choose the one most appropriate for your needs. It is very important to understand that once a pay-out option is chosen and you start receiving payments, that option cannot be changed. Examples of Pay-out Options (Settlement Options)

- Life income (no refund) – Pays an income as long as the annuitant is alive. Payments stop when the annuitant dies.
- Life income with period certain (10, 15 or 20 years) – Pays an income as long as the annuitant is alive. If the annuitant dies before the period certain has expired, payments are made to the beneficiary for the balance of the period.
- Life income with instalments (refund) – This option provides a monthly annuity payment during the lifetime of the annuitant with a guarantee that payments will be made for a certain number of months (not necessarily for the annuitant's full lifetime). The number of months is determined by dividing the accumulated amount of the annuity by the amount of the first monthly annuity payment. Only the number of months is guaranteed so there is no guarantee of a full refund.

### 3.5 Present and Future value of annuities

Present value of annuities the lump-sum amount that equals the value now of a set of equal periodic payments to be paid in the future. Let us look at the present value of ordinary annuity and present value of annuity due.

**Ordinary Annuities**

FV of ordinary annuity; $FV = C\,[\underline{(1+r)^t - 1}\,] = C\,(FVIFA_{r\%,n})$

PV of ordinary annuity; $PV = C\left(\frac{1}{r} - \frac{1}{r(1+r)^t}\right) = C\,(PVIFA_{r\%,n})$

Where

$FVIFA_{r\%,n}$ = future value interest factor of annuity = $\frac{[(1+r)t - 1]}{r}$

$PVIFA_{r\%,n}$ = present value interest factor of annuity = $\left(\frac{1}{r} - \frac{1}{r(1+r)^t}\right)$

C = equal periodic cash flow
r = periodic interest rate
t = time period

**NB**: FVIFA and PVIFA can be obtained using *Annuity Tables* or simply calculating.

**Example 1**

Suppose C = GH¢100,000; t = 5 years; r = 24%. Calculate the PV of cash flows of this annuity.

$PV = 100{,}000(^1/_{0.24} - ^1/_{0.24\ (1+.24)^5})$ = GH¢274,538

Using annuity tables give: GH¢100,000 x 2.7454 = GH¢274,540

**Example 2**

Vincent Kawor has just won the state lottery, paying GH¢50,000 a year for 20 years. He is to receive his first payment a year from now. The state advertises this as the Million Ghana cedis Lottery because GH¢1,000,000 = GH¢50,000 x 20. If the rate of interest is 8%, what is the true value of the lottery?

***Solution***

$PV_{(of\ annuity)} = GH¢50{,}000(^1/_{0.08} - ^1/_{0.08\ (1+0.08)^{20}})$ = GH¢50,000 x 9.8181= GH¢490,905

Where the GH¢50,000 is the periodic payment and 9.8181 is the annuity factor.

**Example 3**
An asset promises to pay GH¢500 at the end of each of the next three years. If you wanted to earn 10% on your money, how much would you offer for this annuity?
***Solution***
PV = C $\{{}^{1}/_{r} - {}^{1}/_{r(1+r)^{n}}\}$
= 500{1/.1 -1/.1(1.1)} $^{3}$
= GH¢1,243.43

**Example 4**
You plan to contribute GH¢2,000 a year into a retirement account paying 8%. If you retire at 30 years, how much will you have?
***Solution***
Future Value Interest Factor of Annuity (FVIFA) = $\left[ \frac{(1+r)^{t} - 1}{r} \right]$
= (future value factor -1)/r
= $(1.08^{30} - 1)/0.08$
= 113.2832
Future Value of Annuity = GH¢2,000 x 113.28
= GH¢226,566.4

Using tables to solve the above problem read 30 years at 8% from the annuity table and multiply by how much you plan to contribute. i.e. GH¢2,000 x 113.28 = $226,566

**Annuity Due**
FV of annuity due;
FV $= C(\frac{(1+r)^{t}-1}{r})(1+r)$ = C * FVIFA*(1+r)

PV of annuity due;
PV = C $\left(\frac{1}{r} - \frac{1}{r(1+r)^{t}}\right)$ (1+r) = C * PVIFA*(1+r)

Where
FVIFA = future value interest factor of annuity = $\frac{(1+r)^{t}-1}{r}$
PVIFA = present value interest factor of annuity = $\left(\frac{1}{r} - \frac{1}{r(1+r)^{t}}\right)$
C = equal periodic cash flow
r = periodic interest rate
t = time period

**Example 5**
Anna deposited GH¢100,000 at the beginning of each year for three consecutive years. Interest is compounded annually at 10%. How much will Anna have in his account after three years?
***Solution***: FVAD = C (FVIFA) (1+r) = GH¢100,000(3.310)(1.10) = GH¢364,100

**Example 6**

A GH¢600 cash flow is starting today. Interest rate is 10%. What is the PV if it continues for 5 years?

***Solution***

PV= 600 $[1/0.1 - 1/0.1(1.1)]^5$ * 1.1 = 600* 3.7908*1.1 = 600 * 4.1699 = GH¢2,501.94

**Summary**

What an interesting time going through this session!! It is wonderful examining how same amount of money could be paid or received for a finite period of time. The application of time value of money in annuities is good and will help us plan for the future. Keeping same amount of money with an insurance company for a period of time, say 10years can really be beneficial. Guess you are considering such an option. Think about it.

**Self-Assessment Questions**

**Exercise 4.3**

## SESSION 4: PERPETUITIES, SINKING FUND AND LOAN AMORTIZATION

**Introduction**

Do you have plans of setting up a business, buying a new car, putting up a house, planning for a marriage in the future, ensuring that you establish a fund that will always take care of a need even when you are no more but do not know how to go about these? We are sure you do. This session on perpetuities and sinking fund helps you achieve that goal. Under the sources of funds, we talked about bank loans. These loans require that you pay a sum of money each month until the last amount is paid. This is loan amortization. We will teach you how to plan for the loans before going for them. Interesting, right? Relax and enjoy this session.

**Objectives**

By the end of this session, you should be able to:

(a) explain perpetuities and how to determine the present value of perpetuities
(b) understand sinking fund and how to establish it
(c) prepare a loan amortization schedule and plan for loans before you take one

Now read on …

### 4.1 Perpetuities

Frequently, you may need to value a stream of equal cash flows and if this stream of equal cash flows lasts forever, it is called a perpetuity. Perpetuities are equal periodic cash flows to be received / paid forever or indefinitely, i.e. cash flows continuing for the foreseeable future. A constant perpetuity pays the owner a constant stream of cash flows forever. Sometime ago the British government borrowed by issuing perpetuities. Instead of repaying these loans, the British government pays the investors holding these securities a fixed annual payment forever. This is called the British consol.

How might we value such a security? Supposed that you could invest GHS1,000 at an interest rate of 10 percent. You would earn interest of 0.10 x 1,000 = GHS100 per year and could withdraw this amount from your investment account each year without ever running down your balance. In order words, a GHS1,000 investment could provide a perpetuity of GHS100 per year. In general:

Cash payment from perpetuity = interest rate x present value

$$C = r \times PV$$

We can rearrange this relationship to derive the present value of a perpetuity, given the interest rate r and the cash payment C:
Present value of perpetuity is PV = C/r,
where C = constant stream of cash flow and r = discount rate.

**Example 1**
Suppose a businessman wishes to endow a chair in finance at the School of Business, University of Cape Coast. The rate of interest is 12%. If the goal is to provide GHS150,000 a year in perpetuity, what is the amount that must be set aside today? (This is a constant perpetuity)
Solution
PV = C/r = GHS150,000/0.12 = GHS1,250,000

A growing perpetuity pays a growing stream of cash flow forever. From the example above, now, suppose that the benefactor suddenly recollects that no allowance has been made for growth in salaries, which will probably average about 5% a year. Therefore, instead of providing GHS150,000 a year in perpetuity, the benefactor must provide GHS150,000 in year one, 1.05 x GHS150,000 in year 2 etc. If we call the growth rate in salaries g, then the present value of this stream of cash flows is:
$PV = C_1 / r\text{-}g = 150{,}000(1.05)/(0.12\text{-}0.05) =$ GHS2,250,000

Therefore, if our benefactor wants to provide perpetually an annual sum that keeps pace with the growth rate of salaries, the amount that must be set aside today is GHS2,250,000. Note: The interest rate r must be greater than the growth rate for the growing perpetuity formula to work.

**4.2 Sinking fund**
This is a fund established to aid in reducing debt by repaying or repurchasing outstanding loan and other debt securities held against the entity. Sinking fund is essentially established for *two (2) main purposes*:

- Repayment of debt
- Replacement of an obsolete or antiquated asset.

**Example 2**
Sefakor, a student borrows GH¢6,000 at a rate of 12% p.a. to fund his education. The debt will be paid back to the lender in one payment at the end of 5 years. She decides to set up a fund for this purpose requiring yearly deposits which will earn 8% interest compounded annually. Assuming that the first payment is made into the fund at the end of the first year; what must this yearly deposit be? Construct a sinking fund schedule.

***Solution***
Loan amount = P= GH¢6,000

FV of loan $= P(1+i)^t$
$= 6000(1 + 0.12)^5$
$= 6000(1.12)^5$
$= 6000(1.7623)$
$=$ GH¢ 10,573.80

This amount is to be provided for through 5 annual deposits commencing at the end of the first year, as such this represents the FV of ordinary annuity;

Therefore,

$$FV = C\left(\frac{(1+i)^t}{i}\right) - 1$$

$$10{,}573.80 = C\left(\frac{(1+0.08)^5}{0.08}\right) - 1$$

$$10{,}573.80 = C\,(5.8667)$$

**C** $= (10573.8 / 5.8667)$
$=$ GH¢**1,802.37**

Therefore, the annual deposit must be GH¢**1,802.37**

**Sinking Fund Table for the repayment of loan**

| Year | LOAN | | SINKING FUND CALCULATION | | |
|---|---|---|---|---|---|
| | Outstanding Debt | Loan Interest @ 12% | Deposit | Cumulative Deposit | Interest @ 8% |
| 1 | 6000.00 | 720.00 | -------- | -------- | -------- |
| 2 | 6720.00 | 806.40 | 1802.37 | 1802.37 | 144.19 |
| 3 | 7526.40 | 903.17 | 1802.37 | 3748.94 | 299.91 |
| 4 | 8429.57 | 1011.55 | 1802.37 | 5851.22 | 468.10 |
| 5 | 9441.12 | 1132.93 | 1802.37 | 8121.69 | 649.74 |
| 6 | **10574.05** * | -------- | 1802.37 | **10573.80** * | -------- |

***** Clearly, it can be seen that the sinking fund is able to provide an amount able to cater for repayment of loan. The final amounts in bold in both cases are supposed to be same value. Negligible difference is due to *rounding off.*

**Example 3**

A Ltd has decided to set up a fund to replace an asset in 6 years' time. The value of the fund after 6 years will be GH¢80,000 and the fund is expected to earn interest at the rate of 8% p.a. The asset to be replaced has been bought for GH¢55,000 and is to be depreciated on a straight-line basis with a scrap value of GH¢7,000. If payment commences one year from now with a final payment in year 6, determine the annual payment into the fund, and prepare a sinking fund schedule.

***Solution***

FV of ordinary annuity =

$$FV = C\left(\frac{(1+i)^t}{i}\right) - 1$$

$$80000 = C\left(\frac{(1+0.08)^6}{0.08}\right) - 1$$

80000 = C (7.3359)

**C** = (80000 / 7.3359)

= GH¢ **10,905.27**

Therefore, the annual sinking fund deposit must be GH¢ **10,905.27**

**Annual Depreciation** = $\frac{\text{cost – scrap value}}{\text{expected useful life}}$

= (55000 – 7000) / 6 years

= GH¢ **8,000 per year**

**Sinking Fund Table for the replacement of obsolete asset**

| **Year** | **ASSET** | | **SINKING FUND CALCULATION** | | |
|---|---|---|---|---|---|
| | Net Book Value | Annual Depreciation | Deposit | Cumulative Deposit | Interest @ 8% |
| 0 | 55000.00 | -------- | -------- | -------- | -------- |
| 1 | 47000.00 | 8000.00 | 10905.27 | 10905.27 | 872.42 |
| 2 | 39000.00 | 8000.00 | 10905.27 | 22682.96 | 1814.64 |
| 3 | 31000.00 | 8000.00 | 10905.27 | 35402.87 | 2832.23 |
| 4 | 23000.00 | 8000.00 | 10905.27 | 49140.37 | 3931.23 |
| 5 | 15000.00 | 8000.00 | 10905.27 | 63976.87 | 5118.15 |
| 6 | **7,000.00** | 8000.00 | 10905.27 | **80,000.00** | -------- |

**4.3 Loan Amortization**

An amortized loan is where the lender may require the borrower to pay instalments over the life of a loan, each of which includes both interest and the repayments of principal. The process of paying off a loan by making regular principal reduction is called amortizing the loan. Mortgaged loans, automobile loans, and many business loans are amortized or liquidated on this type of instalment basis. The amount of the periodic payment can be found using the equation:

$$C = \frac{PVOA}{PVIFA_{i,t}}$$

**Example 4**

Suppose a firm borrows GH¢300,000 from a bank for five years at an interest rate of 10 percent. The loan terms call for the firm to pay the interest on the loan balance each year and to pay the loan in five equal year end installments.

Loan amortization payment = C

$$C = \frac{PVOA}{PVIFA_{i,t}} = \frac{300,000}{3.791} = \text{GH¢ } 79,134.79$$

| Year | Beginning Balance | Instal't Payment | Interest Paid | Principal Paid | Ending Balance |
|---|---|---|---|---|---|
| 1 | 300,000 | 79,134.79 | 30,000 | 49,134.79 | 250,865.21 |
| 2 | 250,865.21 | 79,134.79 | 25,086.52 | 54,048.27 | 196,816.94 |
| 3 | 196,816.94 | 79,134.79 | 19,681.669 | 59,453.10 | 137,363.84 |
| 4 | 137,363.84 | 79,134.79 | 13,736.38 | 65,398.41 | 71,965.43 |
| 5 | 71,965.43 | 79,134.79 | 7,196.54 | 71,938.25 | |

**Example 5**

You took a ¢10m loan today and you have 5 years to pay back. Repayment starts 1 year from now. Interest is 32%. Repayment is in equal instalments. How much should be paid such that after 5 equal payments you would have paid both the interest and principal? Show by table that after 5 years you would have indeed paid off both interest and principal.

***Solution***

$$PV = C\left(\frac{1}{r} - \frac{1}{r(1+r)^t}\right)$$

a. $C = \frac{PV}{\frac{1}{r} - \frac{1}{r(1+r)^t}}$

$$= \frac{10m}{\frac{1}{0.32} - \frac{1}{0.32(1+0.32)^5}} = \text{GH¢4,264,028}$$

If you paid GH¢ 4,264,028 yearly for 5 years, the interest and principal would have been paid.

b.

| **Year** | **Beginning Balance** | **Interest charge** | **Principal Paid** | **Installment payment** | **End of year Balance** |
|---|---|---|---|---|---|
| | A | B=Ax32% | C=D-B | D | E=A-C |

| | | | | | |
|---|---|---|---|---|---|
| 1 | 10,000,000.00 | 3,200,000.00 | 1,064,028.00 | 4,264,028.00 | 8,935,972.00 |
| 2 | 8,935,972.00 | 2,859,511.04 | 1,404,516.96 | 4,264,028.00 | 7,531,455.04 |
| 3 | 7,531,455.04 | 2,410,065.61 | 1,853,962.39 | 4,264,028.00 | 5,677,492.65 |
| 4 | 5,677,492.65 | 1,816,797.65 | 2,447,230.35 | 4,264,028.00 | 3,230,262.30 |
| 5 | 3,230,262.30 | 1,033,683.94 | 3,230,344.06 | 4,264,028.00 | (81.76) |

From the two examples, you will notice that interest payment continues to decrease as you progress through the years. More interest is paid in the earlier years of the loan facility. On the other hand, the principal payment continues to increase as you progress through the years. The interest is charged only on the principal amount (i.e. the beginning balance only).

**Summary**

In this session, we discussed perpetuities, sinking fund and loan amortization. We explained that perpetuities are payments of same amount of money into the future forever. Sinking funds are established to ensure that funds are made available for a particular need. A loan amortization is prepared to know how the principal and the interest are paid off over time. Now you can fulfil those plans of buying a car, building a house or marrying your dream woman by keeping say GHS50 each month for the next five to ten years earning compounding interest. Imagine how much you will have. Stop spending on pizza, air time for your phone and taking beer or any alcoholic drink each evening. Chai, this one you aren't happy. But it is good for you our dear students.

**Self-Assessment Questions**

**Exercise 4.4**

## SESSION 5 VALUATION OF BONDS

**Introduction**

Bonds form a significant portion of the financial market and are a key source of capital for the corporate world. Therefore, every business finance course in the undergraduate program will introduce students to bonds at varying depth. We have provided a quick outline of what a student will need to know to understand bonds and the pricing or valuation of bonds which is the primary focus in the initial Bachelor of Commerce program. More advanced finance courses will introduce students to advanced bond concepts including duration, managing bond portfolios, understanding and interpreting term structures, etc.

**Objectives**

By the end of this session, you should be able to:

(a) understand the meaning of bonds
(b) explain the features of bonds
(c) outline and discuss the issuers of bonds

(d) determine the market price of bonds

Now read on …

**5.1 Meaning of bonds**

Capital comes in two forms: debt capital and equity capital. To raise debt capital the companies, sell bonds to the public. Instead of selling directly to the public, a company usually sells its bonds through an intermediary. An investment bank acts as an agent between the company and the public. Also known as underwriters, they raise the capital for a firm and charge a fee for their services.

Companies sell bonds to borrow money from the investors. As a financial instrument, a bond represents a contractual agreement between the company and the bondholders. Eventually the company has to repay the principal to the investors and pay interest to them in the meantime. A bond is a financial security that obligates the issuer to make specific payments to the bondholder. A bond is a debt instrument that provides a periodic stream of interest payments to investors while repaying the principal sum on a specified maturity date. A bond's terms and conditions are contained in a legal contract between the buyer and the seller, known as the indenture.

The money that companies collect when the bond is issued, or sold to the public, is the amount of the loan. In return, the companies agree to make specific payments to the bondholders, who are the lenders. When you own a bond, you generally receive a fixed interest payment each year until the bond matures.

**5.2 Features of a bond**

Typically, a bond has the following features:

**5.2.1 The face value, F**

The face value of a bond, or its principal, is usually GH¢1,000, which means that the investment in bonds is a multiple of GH¢1,000. The total value of the bonds issued by a company at a certain time could be millions of Ghana cedis.

**5.2.2 The market value, B.**

Although a bond may have a face value of GH¢1000, it may not sell at GH¢1000 in the bond market. If the issuing company is not doing well financially, its bonds may sell for less than GH¢1000, perhaps at GH¢950. If you look up their price on the internet, or some financial newspaper, it is listed as 95. This means that the bond is selling at 95% of its face value, or GH¢950. The bond is selling at a discount. If the market value of the bond is more than GH¢1,000, and then it is selling at a premium. A bond with a market value less than GH¢1,000 is selling at a discount, and a bond, which is priced at its face value, is selling at par.

**5.2.3 The time to maturity, n.**

A bond's maturity is the length of time until the principal is scheduled to be repaid. There is a definite date when a bond matures. At that time, the company must pay the face value of the bonds to the bondholders. This could be from as little as 5 years to as long as 100 years (remember Ghana wanted to borrow for 100 years: century bond?). The short-term bonds are also called notes. The companies that are starting out, do not want to carry a long-term debt burden and so they issue relatively short-term bonds. Well established companies prefer to use long-term debt in their capital, especially when the interest rates are low.

### 5.2.4 The coupon rate, c.

The periodic interest payments promised to bond holders are computed as a fixed percentage of the bond's face value; this percentage is known as the coupon rate. This is the stated rate of interest of the bonds. For example, a bond may be paying 8% interest to the bondholders. The dollar amount of interest C, is the product of the face amount of the bond and the coupon rate. We may write this as:

$$C = cF$$

The 8% bond is paying .08*1000 = GH¢80 per year to the investors. The companies generally pay the interest semi-annually, so the 8% bond really pays GH¢40 every six months. For example, a bond may pay interest on February 15 and August 15 in a calendar year. If an investor buys a bond between the interest payments dates, let us say on May 1, then s/he has to pay the accrued interest, the interest for the period February 16 to May 1, to the seller of the bond.

The interest rate on a bond depends primarily on two factors. First, it depends on the general level of interest rates in the economy. Check the Bank of Ghana website for the interest rates of Government of Ghana bonds. Low interest rates allow companies to borrow money at lower rates enabling them to expand their business easily. At other times, the interest rates may be quite high, partly because of Central Bank's tight money policy. This forces all companies to borrow at a higher rate of interest.

Second, the company, which is issuing bonds, may not be in a strong financial condition. The sales are down, the cash flow is small, and the future prospects of the company are not too bright. It must borrow new money at a higher rate. On the other hand, well-managed companies in a strong financial position can borrow at relatively low interest rates.

### 5.2.5 The indenture

The indenture is the formal contract between the bondholders and the company. Written in legal language, the fine print spells out the rights and responsibilities of both parties. In particular, the indenture requires the company to pay interest to the bondholders whenever it is due. The companies have to pay interest before they pay taxes or dividends on the ordinary shares. This makes the position of the bondholders quite secure. The indenture also spells out the timetable for bond refunding. Another clause in the indenture further strengthens the position of the bondholders. This allows them to force the company into liquidation if the company fails to meet its interest obligations on time.

### 5.2.6 Call Provisions

Many bonds contain a provision that enables the issuer to buy the bond back from the bondholder at a pre-specified price prior to maturity. This price is known as the call price. A bond containing a call provision is said to be callable. This provision enables issuers to reduce their interest costs if rates fall after a bond is issued, since existing bonds can then be replaced with lower yielding bonds. Since a call provision is disadvantageous to the bondholder, the bond will offer a higher yield than an otherwise identical bond with no call provision. A call provision is known as an embedded option, since it can't be bought or sold separately from the bond.

### 5.2.7 Put Provisions

Some bonds contain a provision that enables the buyer to sell the bond back to the issuer at a pre-specified price prior to maturity. This price is known as the put price. A bond containing such a provision is said to be putable. This provision enables bond holders to benefit from rising interest rates since the bond can be sold and the proceeds reinvested at a higher yield than the original bond. Since a put provision is advantageous to the bondholder, the bond will offer a lower yield than an otherwise identical bond with no put provision.

### 5.2.8 Sinking Fund Provisions

Some bonds are issued with a provision that requires the issuer to repurchase a fixed percentage of the outstanding bonds each year, regardless of the level of interest rates. The term sinking fund describes the amount of money that a company puts aside to retire its bonds. A sinking fund reduces the possibility of default; default occurs when a bond issuer is unable to make promised payments in a timely manner. Since a sinking fund reduces credit risk to bond holders, these bonds can be offered with a lower yield than an otherwise identical bond with no sinking fund.

## 5.3 Bond Issuers

Bonds are issued by borrowers to raise funds for long-term investments; the main issuers of bonds in Ghana are:

- The Government of Ghana
- Corporations/Companies
- Municipalities
- Foreign Entities

### 5.3.1 The Government of Ghana

Treasury securities are issued by the Ghana government to finance its deficits. These are free of default risk, which is the risk that the investor will not receive all promised payments.
Ghana Treasuries can be classified by their maturities as follows:

- Treasury Bills-Treasury Bills are short-term debt or money market instruments (securities) issued by the Government of Ghana with a maturity of less than one year. They commonly have maturities of 91 days and 182 days. Treasury Bills are issued at discount from par, which means that rather than making fixed interest payments like conventional bonds, income is earned by the difference between the face value and the cost of purchase. Bank of Ghana also periodically issues short-dated bills to support its monetary operations and Cocobod for the purchase of cocoa in Ghana.
- Treasury notes-Treasury Notes are interest bearing securities which are sold at par and with maturity term ranging from one year to two years. Investors earn interest semi-annually at a fixed rate. On the redemption date the investor received the last coupon payment together with the nominal value of the securities purchased. Another key difference between these securities is that Treasury bills are sold at a discount from their face value and redeemed at face value; Treasury notes and bonds are sold and redeemed at face value and pay semi-annual coupons to investors.
- Treasury Bonds-Treasury Bonds are medium term debt instruments (securities) issued by the Government of Ghana and sold to individuals or companies to raise funds for a specified time at a fixed or variable interest rate. The bonds have maturity periods exceeding two years. Presently the 7 Year bond is the longest-term security issued in Ghana. Bonds normally make fixed interest payments bi-annually (i.e. every six months) to the holder.

### 5.3.2 Corporations

Corporations can raise funds by issuing debt in the form of corporate bonds. Corporate Bonds are debt securities issued by organisations so as to raise funds for several reasons depending on the organisation's objectives to finance an ongoing task, to purchase new equipment or to expand business etc. Corporate Bonds normally have a redemption period exceeding one year and can either have a floating or fixed coupon rate.

These bonds offer a higher promised coupon rate than Treasuries, but expose investors to default risk. Ratings agencies, such as Standard and Poor's and Moody's, rank corporate issuers according to their likelihood of default. The riskiest corporations offer the highest coupon rates to investors as compensation for default risk.

### 5.3.3 Municipalities

A municipal bond is issued by a state or local government; as a result, they carry little or no default risk. Municipal bonds offer an extremely favourable tax treatment to investors. They are not taxed by state or local governments as long as the bond holder lives in the municipality in which the bonds were issued. As a result, municipal bonds can be issued with very low yields. Unfortunately, in Ghana, we do not see the use of this bonds by our MMDAs.

### 5.3.4 Foreign Entities

Foreign bonds are issued by foreign governments and corporations and are denominated in dollars. (If they are denominated in a foreign currency, they are known as Eurobonds.) Dollar-denominated bonds issued by Ghana entities in the U.S. are known as Yankee Bonds.

## 5.4 Bond Valuation

We can evaluate a bond by finding the present value of the interest payments and that of the principal. The proper discount rate that calculates the present value depends on the risk of the bonds. The risky bonds have a relatively higher discount rate. Further, the discount rate is also the rate of return required by an investor buying that bond. The basic financial principle is:

The present value of a bond is simply the present value of all future cash flows from the bond, properly discounted.

We may express the above statement as follows:

$$\text{Market Value or Price} = P_o = \frac{Int_1}{(1+r)} + \frac{Int_2}{(1+r)^2} + \frac{Int_3}{(1+r)^3} + \cdots + \frac{(Int_t + P_t)}{(1+r)^t} \qquad \text{Equation 1}$$

Since the interest on the bond is generally fixed, i.e. $Int_1 = Int_2 = Int_3 = \text{Int}$ we can apply the annuity formula as:

$$\text{Market Value or Price} = P_o = Int\left(\frac{1}{r} - \frac{1}{r(1+r)^t}\right) + \frac{P_t}{(1+r)^t} \qquad \text{Equation 2}$$

**NB**: Either *Equation 1* or *2* can be used. However, where ***t*** is relatively large, say *100 years*, only *Equation 2* can be possibly used because it will be time-consuming and cumbersome to use Equation 1.

Market Value or Price = $P_o = \sum_{i=1}^{n} \frac{C}{(1+t)^i} + \frac{F}{(1+t)^n}$ Equation 3

The first term on the right side is the present value of the coupon payments, or the interest payments in Ghana Cedis or dollars. The second term is the present value of the face amount of the bond in Ghana Cedis or dollars. This resembles equation (2).

**Example 1**

The Government in 2000, raised money by selling 8% coupon, 2010 maturity bonds. Each bond has a face value of $1000 and the coupon rate is 8%. From the above, the Government of Ghana is under obligation to pay an 8% of $1000 or $80 to bond holders until 2010. Note that the Government will have to pay the face value of the bond, $1000, at maturity. But what is the market value of this stream of cash flows, assuming we are in 2005.

First, we need to find the return on bond's with similar features. Let's assume bonds of similar features offered a return of 5.1%. This is what investors could have earned on alternative investments of comparable risk.

Therefore, the value of this bond in 2005 is:

Market Value or Price = $P_o = \frac{Int_1}{(1+r)} + \frac{Int_2}{(1+r)^2} + \frac{Int_3}{(1+r)^3} + \frac{Int_4}{(1+r)^4} + \frac{(Int_t+P_t)}{(1+r)^5}$

Market Value or $P_o = \frac{80}{(1+0.051)} + \frac{80}{(1+0.051)^2} + \frac{80}{(1+0.051)^3} + \frac{80}{(1+0.051)^4} + \frac{1080}{(1+0.051)^5}$

$P_o$ = 76.12 + 72.42 + 68.91 + 65.57 + 842.19

$P_o$ = $1125.21

The bond is worth 112.52% of its face value i.e. {(1125.21-1000)/1000} (100) + 100. This bond is selling at a premium. Note that bond prices are usually expressed as a percentage of their face value.

**Example 2**

KK Brothers Company Ltd issued a bond in 1984, with maturity date of 2014 and a stated coupon rate of 6.875%. In 1997, with 17 years left to maturity, investors owning the bonds were requiring an 8% rate of return. Calculate the value of the bonds to these investors.

**Solution**

Note that bondholders receive a level stream of coupon payments for each of the years to maturity; hence coupon payments on a bond are an annuity.

Market Value or Price = $P_o = Int\left(\frac{1}{r} - \frac{1}{r(1+r)^t}\right) + \frac{P_t}{(1+r)^t}$

Market Value or Price = $P_o = 68.75\left(\frac{1}{0.08} - \frac{1}{0.08(1+0.08)^{17}}\right) + \frac{1000}{(1+0.08)^{17}}$ = $897.38

This bond is selling at 89.74% of face value; hence we say the bond is selling at a discount. This is how much investors are prepared to pay or lend in return for the procured $1000 repayment.

**Example 3**

A Company issues a bond with 10 years maturity. Annual coupon is $80. Similar bonds have a yield of maturity of 8%. What is the value of this bond?

Market Value or Price = $P_o = Int\left(\frac{1}{r} - \frac{1}{r(1+r)^t}\right) + \frac{P_t}{(1+r)^t}$

Market Value or Price = $P_o = 80\left(\frac{1}{0.08} - \frac{1}{0.08(1+0.08)^{10}}\right) + \frac{1000}{(1+0.08)^{10}}$ = $1,000

This bond is selling at par value.

**Example 4**
In 1990, the government of Ghana issued a 7-year bond. Interest payments were 14% of face value, paid semi-annually. The yield to maturity is quoted at 16%. What is the bond's price?

**Solution**
This bond will sell at a discount because it has a coupon rate of 7% every six months, when the market requires 8% every six months.

$P_o = \frac{\$1000}{(1+0.08)^{14}}$ = $340.46

We use the annuity present value to estimate the PV of coupons.

Market Value or Price = $P_o = \$70\left(\frac{1}{0.08} - \frac{1}{0.08(1+0.08)^{14}}\right)$ = $70 (8.2442) = $577.10

Total $P_o$ = $577.10 + 340.46 = 917.56

Or

Market Value or Price = $P_o = 70\left(\frac{1}{0.08} - \frac{1}{0.08(1+0.08)^{14}}\right) + \frac{1000}{(1+0.08)^{14}}$ = $577.10

Perpetual bonds have an infinite life span. In essence, they are perpetuities. The bondholders continue to receive interest payments and if they want to, they can always sell the bonds to other investors. Since the bond is never going to mature, the implicit assumption is that the investors will never receive the face amount of such a bond. From (Equation 2), when *t* approaches infinity, the summation becomes C/r. The second term for the present value of the face amount also approaches zero. We get the simple formula for perpetual bonds.

$$B = C/r$$

Some companies try to conserve cash and they may sell zero-coupon bonds. These bonds make no periodic interest payments and they pay the entire accumulated interest and the principal at the maturity of the bond. Because of this feature, these bonds sell at a substantial discount from their face value. For instance, General Motors issued zero coupon bonds in 1996 due to mature in 2036. In January 2007, these bonds were selling at 38.11, or $381.10 per $1000 bond. For zero-coupon bonds, the first term in (Equation 3) is zero because C is zero. This leaves only the second term for the valuation of zero-coupon bonds as follows:

Market Value or Price = $P_o = \frac{F}{(1+t)^n}$

**Yield-to-Maturity (YTM)**
This is the required rate of return by bondholders, or a discount rate, or expected rate of return. In other words, the YTM is the discount rate that equates the present value of all interest payments plus the repayment of principal from a bond with the present bond price. In finding/calculating the YTM, we use ***trial and error*** by using the formula shown below to approximate the yield to maturity.

$$\text{YTM} = \frac{I + \frac{Par\ Value - Bond's\ Selling\ Price}{n}}{\frac{Par\ Value + Bond's Selling\ Price}{2}}$$

Where I = Coupon Payments

**Example 5**
Suppose you were interested in a six-year, 8% coupon bond. A broker quotes a price of 955.14. What is the yield on this bond?

**Solution**:
Using the **Trial & Error Technique**;

$$\text{YTM} = \frac{I + \frac{Par\ Value - Bond's\ Selling\ Price}{n}}{\frac{Par\ Value + Bond's Selling\ Price}{2}}$$

$$\text{YTM} = \frac{\$80 + \frac{\$1000 - \$955.14}{6}}{\frac{\$1000 + \$955.14}{2}}$$

$$\text{YTM} = \frac{\$80 + \frac{\$44.86}{6}}{\frac{\$1955.14}{2}}$$

$$\text{YTM} = \frac{\$80 + \$7.4767}{\$977.57}$$

$$\text{YTM} = \frac{\$87.4767}{\$977.57}$$

$$\text{YTM} = 0.08948$$
$$\text{YTM} = 8.95\%$$

**Example 6**

A corporate bond with a par value of $1000 and a coupon rate of 10% matures in 8 years. The price of the bond is $1,114.70. Calculate YTM.

**Solution**

$$YTM = \frac{100 + \frac{1000 - 1114.70}{8}}{\frac{1000 + 1114.70}{2}}$$

$$YTM = \frac{100 - 14.34}{1057.35}$$

$$YTM = 0.081$$
$$YTM = 8.1\%$$

**Current Yield**

This is the ratio of the annual interest payment to the bond's current market price. It is the rate of return available from a bond on its current price in the secondary market. It fluctuates depending on the bond price. It only measures the annual rate of return accruing to an investor who purchase the bond from the secondary market.

**Example 7**

What is the current yield on a bond with 8% coupon and par value of $1000, if the market price is $700?

$$\text{Current Yield} = \frac{\text{Annual interest payments}}{\text{Market price of the bond}}$$

$$= \frac{0.08 * 1000}{700} = \frac{\$80}{\$700} = 11.4\%$$

Note however, that this is not equal to YTM (unless the bond is selling at par).

**Bond Ratings**

Bond ratings are used to judge the safety of bonds. It is an assessment of the creditworthiness of the corporate issuer. The two leading bond rating firms are Moody's and Standard and Poor's (S&P). Credit worthiness implies how likely the firm is to default and the protection creditors have in the event of a default. The ratings are shown below:

| S and P | Moody's |
|---|---|
| AAA | Aaa |
| AA | Aa |
| A | A |
| BBB | Baa |
| BB,B | BaB |
| CCC | Caa |
| CC | Ca |
| C | C |
| D | D |

Bonds rated BBB or Baa and above are called investment grade, while those with a rating of BaB or below are known as speculative grade or junk bonds. Junk bonds are also referred to as “falling angels”.

**Summary**

In this session, examined bonds, the features of bonds, the issuers of bonds and how to calculate the market value or price of a bond. We also discussed the yield to maturity of a bond. Bond rating was not left out. We are sure you asking why MMDAs are not taking advantage of the bonds so as to establish factories in the various areas. For now, we only have some companies issuing bonds and the Government of Ghana issuing bonds to raise funds for various projects.

**Self-Assessment Questions**

**Exercise 4.5**

## SESSION 6 VALUATION OF SHARES

### Introduction

The two principal components of the capital structure of a company are its equity and debt. A corporation or company sells its stock/shares to the investors to raise equity capital. The financial markets ultimately determine the value of a share or stock. If the company is in strong financial condition and it has good earnings prospects, then the investors will aggressively buy its stock or share and raise the price per share. The market value of a stock/share could be quite different from its book value or its accounting value. The value of the stock/shares depends upon the expectations of the investors regarding the future earnings and growth possibilities of the firm. In this session, we shall examine the value of different types of shares.

### Objectives

By the end of this session, you should be able to:

(a) understand shares and how share prices are quoted
(b) explain the different types of shares and their characteristics
(c) determine the value of ordinary shares and preference shares

Now read on …

### 6.1 Stating share prices

Share prices can be stated in two basic ways namely *Ex-dividend share price* and *Cum-dividend share price*. *Ex-dividend share price* is that price of an ordinary share that is determined immediately after a pending or immediate dividend is being paid. *Cum-dividend share price* is that price of an ordinary immediately prior to the payment of immediate or pending dividend.

By computation: Ex-dividend share price = Cum-dividend share price – dividend due

### 6.2 Ordinary Share (Common Stock) Valuation

Ordinary shares are held by the owners of the firm. Ordinary share valuation is complicated by the uncertainty of future returns. There are two important differences between the factors for valuing ordinary share (common stock) and those for valuing bonds.

**First**, the horizon, or life of the investment, is infinite rather than finite. The horizon is finite when the investor plans to purchase an ordinary share and hold it for a specific length of time whereas the infinite period assumes that an investor plans to purchase an ordinary share and hold it indefinitely.

**Second**, the future cash flows are not explicitly promised. Future cash flows must be estimated on the basis of expectations about the firm's future earnings and dividend policy. Thus, applying the fundamental valuation equation to the case of ordinary shares can pose certain problems.

a. The future stream of cash flows is more uncertain compared to other financial assets, such as bonds.
b. These cash benefits to owners occur in perpetuity because the firm is viewed as a going concern.

It is the receipt of dividends and the price expected when the share is sold at a future date that induce investors to buy stocks. Cash payment to shareholders comes in two forms:

(a) Cash dividend,
(b) Capital gains or loss.

The rate of return that investors expect from their share over the next year is:

$$\text{Expected return} = r = \frac{Div_1 + P_1 - P_0}{P_0}$$

Where = $P_0$ = Current Price of share
$P_1$ = Expected Price a year from now
$Div_1$ = Expected dividend per share

**Example 1**
Suppose TGF's share sells for GH¢70 a share. Investors expect a GH¢3 cash dividend over the next year. They also expect the share to sell for GH¢85 a year hence. What is the expected return to shareholders?

$$\text{Expected return} = r = \frac{70+85-70}{70} = 25.3\%$$

Note that actual return may turn out to be more or less than what investors expected.

Take a second look at the expected return again. Did you notice that it comes in two parts? The dividend yield and capital appreciation?
Expected return = r = Expected Dividend yield + Expected Capital Appreciation

$$\text{Expected return} = r = \frac{Div_1}{P_0} + \frac{P_1 - P_0}{P_0}$$

$$\text{Expected return} = r = \frac{3}{70} + \frac{85 - 70}{70}$$

Expected return = r = 0.043 + 0.21
Expected return = r = 25.3%

**6.3 Ordinary Share Valuation Models**
Although there are numerous ordinary shares or common stock valuation models, the models to be discussed in Business Finance are dependent on *holding periods* (the length of time that an investor expects to own or hold a security) and *rates of dividend growth*.
To discount estimated cash flows in respect of common stock, it is necessary to make assumptions regarding the *time* at which the shares will be sold. For this purpose, the share or stock valuation models are categorized into *finite* and *infinite period valuation models.*

### 6.3.1 Finite-Period Valuation Model

This is valuation model for which an investor plans to purchase an ordinary share or a common stock and hold it for a specific length of time. During the holding period, the investor expects to receive cash dividends and also sell the share or stock for a price at the end of the holding period.

The market value of shares, $P_0$, can also be explained in terms of investors' forecasts of dividend and price and the expected return offered by others equally risky shares or stocks. This represents the PV of the cash flows the share will provide to its owner:

For a *one-year holding period*:

$$\text{Price today} = P_0 = \frac{Div_1 + P_1}{1+r}$$

Where r = required return in the market on this investment.

Using figures from the previous example, we have:

$$\text{Price today} = P_0 = \frac{3+85}{1+0.253} = 70$$

If the formula $P_0 = \frac{Div_1 + P_1}{1+r}$ holds now, it ought to hold next year too.

$$\Rightarrow P_1 = \frac{Div_2 + P_2}{1+r}$$

We can turn the *one-period* stock valuation model;

$P_0 = \frac{Div_1 + P_1}{1+r}$ and

$P_1 = \frac{Div_2 + P_2}{1+r}$ into a *two-period model*:

$$P_0 = \frac{Div_1 + P_1}{1+r} + \frac{Div_2 + P_2}{(1+r)^2}$$

The general formula under a *finite-period* for valuing a share irrespective of the time horizon is

$$P_0 = \frac{Div_1 + P_1}{1+r} + \frac{Div_2 + P_2}{(1+r)^2} + \frac{Div_3 + P_3}{(1+r)^3} + \ldots + \frac{Div_t + P_t}{(1+r)^t}$$

**Example 2**

BP is increasing next year's dividend to GH¢5 per share. The forecast share price next year is GH¢105. Equally risky shares of other companies offer expected rates of return of 10%. What should BP's ordinary share sell for?

**Solution**

Price today = $P_0 = \frac{Div_1+P_1}{1+r} = P_0 = \frac{5+105}{1+0.10}$ = GH¢100

**NB**: Regardless of the time horizon, the share value will be the same. Let us confirm this with another example.

**Example 3**

Princess & Angel Limited is growing steadily and investors expect both the share price and the dividend to increase at 8% per year. Three rational investors – Kojo, Kosi and Ami plan to hold Princess and Angel Limited's shares for 1, 2 and 3 years respectively. Next year's share price is estimated at GH¢81. Princess and Angel pays GH¢3 dividend. Calculate the PV of the share at each of the time horizon; assuming the expected return required by the investors is 12%.

**Solution**

Kojo's 1-year horizon: PV $= \frac{Div_1+P_1}{1+r} = \frac{3+81}{1+0.12}$ = GH¢75

Kosi's 2-years horizon: PV $= \frac{Div_1+P_1}{1+r} + \frac{Div_2+P_2}{(1+r)^2}$

$$PV = \frac{3}{1+0.12} + \frac{3.24+87.48}{(1+0.12)^2} = 2.68 + 72.32 = \text{GH¢}75$$

Ami's 3-years horizon: PV $= \frac{Div_1+P_1}{1+r} + \frac{Div_2+P_2}{(1+r)^2} + \frac{Div_3+P_3}{(1+r)^3}$

$$PV = \frac{3}{1+0.12} + \frac{3.24}{(1+0.12)^2} + \frac{3.50+94.48}{(1+0.12)^3}$$

$$PV = \frac{3}{1.12} + \frac{3.24}{(1.12)^2} + \frac{97.98}{(1.12)^3}$$

PV = GH¢2.68 + GH¢2.58 + GH¢69.74 = GH¢75

This is the *dividend discount model*.

**6.3.2 Infinite-Period Valuation Model**

Under infinite – period valuation, the investor plans to purchase a common stock or an ordinary share and hold it indefinitely. Hence, the returns are only in the form of dividends over multiple periods.

Although it is difficult to value ordinary shares, we can reduce the completeness in the simplifying assumptions about the pattern of future dividends. The three cases are:

- The dividend has a zero-growth rate,

- The dividend grows at a constant rate, and
- The dividend grows at varied rate.

**6.3.2.1 The dividend discount model with no growth:**
Some businesses pay out all its earnings to ordinary shareholders. Such companies could not grow because it did not reinvest in the business, hence dividend payments would be same for years. The price of such shares Po is calculated by using the formula for Perpetuities:

$$P_0 = \frac{Div_1}{r} = \frac{EPS_1}{r}$$

**Example 4**
A company pays a constant dividend of GH¢100 per share. If the market capitalization rate is 25%, how much will you pay for a share of the company today?
Note: Market capitalization rate = Cost of Capital.

**Solution**

$$P_0 = \frac{100}{0.25} = \text{GH¢}400$$

**6.3.2.2 The Dividend Discount Model with Constant Growth**
If dividends grow at a constant rate into the future, then instead of forecasting an infinite number of dividends, we forecast only the next dividend and the dividend growth rate. The formula therefore is:

$$P_0 = \frac{Div_1}{r-g}$$

This model is also known as the Gordon Growth Model after Myron Gordon, who popularized it. However, note that if there is an immediate dividend $D_0$, the formula changes to:

$$P_0 = D_0 + \frac{Div_1}{r-g}$$

where $P_0$ = cum-dividend share price, and $P_0$-$D_0$ = ex-dividend share price

$D_0$ is ignored if it has been paid out. In this case, $P_0$ becomes the ex-dividend share price.

**Example 5**
Princess and Angel Limited has just paid dividend. Next year's dividend is forecast at GH¢3 and expected to grow at 8% hence. Cost of capital is 12%. What is the share price?

**Solution**

$$P_0 = \frac{Div_1}{r-g} = P_0 = \frac{3}{0.12-0.08} = \text{GH¢}75$$

**Note that r** $= \frac{Div_1}{P_0} + g \Rightarrow \frac{3}{75} + 0.09 = 12\%$

**Example 6**

Kazookawor Limited is growing at 5% per year into the indefinite future. It is selling at GH¢50 and next year's dividend is forecast at GH¢5. What is the expected rate of return from investing in BT Limited? BT Limited and Kazookawor Limited's shares are equally risky.

**Solution:**

The 2 firms have equal risk.

$$\mathbf{r} = \frac{Div_1}{P_0} + g \Rightarrow \frac{5}{50} + 0.05 = 15\%$$

**Example 7**

SK Plc is about to pay a dividend of 15 pesewas. Shareholders expect dividends to grow at 6% p.a. The company's share price is GH¢1.25. Calculate the company's cost of capital.

**Solution**:

Because company is about to pay dividend, => impending dividend = GH¢0.15

Cum-dividend share price = $P_0$ = GH¢1.25

Ex-dividend share price = $P_0$-$D_0$= GH¢1.25 - GH¢0.15 = GH¢1.10

$$\mathbf{r} = \frac{Div_1}{P_{0-}D_0} + g \Rightarrow \frac{0.15\ (1+0.06)}{1.25-0.15} + 0.06 = 20.5\%$$

**6.4 Preference Shares Valuation**

A preference share is a share that has a claim against income and assets of a firm before ordinary shares but after debt. Remember you treated this under Company and Partnership Law? Excellent you do remember. Preference share possesses characteristics of both debt and equity, hence a hybrid security. Preferred share pays fixed dividends at the end of each year.

Dividends on preference shares are either cumulative or non-cumulative. If preference dividends are cumulative and are not paid in a particular year, they will be carried forward. Usually, both the accumulated preference dividends plus the current preferred dividends must be paid before its ordinary shareholders can receive anything.

**6.4.1 General Characteristics of Preference Shares**

**Preference:** the shareholders have preference over ordinary shareholders in the distribution of dividends and in the firm's asset in the event that the firm fails and/or is in liquidation, hence the term "preferred".

**Par Value**: preference shares have a fixed amount of dividend which is usually expressed as a percentage of par–value (its dividend yield). This preference dividend yield is set when the shares is issued and is paid on a certain date every year. The dividend is usually cumulative. If they are not paid on time, the cumulative unpaid dividends must be paid out before ordinary shareholders can receive any dividends.

**No maturity:** preference shares have no maturity date; however, some preference shares may include the call provision which allows the firm to redeem (retire) the shares, at a

predetermined time and price. In contrast to ordinary shares, preference shares are much easier to value because their cash benefits are regular, fixed dividends.

### 6.4.2 Valuation of Preference Shares

If preference share pays fixed dividends at the end of each year, then it can be treated as perpetuity. The discount rate would be the investors required rate of return.

The formula is $P_0 = \frac{Div}{r}$

**Example 8**

ECG has an issue of preference shares outstanding that pays a yearly dividend of GH¢5.60. Investors require a 9% return on this preferred stock. What is the PV of this shares?

**Solution:**

$P_0 = \frac{Div}{r} = \frac{5.60}{0.09} =$ GH¢62.22

**Summary**

In this session we discussed shares. We learnt how to compute the value of preference shares and ordinary shares under different time periods. The characteristics of these shares were all discussed. Remember one major source of financing for businesses is to issue shares and its valuation is crucial in times of disputes and when there is change in ownership of a business.

**Self-Assessment Questions**

**Exercise 4.6**

## UNIT 5: CAPITAL BUDGETING

**Unit Outline**

Session 1: Meaning and Process of Capital Budgeting
Session 2: Types of Capital Expenditures and Proposals
Session 3: Determination of Cash Flows
Session 4: Appraisal Techniques: Pay-Back Period and Accounting Rate of Return
Session 5: Appraisal Techniques: Net Present Value and Profitability Index
Session 6: Appraisal Techniques: Internal Rate of Return and Modified Internal Rate of Return

**Overview**

Any investment decision depends upon the decision rule that is applied under circumstances. However, the decision rule itself considers following inputs: Cash flows, Project Life and Discounting Factor. The effectiveness of the decision rule depends on how these three factors have been properly assessed. Estimation of cash flows require immense understanding of the project before it is implemented; particularly macro and micro view of the economy, policy and the company. Project life is very important, otherwise it will change the entire perspective of the project. So great care is required to be observed for estimating the project life. Cost of capital is being considered as discounting factor which has undergone a change over the years. Cost of capital has different connotations in different economic philosophies. Particularly, Ghana has undergone a change in its economic ideology from a closed-economy to open-economy. Hence determination of cost of capital would carry greatest impact on the investment evaluation. This Unit is focusing on various techniques available for evaluating capital budgeting projects. We shall discuss all investment evaluation criteria from its economic viability point of view and how it can help in maximizing shareholders' wealth.

**Objectives**

By the end of this Unit, you should be able to:

(a) be aware of the basic terminology
(b) be aware of the importance of capital budgeting to the firm
(c) be able to recognize and distinguish the main types of investment projects

(d) define the capital budgeting process, explain the administrative steps of the process, and categorize the capital projects that can be evaluated.
(e) Summarize and explain the principles of capital budgeting, including the choice of the proper cash flows and the identification of the proper discount rate.
(f) Calculate and interpret the results produced by each of the following methods when evaluating a single capital project: net present value (NPV), internal rate of return (IRR), payback period, discounted payback period, accounting rate of return (ARR), and profitability index (PI).
(g) Explain the NPV profile, compare and contrast the NPV and IRR methods when evaluating more than one capital project, and describe the multiple IRR and no IRR problems that can arise when calculating an IRR.
(h) Describe the relative popularity of the various capital budgeting methods

## SESSION 1 MEANING AND PROCESS OF CAPITAL BUDGETING

**Introduction**

We warmly welcome you to this session. The beginning session on capital budgeting takes you through the meaning of capital investments, the persons to be involved in capital investment decisions, the purpose of capital investment decisions and the process we need to follow in making capital investment decisions. Read on and have fun whilst you read.

**Objectives**

By the end of this session, you should be able to:
(a) define the capital budgeting process
(b) explain the administrative steps of the process, and categorize the capital projects that can be evaluated.
(c) identify those that must be involved in capital budgeting decisions
(d) outline the objectives of capital investments

Now read on …

### 1.1 Definition of capital investments

Capital Budgeting is also known as "Investment Decision Making or Capital Expenditure Decisions" or "Planning Capital Expenditure" or “Capital Investments” etc. Normally such decisions where investment of money and expected benefits arising therefrom are spread over more than one year, it includes raising of long-term funds as well as their utilization.

Charles T. Horngren has defined capital budgeting as "Capital Budgeting is long term planning for making and financing proposed capital outlays." In other words, capital budgeting is the decision-making process by which a firm evaluates the purchase of major non-current assets including building, machinery and equipment. According to Hamption, John. "Capital budgeting is concerned with the firm's formal process for the acquisition and investment of capital.” According to the definition of Richard and Green law, “capital budgeting is acquiring inputs with long-term return”.

Capital budgeting can also be defined as the process of planning and managing a firm's long-term investment. In other words, it is the process of planning for purchases of assets whose cash flows are expected to continue beyond one year.

In capital budgeting, the financial manager tries to identify investment opportunities that are worth more to the firm than they cost to acquire. The process of allocating or budgeting capital is usually more involved than just deciding on whether or not to buy a particular fixed asset. Decisions such as these will determine the nature of the firm's operations and products for years to come, primarily because non-current asset investments are generally long-lived and not easily reversed, once they are made. Hence, the capital budgeting question is probably the most important issue in corporate finance because the non-current assets define the business of the firm. The result of capital budgeting decisions continues for many years and hence the firm loses some of its flexibility. Airlines, for example, are airlines because they operate airplanes, regardless of how they are financed and State Transport Company of Ghana is a transport company because of the buses they operate. Also, the capital budgeting decisions defines the strategic direction of the business.

**1.2 Who is involved in making capital investment decisions?**

There is no one answer to this question. In practice and depending on the type of organization, the following are involved:

- Accountants
- Engineers
- Operational managers
- Line staff (who work with capital assets)
- Specialist capital investment officers
- General managers and board of directors (who are responsible for the final decision)

**1.3 Why capital investment decisions?**

Capital budgeting is important because of the following reasons:

(i) Capital budgeting decisions involve long-term implication for the firm, and influence its risk complexion.
(ii) Capital budgeting involves commitment of large amount of funds.
(iii) Capital decisions are required to assessment of future events which are uncertain.
(iv) Wrong sale forecast; may lead to over or under investment of resources.
(v) In most cases, capital budgeting decisions are irreversible. This is because it is very difficult to find a market for the capital goods. The only alternative available is to scrap the asset, and incur heavy loss.
(vi) Capital budgeting ensures the selection of right source of finance at the right time.
(vii) Many firms fail, because they have too much or too little capital equipment.
(viii) Investment decision taken by individual concern is of national importance because it determines employment, economic activities and economic growth.

**1.4 Objectives of Capital Investment**

The following are the important objectives of capital budgeting:

(i) To facilitate co-ordination of inter-departmental project funds among the competing capital projects.
(ii) To ensure the selection of the possible profitable capital projects.

(iii) To ensure the effective control of capital expenditure in order to achieve organizational goals by forecasting the long-term financial requirements.
(iv) Determining the required quantum of money needed and how it takes place as per authorization and sanctions.
(v) To make estimation of capital expenditure during the budget period and to see that the benefits and costs may be measured in terms of cash flow.
(vi) To ensure maximization of profit by allocating the available investible.

**1.5 Capital budgeting process**

There is no universal view on the stages of the capital budgeting process. Some researchers define three stages while others call for up to five stages. However, most authors generally identify the same critical phases but separate them slightly differently. Thus, in the following, a five-stage approach will be used with stages identified by the authors as identification and filtering; selection; authorization; implementation; and control.

**Stage 1: Identification and filtering of investment proposals**

This stage is seen by many practitioners as the most critical one. Project ideas can be developed in two ways: first, ideas can emerge bottom-up or top-down and, second, they can be driven by an opportunity or by a need for an investment. The bottom-up investment ideas mirror the opportunities discovered by operations managers. Thereby, middle management cannot be expected to suggest strategic proposals. Instead, such ideas for strategic investments with high value will rather come from senior management having an overview of the company and its development. Furthermore, investments need to be made e.g. for replacement or expansions to new markets. Both investment opportunities and needs can originate both ways, top-down or bottom-up.

After the identification of proposals these undergo a preliminary screening and filtering among others for inconsistencies with strategic goals (if originated bottom-up), inadequate hurdle rate, risk levels, and feasibility. Here, the data-gathering efforts of the company are crucial i.e. whether accounting or cash-flow methods are used, the state of the decision support system as well as how senior management deals with forecast bias.

**Stage 2: Selection**

In this stage, survived proposals are thoroughly examined including projection of cash flows (which is perceived as the most difficult part), risk, demand for, and cost of capital, timing of investments, personnel involved, and a first implementation plan. As a result, the best projects are selected and forwarded to top management for approval and authorization. The mostly used algorithms or techniques for evaluation are the net present value, followed by internal rate of return and payback period, weighted average cost of capital for determination of the cost of capital or hurdle rate, with capital asset pricing model for determination of the cost of equity.

**Stage 3: Authorization**

Given capital constraints of companies, at this stage capital rationing and previously defined capital budgeting targets eventually decide which projects are realized. Therefore, the capital demand of the proposals is compared with the company's internal and external sources of capital supply such as depreciation reserves and retained earnings, and loans, corporate bonds and shares, respectively. The crucial factor here is the cost of capital – e.g., companies cannot control their cost of debt because it is influenced by financial market conditions. After solving the question of financing, projects are ranked by e.g. strategic importance, return, and

risk. A number of risks such as general risks (market risk, inflation, interest rate, foreign exchange rate risk etc.) and specific risks i.e. those inherent to a particular project must be assessed. That can be done e.g. by means of sensitivity analysis or a risk map ranking the risks by their impact and probability. Then, corresponding risk responses (e.g. risk adjusted discount rate or cash flows) for the setup of a risk management system can be provided. Finally, after answering the questions of financing, prioritization, and a first implementation plan, projects perceived as the best are authorized for implementation.

**Stage 4: Implementation**
In the implementation phase, a detailed implementation plan is set up and cascaded down the organization since the implementation itself is essentially the task of operations management while it is up to senior management to monitor it. This stage can follow the common practice of project management. That means, first a work breakdown structure has to be installed. It breaks up the project into work packages and individual activities or tasks to be performed. Then, for each of the tasks a responsible person as well as time frames and a budget are assigned. Finally, milestones, meaning meetings or deadlines until which certain deliverables have to be performed, are set. Mostly, a project management committee is created in charge of planning, implementation, and reporting.

**Stage 5: Performance measurement and control**
Three kinds of measurement of a project's performance exist. First, monitoring shortly before and after the start of implementation to detect and counteract previously unforeseen problems. Second, monitoring during the implementation in order to oversee overruns in timing and expenditures and to adequately meet problems. And finally, after the completion of the project (post-audit), mainly to gather lessons for the coming projects but also, in a limited way, to examine the quality of forecasts made by project initiators. To audit the results, usually, estimates are compared to actual results such as profits, costs (initial expenditure or operating cash outflows), volumes, time, or rates of return, respectively.

Even though the importance of performance measurement and control may seem obvious, surprisingly little is being done in this area. Gordon and Myers found in 1991 that, although 76% of their survey respondents performed post-audits, those were neither regular, nor risk-adjusted or thoroughly documented thus not being a standard capital budgeting procedure. This to us in Business Finance is crucial and need to be taken seriously. Performance measurement is very very important and should not be taken for granted. It helps improve the future operations and profitability of the firm. We, therefore, encourage our distance education students not to forget this valuable aspect of the entire capital budgeting process.

**Summary**
In a nutshell, we can conclude that capital investment decisions are important for the survival of businesses and must be properly done. We defined capital investments, the need for capital investments and the persons to be involved in capital investment decision-making. We ended the session with the process that must be followed when making capital budgeting decisions. These we explained should not be mixed up and must be followed in order to arrive at an appropriate decision for the company.

**Self-Assessment Questions**
**Exercise 5.1**

## SESSION 2: TYPES OF CAPITAL EXPENDITURES AND PROPOSALS

### Introduction

You are welcome to this session that deals with types of capital expenditure. A management has taken number of decisions with regard to short term investment and long-term investment. Whatever be and form of investment, the profitability of the company should be increased or at least maintained. In the case of long-term investment, huge amount is necessary. Such an amount is used for acquiring non-current assets i.e. plant and machinery, land and building, introducing new products, starting of one more unit, expansion, improvement or alteration of existing non-current assets, replacement of existing non-current assets, research and development and the like. The management is going to take a decision for this purpose i.e. capital expenditure decision. These are explained in this session. Have fun as you read.

### Objectives

By the end of this session, you should be able to:

(a) define capital expenditure
(b) describe the types of capital expenditure
(c) discuss the types of capital budgeting decisions

Now read on …

## 2.1 Capital Expenditure

Capital expenditure is classified into three main forms viz:

- Expenditure made to reduce costs; Such a capital expenditure reduces the cost of present product and thereby increases the profitability of existing operations. It can be done by replacement of old machine by a new one.
- Expenditure made to increase revenue; It is the expenditure which brings more revenue to the firm either by expanding the existing production facilities or development of new production line.
- Expenditure which is justified on non-economic grounds.

With exercise control over capital expenditure in any of the above categories, the capital expenditure analysis should concentrate on three types of outlays viz:

- Major projects,
- Routine expenditure,
- Replacement.

### 2.1.1 Capital Expenditure analysis in major projects

As regards major projects, strategic investment may be made for expansion of productive capacity or achieving product innovation or preparing barriers against capital fluctuations.

### 2.1.2 Analysis in routine expenditures

In the second type of outlay, routine expenditure may be working condition improvement, maintenance expenditure, competition-oriented expenditure etc.

### 2.1.3 Replacement in capital expenditure analysis

Thirdly, replacement need may arise to avoid capital wastage for existing equipment to check its disposal value or it may be obsolescence replacement.

In all circumstances, proper attention is to be devoted in analysing the need for the capital expenditure so that it would be curtailed to the minimum required.

## 2.2 Generation of investment projects

Investment projects can be generated by:

- Growth opportunities. Expansion of existing business into new products and markets, R&D, exploration etc.
- Cost reduction opportunities. Replacement of plants, equipment and buildings.
- Meeting legal requirements and health and safety standards. Pollution control, ventilation and fire protection equipment. Projects here are treated as contingent upon other projects.

## 2.3 Types of capital budgeting decisions

Generally, the business firms are confronted with three types of capital budgeting decisions.

- The accept-reject decisions or independent proposals;
- mutually exclusive decisions; and
- capital rationing decisions or dependent proposals.

### 2.3.1 Accept-reject decisions or independent proposals

Business firm is confronted with alternative investment proposals. If the proposal is accepted, the firm incur the investment and not otherwise. Broadly, all those investment proposals which yield a rate of return greater than cost of capital are accepted and the others are rejected. Under this criterion, all the independent proposals are accepted.

These proposals are said to be economically independent when accepted or rejected on the basis of minimum return on investment required. Independent proposals do not depend upon each other. Independent projects are those projects in which a decision to invest in one project has no bearing on the decision to make investment in another. Examples would include:

- The introduction of a new product line (soap) and at the same time the replacement of a machine, which is currently producing a different product (plastic bottles).
- The installation of a new air conditioning system and the commissioning of a new advertising campaign for a product currently sold by the firm.

**2.3.2 Mutually exclusive decisions**

It includes all those projects which compete with each other in a way that acceptance of one precludes the acceptance of other or others. Thus, some technique has to be used for selecting the best among all and eliminates other alternatives.

Mutually Exclusive Proposals refer to the acceptance of one proposal results is the automatic rejection of the other proposal. In other words, one can be rejected and the other can be accepted. It is easier for a firm to take capital budgeting decisions on such projects. Stated differently, mutually-exclusive projects refer to a set of projects out of which only one project can be. Examples would include:

- A firm may own a block of land, which is large enough to establish a shoe manufacturing business or a steel fabrication plant. The selection of one will exclude the acceptance of the other.
- A car manufacturing company considering to establish one of its manufacturing complexes can locate it in Ghana, Nigeria or Benin. If the company chooses Ghana, the other two locations are ruled out.

**2.3.3 Capital rationing decisions**

Capital budgeting decision is a simple process in those firms where fund is not the constraint, but in majority of the cases, firms have fixed capital budget. So large number of projects compete for these limited budgets. So, the firm rations them in a manner so as to maximize the long run returns. Thus, capital rationing refers to the situations where the firm has more acceptable investment requiring greater amount of finance than is available with the firm. It is concerned with the selection of a group of investment out of many investment proposals ranked in the descending order of the rate or return.

When the acceptance of one proposal is contingent upon the acceptance of other proposals. It is called as "Dependent or Contingent Proposals. The acceptance or rejection of one is dependent on the decision to accept or reject one or more other projects. Contingent projects may be complementary or substitute. Example of each would include:

- Complementary: The decision to start a pharmacy may be contingent upon a decision to establish a doctors' surgery in an adjacent building. The cash flows of the pharmacy will be enhanced by the existence of a nearby surgery and vice versa.
- Complementary: The introduction of a water recycling plant may increase the profitability of several other projects.

- Substitutes: customers visiting a shopping complex may treat Chinese and Thai food as close substitutes. Consequently, if the firm establishes both restaurants, none of the restaurants may be profitable due to the distribution of a given number of customers between the two restaurants. However, if only one restaurant is established, it may generate net present value. Therefore, the acceptance of one is contingent upon the non-acceptance of the other project.
- Substitutes: two different brands of butter as two different projects. The success of one project may depend on the non-acceptance of the other project.

**Summary**

In this session we examined the different types of capital expenditure. We explained that generally, the business firms are confronted with three types of capital budgeting decisions. These are the accept-reject decisions or independent proposals; mutually exclusive decisions; and capital rationing decisions or dependent proposals.

**Self-Assessment Questions**

**Exercise 5.2**

## SESSION 3 DETERMINATION OF CASH FLOWS

**Introduction**

The capital budgeting process is concerned primarily with the estimation of the cash flows associated with a project, not just the project's contribution to accounting profits. Typically, a capital expenditure requires an initial cash outflow, termed the net investment. Thus, it is important to measure a project's performance in terms of the net (operating) cash flows it is expected to generate over a number of future years. In this session, we discuss how to generate the initial cash outflow or outlay and the expected cash inflows for the project. Pay particular attention to details.

**Objectives**
By the end of this session, you should be able to:
(a) describe the principles of cash flow estimation
(b) understand discounted cash flow model
(c) determine initial cash outlay, operating cash inflows and terminal cash flows

Now read on …

## 3.1 Principles for Cash Flow Estimation
Cash flow estimation is an integral part of the valuation and capital budgeting process. Cash flow estimation is a necessary step for assessing investment decisions of any kind. Project cash flows consider all kinds of inflows of cash. The estimation of cash flows is done through the coordination of wide range of professionals involved in the project. The engineering department is responsible for forecasting of capital outlays. The production team is responsible for forecasting operational costs. The marketing team is basically involved in forecasting revenues. The finance manager has the responsibility to collect data and set norms for better estimation. Forecasting cash flows is one of the major challenges faced by valuation professionals. Cash flow estimation is based on a number of principles.

### 3.1.1 Consistency Principle
The consistency principle states that consistency has to be maintained between the flow of cash in a project and the rates of discount that are applicable on the cash flows.

### 3.1.2 Separation Principle
The separation principle suggests that project cash flows can be segregated into investment and financing flows.

### 3.1.3 Post-tax Principle
The post-tax principle suggests that the forecast of cash flows should be based on after-tax method.

### 3.1.4 Incremental Cash Flow Principle
The incremental cash flow principle suggests that only cash flows relevant to the valuation of a project are the incremental cash flows resulting from it.

## 3.2 Discounted Cash Flow Model (DCF)
The discounted cash flow (DCF) model is based on the assumption that a company's value is basically equal to the present value of its estimated future cash flows. In this DCF model, the major step of valuation is the estimation of future cash flows. The most important variable in estimating cash flows are the firm's future sales growth and profit margins. Hence, estimation of growth rates is an important determinant of cash flow estimation. A firm's revenue growth rate is based on a number of factors like industry trends, economic environment, and company's competitive advantage. The costs incurred by firms are a major factor for determining the company's future operating profits. Companies that are heavily dependent on oil and natural gas face wide swings in profit margin if the price of the raw materials increases. Operating leverage is a factor that determines the profitability position of a firm. As a company grows, it must be able to spread its fixed costs across a broader base of production.

DCF model focusses on free cash flow, which is defined as operating cash flow minus capital expenditures and non-cash working capital. Free cash flow represents the residual cash flow after all expenses necessary to keep the firm growing at its current rate. It is vital to estimate how much the company reinvests in itself every year with respect to capital expenditures and working capital. DCF consists of two models: the free cash flow to equity and the free cash flow to firm model.

**3.3 Initial outlay**

For expansion projects, this will consist of the cash flows resulting from acquiring the new asset and will consist of:

a. The Purchase price of the new asset
b. Installation costs of the new asset e.g. transportation, shipping, handling etc.
c. Increases in working capital requirements e.g. inventory i.e. raw materials, finished goods etc
d. After-tax non-capital expenditures e.g. costs to train employees to operate asset
e. Investment tax credit (ITC) given by government due to the nature of the company's business. (This ITC could be given at any time during the project's life).

Note - the sum of the cash flows at 'a' and 'b' above is equal to the recorded value of the new asset in the company's books. (i.e. the value at which the asset will be depreciated).

For the replacement project, we would also have to incorporate:

f. those after-tax cash flows resulting from the sale of the existing asset. This is determined as follows:

| | |
|---|---|
| Proceeds of sales | xxx |
| Current book value | (xxx) |
| Profit or loss | xxx/(xxx) |
| Less tax/tax credit | xxx/(xxx) |
| Profit or loss after tax | xxx/(xxx) |
| Add book value | xxx |
| After tax cash inflows | xxx |

**3.4 Annual after-tax operating cash flows**

These cash flows occur on a yearly basis throughout the life of the new project. In practice, the yearly cash flows will not be constant but will more likely increase due to inflation. However, for our purpose, unless otherwise stated, we are assuming that the yearly cash flows will be constant.

| | |
|---|---|
| Revenue/Sales | xxx |
| Cost/ Expenses | (xxx) |
| Depreciation | (xxx) |
| Profit before tax | xxx |
| Less tax | (xxx) |
| Profit after tax | xxx |
| Add Depreciation | xxx |
| After tax cash flows | xxx |

Note - Interest expenses are not to be included as costs. In other words, they are not to be deducted from Revenues when determining Profit before tax (as normal accounting rules stipulate). To do this would be double counting as interest expense (cost of debt) is already accounted for in the cost of capital (which is used to discount the cash flows).

### 3.5 Terminal cash flows

For both the expansion and replacement projects this will comprises of those cash flows that occur as a result of termination of the new project. These may include:

- The after-tax cash flows resulting from the sale of the new asset (calculation is similar to above, see Initial outlay)
- Recovery of working capital, i.e. the working capital cash outflows experienced in the initial outlay will be recovered
- Any other after-tax clean-up costs

### 3.6 Other types of cash flows

#### 3.6.1 Sunk costs

These are cash outflows that have already occurred and therefore do not affect the capital budgeting decision and should not be included in the determination of cash flows.

#### 3.6.2 Opportunity costs

In Economics, this is referred to as benefits foregone. In other words, opportunity costs are benefits that are not going to be achieved due to a particular action/decision. These must be accounted for in the capital budgeting decision as cash outflows, on an after-tax basis.

E.g. In a replacement project analysis, we assume that the existing asset will be sold and the new asset bought. However, if that 'old' asset were kept in operation until the end of its useful life, the company may have been able to receive some cash by selling it afterwards. If such a salvage value existed, then the company would not be able to fetch this amount if they went ahead and replaced the old asset with the new one. Therefore, the after-tax effects of this opportunity cost would have to be incorporated in the analysis. Using this example of the asset's foregone salvage value, the procedure for determining the after-tax effect would be identical to that mentioned earlier (see Initial outlay above), except that the result would be an after-tax cash outflow.

#### 3.6.3 Externalities

In Economics, this refers to the effects of a project on other parts of the firm/company. If the increased revenues or the cost savings derived by a new project result in decreased revenues for other existing projects/products in a company, this effect must be factored into the capital budgeting process. To account for this, the decreased revenues of the existing project/products must be applied to offset/reduce the increased revenues of the new project (or be treated as increased costs of the new project).

Alternatively, the increased revenues or the cost savings derived by the new project could result increased revenues for other existing projects/products in a company. In this case, the increased revenues of the other existing projects/products would be treated as additional increased revenues for the new project.

### 3.7 Illustration

There are two mutually exclusive projects under active consideration of a company. Both the

projects have a life of 5 years and have initial cash outlays of GH¢100,000 each. The company pays tax at 50% rate and the maximum required rate of the company has been given as 10%. The straight-line method of depreciation will be charged on the projects. The projects are expected to generate a net cash inflow before taxes as follows:

| Year | Project Kasama | Project Kpandu |
|---|---|---|
| | GH¢ | GH¢ |
| 1 | 40,000 | 60,000 |
| 2 | 40,000 | 30,000 |
| 3 | 40,000 | 20,000 |
| 4 | 40,000 | 50,000 |
| 5 | 40,000 | 50,000 |

With the help of the above given information you are required to determine the cash inflows after tax for the evaluation of these projects.

**Solution**

In solving this, we first need to find our depreciation. You studied how to calculate depreciation in your course in Foundation of Accounting where different methods were used. The most popular of all the methods is the straight-line method. In calculating depreciation using the straight-line method, we use the following formula:

$$\text{Depreciation} = \frac{Cost - Residual/Scrapvalue}{Number\ of\ years}$$

From our illustration, Cost = initial outlay = GH¢100,000; Residual/Scrap value = GH¢0.00 and Number of years = 5. Therefore, our depreciation for each year is:

$$\text{Depreciation} = \frac{\text{GH¢}100{,}000 - \text{GH¢}0.00}{5} = \text{GH¢}20{,}000$$

| | Project Kasama | | | | | Project Kpandu | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| YEAR | 1 | 2 | 3 | 4 | 5 | 1 | 2 | 3 | 4 | 5 |
| CF Befor tax | 40,000.00 | 40,000.00 | 40,000.00 | 40,000.00 | 40,000.00 | 60,000.00 | 30,000.00 | 20,000.00 | 50,000.00 | 50,000.00 |
| Depreciation | (20,000.00) | (20,000.00) | (20,000.00) | (20,000.00) | (20,000.00) | (20,000.00) | (20,000.00) | (20,000.00) | (20,000.00) | (20,000.00) |
| Profit before tax | 20,000.00 | 20,000.00 | 20,000.00 | 20,000.00 | 20,000.00 | 40,000.00 | 10,000.00 | - | 30,000.00 | 30,000.00 |
| Tax (50%) | (10,000.00) | (10,000.00) | (10,000.00) | (10,000.00) | (10,000.00) | (20,000.00) | (5,000.00) | - | (15,000.00) | (15,000.00) |
| Profit after tax | 10,000.00 | 10,000.00 | 10,000.00 | 10,000.00 | 10,000.00 | 20,000.00 | 5,000.00 | - | 15,000.00 | 15,000.00 |
| Add Depreciation | 20,000.00 | 20,000.00 | 20,000.00 | 20,000.00 | 20,000.00 | 20,000.00 | 20,000.00 | 20,000.00 | 20,000.00 | 20,000.00 |
| **CF after tax** | **30,000.00** | **30,000.00** | **30,000.00** | **30,000.00** | **30,000.00** | **40,000.00** | **25,000.00** | **20,000.00** | **35,000.00** | **35,000.00** |

**Summary**

Determination of cash flows is crucial for any project valuation. In this session, we took you through the various processes that must be followed in determining the initial cash outlay, the operating cash flows, terminal cash flows and issues of externalities that affect the determinations. We encourage you to continue practicing and paying particular attention to details.

**Self-Assessment Questions**

**Exercise 5.3**

## SESSION 4: APPRAISAL TECHNIQUES: PAY-BACK PERIOD AND ACCOUNTING RATE OF RETURN

### Introduction

There are number of appraisal methods which may be recommended for evaluating the capital

investment proposals. We shall discuss the most widely accepted methods. These methods can be grouped into the following categories: Traditional or non-discounted cash flow methods and Time adjusted methods or discounted cash flow methods. Pay-back method and accounting rate of return form part of the traditional methods. These are discussed in this session. Take your time and read, write down questions that you think need answers and ask them during your face-to-face sessions. We can see you smiling. That's a good start.

**Objectives**

By the end of this session, you should be able to:

(a) explain pay-back period method and its advantages and disadvantages
(b) calculate the pay-back period for a single project and for a multiple project
(c) explain accounting rate of return, advantages and disadvantages
(d) calculate the accounting rate of return for a single project and for a multiple project
(e) explain the decision rules on pay-back period and ARR

### 4.1 Payback Period Method

Pay-back period is also termed as "Pay-out period" or Pay-off period. Pay-out Period Method is one of the most popular and widely recognized traditional method of evaluating investment proposals. It is defined as the number of years required to recover the initial investment in full with the help of the stream of annual cash flows generated by the project.

The payback period is the most basic and simple decision tool. T. Lucy (1992) defined pay-back period as the period, usually expressed in years which it takes for the project's net cash inflows to recoup the original investment. The usual decisions rule is to accept the project with the shortest pay-back period. Pay-back period method does not measure overall project worth because it does not consider cash flows after the pay-back period. According to T. Lucy, (1992) pay-back period provides only a crude measure of the timing of project cash flows. The payback period is probably best served when dealing with small and simple investment projects.

We observed that the simplicity of pay-back period method should not be interpreted as ineffective. If the business is generating healthy levels of cash flow that allow a project to recoup its investment in a few short years, the pay-back period can be a highly effective and efficient way to evaluate a project. When dealing with mutually exclusive projects, the project with the shorter pay-back period is selected.

Pay-back period can be calculated using the following two different situations:

- In the case of constant annual cash inflows.
- In the case of uneven or unequal cash inflows.

#### 4.1.1 In the case of constant annual cash inflows

If the project generates constant cash flow the Pay-back period can be computed by dividing cash outlays (original investment) by annual cash inflows. The following formula can be used to ascertain pay-back period:

$$\text{Pay-back Period} = \frac{Cash\ Outlay\ (Initial\ Investment)}{Annual\ Cash\ Inflows}$$

**Example 1**

A project requires initial investment of GH¢400,000 and it will generate an annual cash inflow of GH¢80,000 per year for 6 years. You are required to find out pay-back period.

Solution
Calculation of Pay-back Period:

$$\text{Pay-back Period} = \frac{Cash\ Outlay\ (Initial\ Investment)}{Annual\ Cash\ Inflows}$$

$$\text{Pay-back Period} = \frac{\text{GH¢400,000}}{\text{GH¢80,000}} = 5 \text{ years}$$

Pay-back period is 5 years, i.e., the investment is fully recovered in 5 years.

**4.1.2 In the case of uneven or unequal cash inflows**

In the case of uneven or unequal cash inflows, the Pay-back period is determined with the help of cumulative cash inflow. It can be calculated by adding up the cash inflows until the total is equal to the initial investment. We can use the following formula as a guide:

$$\text{Pay-back Period} = \text{Year before full recovery} + \frac{Unrecovered\ cost\ at\ start\ of\ years}{Cash\ flow\ during\ the\ year}$$

**Example 2**

An organisation wants to embark on an investment. It has been presented with the cashflows for two projects and would want to decide with of the two projects to invest in. Using pay-back period method, help the organisation decide.

| Year | Project A Cash flows | Project B Cash flows |
|---|---|---|
| 1 | GHS12,500 | GHS5,000 |
| 2 | 12,500 | 10,000 |
| 3 | 12,500 | 15,000 |
| 4 | 12,500 | 15,000 |
| 5 | 12,500 | 25,000 |
| 6 | 12,500 | 30,000 |
| | Net Investment = GHS50,000 | Net Investment = GHS50,000 |

**Solution**

For Project A, the cash flows are even or equal so we shall use the formula we looked at earlier.

$$\text{Pay-back Period} = \frac{Cash\ Outlay\ (Initial\ Investment)}{Annual\ Cash\ Inflows}$$

$$\text{Pay-back Period} = \frac{GHS50{,}000}{GHS12{,}500} = 4 \text{ years}$$

It means that if the organisation wants to invest in Project A it will take 4 years to recover the initial investment.

For Project B, the cash flows are not equal, hence we shall use the cumulative cash flows:

| Year | Project B Cash flows | Cumulative Cash inflows |
|---|---|---|
| 1 | GHS5,000 | GHS5,000 |
| 2 | 10,000 | 15,000 |
| 3 | 15,000 | 30,000 |
| **4** | **15,000** | **45,000 *(50,000 – 45,000)*** |
| 5 | 25,000 | |

$$\text{Pay-back Period} = \text{Year before full recovery} + \frac{Unrecovered\ cost\ at\ start\ of\ years}{Cash\ flow\ during\ the\ year}$$

$$\text{Pay-back Period} = 4 \text{ years} + \frac{GHS50{,}000 - GHS45{,}000}{GHS25{,}000}$$

$$\text{Pay-back Period} = 4 \text{ years} + \frac{GHS5{,}000}{GHS25{,}000}$$

Pay-back Period = 4 years + 0.2 = 4.2 Years

This means it will take the organisation 4.2 years to recoup its initial investment.

Considering the two projects the organisation will opt for Project A because it has shorter payback period.

### 4.2 Accept or Reject Criterion

Investment decisions based on pay-back period used by many firms to accept or reject an investment proposal. Among the mutually exclusive or alternative projects whose pay-back periods are lower than the cut off period the project would be accepted; if not it would be rejected. The usual decisions rule is to accept the project with the shortest pay-back period.

### 4.3 Advantages of Pay-back Period Method

- It is an important guide to investment policy.
- It is simple to understand and easy to calculate.
- It facilitates to determine the liquidity and solvency of a firm.
- It helps to measure the profitable internal investment opportunities.
- It enables the firm to select an investment which yields a quick return on cash funds.
- It used as a method of ranking competitive projects.
- It ensures reduction of cost of capital expenditure.

### 4.4 Disadvantages of Pay-back Period Method

- It does not measure the profitability of a project.
- It does not value projects of different economic lives.
- This method does not consider income beyond the pay-back period.

- It does not give proper weight to timing of cash flows.
- It does not indicate how to maximize value and ignores the relative profitability of the project.
- It does not consider cost of capital and interest factor which are very important factors in taking sound investment decisions.

**4.5 Accounting Rate of Return (ARR) Method**

Accounting Rate of Return Method is also termed as Average Rate of Return Method. This method focuses on the average net income generated in a project in relation to the project's average investment outlay. This method involves accounting profits not cash flows and is similar to the performance measure of return on capital employed. The accounting rate of return can be determined by:

$$\text{ARR} = \frac{Average\ Income}{Average\ Investments} x\ 100$$

Or

$$\text{ARR} = \frac{Average\ Annual\ Profit - Depreciation\ and\ Taxes}{Average\ Investments} x\ 100$$

Where,

Average investment would be equal to the Original investment plus salvage value divided by Two

$$\text{Average Investment} = \frac{Original\ Investment}{2}$$

Or

$$\text{Average Investment} = \frac{Original\ Investment + Scrap\ Value\ of\ the\ project}{2}$$

**Example 3**

From the following information you are required to calculate Accounting Rate of Return:
An investment with expenditure of GHS100,000 is expected to produce the following profits (after deducting depreciation)

| | |
|---|---|
| 1st Year | GHS8,000 |
| 2nd Year | GHS16,000 |
| 3rd Year | GHS18,000 |
| 4th Year | GHS6,000 |

If the desired rate of return is 12%, which project should be accepted?

Solution

$$\text{ARR} = \frac{Average\ income}{Average\ investment} \times 100$$

$$\text{Average profit/income} = \frac{Total\ profit}{number\ of\ periods}$$

$$= \frac{8{,}000+16{,}000+18{,}000+6{,}000}{4} = \text{GHS}120{,}00$$

Average Investment $= \frac{Original\ Investment}{2}$

$$= \frac{100{,}000}{2} = \text{GHS}50{,}000$$

ARR $= \frac{Average\ income}{Average\ investment} \times 100$

$$= \frac{GHS12{,}000}{GHS50{,}000} \times 100$$

$$= 24\%$$

Since the ARR is greater than the required rate of return, the project should be accepted.

**Example 4**

A project costs GHS500,000 and has a scrap value of GHS100,000 after 5 years. The net profit before depreciation and taxes for the five years period are expected to be GHS100,000; GHS120,000; GHS140,000; GHS160,000 and GHS200, 000. You are required to calculate the Accounting Rate of Return, assuming 20% rate of tax and depreciation on straight line method.

Solution

Depreciation $= \frac{cost - scrap\ value}{Estimated\ useful\ life}$

$$= \frac{500{,}000 - 100{,}000}{5}$$

$$= 80{,}000$$

| Year | 1 | 2 | 3 | 4 | 5 | **Total** |
|---|---|---|---|---|---|---|
| Net profit before depreciation | 100,000.00 | 120,000.00 | 140,000.00 | 160,000.00 | 200,000.00 | 720,000.00 |
| Depreciation | (80,000.00) | (80,000.00) | (80,000.00) | (80,000.00) | (80,000.00) | (400,000.00) |
| Profit after depreciation | 20,000.00 | 40,000.00 | 60,000.00 | 80,000.00 | 120,000.00 | 320,000.00 |
| Tax (20%) | (4,000.00) | (8,000.00) | (12,000.00) | (16,000.00) | (24,000.00) | (64,000.00) |
| Net profit after tax | **16,000.00** | **32,000.00** | **48,000.00** | **64,000.00** | **96,000.00** | **256,000.00** |

Average Investment $= \frac{Original\ Investment + scrap\ value}{2}$

$= \frac{500{,}000 + 100{,}000}{2}$

= GHS300, 000

Average profit/income $= \frac{256000}{5}$

= 51,200

ARR $= \frac{Average\ income}{Average\ investment} \times 100$

$= \frac{GHS51{,}200}{GHS300{,}000} \times 100$

= 17%

**4.6 Decision Rule**

A project is accepted if the ARR calculated meets the minimum rate of return. If mutually exclusive, you accept the project with the highest ARR.

**4.7 Advantages**

- It considers all the years involved in the life of a project rather than only pay-back years.
- It applies accounting profit as a criterion of measurement and not cash flow.
- It is easy to calculate and simple to understand.
- It considers the total benefits associated with the project.

**4.8 Disadvantages**

- It applies profit as a measure of yardstick not cash flow.
- The time value of money is ignored in this method.
- Yearly profit determination may be a difficult task.

**Summary**

This session looked interesting. See how you are smiling. In this session, we discussed the traditional methods of pay-back period and accounting rate of return in valuing capital investment projects. We again looked at the decision rules of the two methods, advantages and disadvantages. These methods are still relevant and used in industry. Keep practicing.

**Self-Assessment Questions**

**Exercise 5.5**

## SESSION 5: APPRAISAL TECHNIQUES: NET PRESENT VALUE AND PROFITABILITY INDEX

### Introduction

There are number of appraisal methods which may be recommended for evaluating the capital
investment proposals. We shall discuss the most widely accepted methods. These methods can be grouped into the following categories: Traditional or non-discounted cash flow methods and Time adjusted methods or discounted cash flow methods. Net present value and profitability index are part of the time adjusted methods. This is where we also apply the concept of time value of money. Take your time and read, write down questions that you think need answers and ask them during your face-to-face sessions. We can see you smiling. That's a good start.

### Objectives

By the end of this session, you should be able to:

(a) explain net present value method and its advantages and disadvantages
(b) calculate the net present value for a single project and for a multiple project
(c) explain profitability index and its advantages
(d) calculate the profitability index for a single project and for a multiple project
(e) explain the decision rules on net present value and profitability index

Now read on …

### 5.1 Net Present Value (NPV)

The net present value decision tool is a more common and more effective process of evaluating a project. The NPV tool is effective because it uses discounted cash flow analysis, where future cash flows are discounted at a discount rate to compensate for the uncertainty of those future cash flows. Independent projects are accepted when NPV is positive and rejected when NPV is negative. In the case of mutually exclusive projects, the project with the highest NPV is accepted. Despite a strong academic preference for NPV, surveys indicate that executives prefer IRR over NPV although they should be used in concert. In a budget-constrained environment, efficiency measures should be used to maximize the overall NPV of the firm.

From a purely financial viewpoint, the NPV rule is consistent with the shareholder's objective of wealth maximization, because it exclusively uses cash flows in the calculations as well as considers the time value of money. It evaluates investment projects in the way as investors do (Shapiro, 2005). The NPV has several strengths and weaknesses. Though the method has a number of strength (Brealey, 2006, Shapiro, 2005, Ansari, 2000) the concept may be hardly understood due to its complexity (Shapiro, 2005). Selecting a suitable discount rate based on assumptions about a potential investment and considering an investment's risk may be difficult to comprehend for individuals without any financial training, background or experience (Ansari, 2000). The model gives a false sense of accuracy, since the computed present value is based on estimated and uncertain cash flows (Ansari, 2000).

**5.2 Calculating NPV of a project**

Net Present Value (NPV) is the difference between the present value of cash inflows and the present value of cash outflows over a period of time. It is used in capital budgeting and investment planning to analyse the profitability of a projected investment or project. The concept is similar to Intrinsic Value of a security but subtracts cost of the project. A positive net present value shows that the projected earnings generated by a project exceeds the initial costs of the project. It is assumed that an investment with a positive NPV will be profitable while an investment with a negative NPV will result in a loss.

NPV = PV of Inflows - Initial Outlay

$$\text{NPV} = \left[ \frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \ldots + \frac{CF_n}{(1+r)^n} \right] - I_0$$

Where:

NPV = Net Present Value
CF = Future Cash Inflows at different times
r = Cost of Capital or Cut-off rate or Discounting Rate
$I_0$ =Cash outflows at different times

**Example 1**

Calculate the Net Present Value of the following project requiring an initial cash outlay of GHS20,000 and has a no scrap value after 6 years. The net profits after depreciation and taxes for each year of GHS6,000 for six years. Assume the present value of an annuity of GHS1 for 6 years at 8% p.a. interest is 4.623.

Solution:

Calculation of Net Present Value

Initial cash outlay = GHS20,000
Present value of cash inflows = GHS6,000 x 4.623
= GHS27,738
Net Present Value = Present Value of Cash Inflows – Initial Cash Outlay
= GHS27,738 – GHS20,000
= GHS7,738
Net Present Value = GHS7,738

**Example 2**

An investor has GHS10,000 to under a project. He has been presented with two possible projects and their expected cash inflows. Assuming that the rate of return is 15%, calculate the NPV for the projects and decide on which one to invest in.

Project A Project B

| | | |
|---|---|---|
| Year 1 | GHS2,500 | GHS3,750 |
| Year 2 | 2,000 | 3,750 |
| Year 3 | 1,875 | 2,500 |
| Year 4 | 1,250 | 2,250 |
| Year 5 | 3000 | 2,250 |

Solution

| YEAR | CASHFLOW (PROJECT A) | CASHFLOW (PROJECT B) | DISC. FACTORS (15%) | PRESENT VALUE (PROJECT A) | PRESENT VALUE (PROJECT B) |
|---|---|---|---|---|---|
| 0 | (10,000.00) | (10,000.00) | 1.0000 | (10,000.00) | (10,000.00) |
| 1 | 2,500.00 | 3,750.00 | 0.8696 | 2,173.91 | 3,260.87 |
| 2 | 2,000.00 | 3,750.00 | 0.7561 | 1,512.29 | 2,835.54 |
| 3 | 1,875.00 | 2,500.00 | 0.6575 | 1,232.84 | 1,643.79 |
| 4 | 1,250.00 | 2,250.00 | 0.5718 | 714.69 | 1,286.44 |
| 5 | 3,000.00 | 2,250.00 | 0.4972 | 1,491.53 | 1,118.65 |
| | | | | **(2,874.73)** | **145.29** |

The NPV of Project A is (GHS2,874.73) and Project B is GHS145.29. If the projects are mutually exclusive projects, then the investor should undertake Project B because it has positive NPV which means that he will make some amount of returns.
You can equally use the formula given above. You will arrive at the same answer just that it is time consuming.

$$NPV = [\frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \frac{CF_4}{(1+r)^5} + \frac{CF_5}{(1+r)^5}] - I_0$$

$$NPV_A = [\frac{2,500}{(1+0.15)} + \frac{2,000}{(1+0.15)^2} + \frac{1,875}{(1+0.15)^3} + \frac{1,250}{(1+0.15)^5} + \frac{3,000}{(1+0.15)^5}] - I_0$$

$$NPV_A = [\frac{2,500}{(1.15)} + \frac{2,000}{(1.15)^2} + \frac{1,875}{(1.15)^3} + \frac{1,250}{(1.15)^5} + \frac{3,000}{(1.15)^5}] - I_0$$

$NPV_A$ = [ 3,260.87 + 2,835.54 + 1,643.79 + 1,286.44 + 1,118.65 ] – 10,000

$NPV_A$ = -2874.72

Using the same approach, calculate the NPV of Project B ($NPV_B$).

You noticed that the discount factor was from the Present Value Table. If the expected rate of return is different, you will have different discount factors all from the Table. This was

discussed extensively under time value of money. What happens if you do not have the Table? As students, we may sometimes forget the Tables at home but will remember to take our calculators with us. When this happens, we can use our calculators in computing our discount factor. In doing so we need to remember this:

$$\text{d. f. period 1} = \frac{1}{1+r}$$

The denominator "1+ r" will be the same throughout the period. For our question r = 15% = 0.15. The discount factor for period 1 = $\frac{1}{1+0.15} = \frac{1}{1.15}$ = 0.8696. The discount factor for period 2; we divided the discount factor for period 1 by 1.15. Discount Factor for period 2 = $\frac{0.8696}{1.15}$ = 0.7561. Discount Factor for period 3 = $\frac{0.7561}{1.15}$ = 0.6575. Now, wait, all you need to do is to press the equal to sign for the next value, press it again for the next, press again for the next. Excellent. You got it right. This means that without the Table, you can use your calculator to find the discount factor. But always remember that the discount factor for period 0 = 1.0000. This 1 is what is the numerator you find above.

**5.3 Rules of Acceptance**

If the rate of return from a project is greater than the return from an equivalent risk investment
in securities traded in the financial market, the Net Present Value will be positive. Alternatively, if the rate of return is lower, the Net Present Value will be negative. In other words, if a project has a positive Net Present Value it is considered to be viable because the present value of the inflows exceeds the present value of the outflows. If the projects are to be ranked or the decision is to select one or another. the project with the greatest Net Present Value should be chosen. Symbolically the accept or reject criterion can be expressed as follows:

Where

NPV > Zero Accept the proposal
NPV < Zero Reject the Proposal

**5.4 Advantages**

- It recognizes the time value of money.
- It considers the total benefits arising out of the proposal.
- It is the best method for the selection of mutually exclusive projects.
- It helps to achieve the maximization of shareholders' wealth.

**5.5 Disadvantages**

- The NPV approach is **difficult to apply when comparing projects that have different life spans.**
- The NPV method is not applicable when comparing projects that have differing investment amounts.

**5.6 Profitability Index Method**

Profitability Index is also known as Benefit Cost Ratio. It gives the present value of future benefits, computed at the required rate of return on the initial investment. Profitability Index may either be Gross Profitability Index or Net Profitability Index. Net Profitability Index is the Gross Profitability Index minus one. The Profitability Index can be calculated by the following equation:

$$\text{Profitability Index} = \frac{Present\ Value\ of\ Cash\ Inflows}{Initial\ Cash\ Outlay}$$

Or

$$\text{Profitability Index} = \frac{Net\ Present\ Value}{Initial\ Cash\ Outlay} + 1$$

**Example 3**

AnnSey Limited is considering the purchase of a new photocopier machine to increase its production. The company has the policy of charging depreciation on straight-line basis. Assuming the company's weighted cost of capital (WACC) is 10% and a tax rate of 20%. The following information relates to the project.

| | |
|---|---|
| Initial cash outlay | GHS100,000 |
| Estimated useful life | 5 years |
| Residual value | GHS10,000 |

Profit before depreciation and tax:

| Year | GHS |
|---|---|
| 1 | 26,000 |
| 2 | 19,000 |
| 3 | 24,000 |
| 4 | 20,000 |
| 5 | 2,000 |

Calculate the Profitability Index and advise the company whether to buy the machine.

Solution

Depreciation on straight-line method

$$\text{Depreciation} = \frac{cost - residual\ value}{estimated\ useful\ life} = \frac{100{,}000 - 10{,}000}{5} = \text{GHS18, 000}$$

The scrap value will be added to the last year inflow because it is an income to be recouped in the year.

| YEARS | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Profit before depreciation and tax | 26,000.00 | 19,000.00 | 24,000.00 | 20,000.00 | 12,000.00 |
| Depreciation | (18,000.00) | (18,000.00) | (18,000.00) | (18,000.00) | (18,000.00) |
| Profit before a tax | 8,000.00 | 1,000.00 | 6,000.00 | 2,000.00 | (6,000.00) |
| Tax (20%) | (1,600.00) | (200.00) | (1,200.00) | (400.00) | - |
| Profit after tax | 6,400.00 | 800.00 | 4,800.00 | 1,600.00 | (6,000.00) |
| Depreciation | 18,000.00 | 18,000.00 | 18,000.00 | 18,000.00 | 18,000.00 |
| Cash flow | 24,400.00 | 18,800.00 | 22,800.00 | 19,600.00 | 12,000.00 |

| Discount factors (10%) | 0.9091 | 0.8264 | 0.7513 | 0.6830 | 0.6209 |
|---|---|---|---|---|---|
| Present value | **22,181.82** | **15,537.19** | **17,129.98** | **13,387.06** | **7,451.06** |

Total Present Value = GHS22,181.82 + 15,537.19 + 17,129.98 + 13,387.06 + 7,451.06
= GHS75,687.11

Profitability Index $= \frac{Present\ Value\ of\ Cash\ Inflows}{Initial\ Cash\ Outlay}$

Profitability Index $= \frac{GHS75{,}687.11}{GHS100{,}000} = 0.76$

Since the profitability index is less than 1, the photocopier should not be purchased.

Using the second formula:

Profitability Index $= \frac{Net\ Present\ Value}{Initial\ Cash\ Outlay} + 1$

Profitability Index $= \frac{GHS75{,}687.11 - GHS100{,}000}{GHS100{,}000} + 1$

Profitability Index $= \frac{-24{,}312.89}{GHS100{,}000} + 1$

Profitability Index $= -0.24 + 1 = 0.76$

The Profitability Index indicates less than one, the project is not beneficial and should not be accepted.

**5.7 Rule of Acceptance**

As per the Benefit Cost Ratio or Profitability Index a project with Profitability Index greater than one should be accepted as it will have Positive Net Present Value. Likewise, if Profitability Index is less than one the project is not beneficial and should not be accepted. If the Profitability Index is equal to 1, the project breaks even and the company is indifferent between proceeding and not proceeding with the project.

**5.8 Advantages of Profitability Index**

- It duly recognizes the time value of money.
- For calculations when compared with internal rate of return method it requires less time.
- It helps in ranking the project for investment decisions.
- As this method is capable of calculating incremental benefit cost ratio, it can be used to choose between mutually exclusive projects.

**Summary**

In this session, we examined net present value and profitability index. These are methods of appraising investment projects using the concept of time value of money. For NPV we said we should accept a project with a positive value and reject a project with a negative value and if the projects are mutually exclusive, we accept the project with the highest NPV. In the case

of profitability index, we concluded that projects with profitability index greater than one should be accepted and those with profitability index of less than one should be rejected. You did well. Shine for yourself.

**Self-Assessment Questions**
**Exercise 5.5**

## SESSION 6 APPRAISAL TECHNIQUES: INTERNAL RATE OF RETURN AND MODIFIED INTERNAL RATE OF RETURN

### Introduction

There are number of appraisal methods which may be recommended for evaluating the capital
investment proposals. We shall discuss the most widely accepted methods. These methods can be grouped into the following categories: Traditional or non-discounted cash flow methods and Time adjusted methods or discounted cash flow methods. Internal rate of return and modified internal rate of return are part of the time adjusted methods. They also apply the concept of time value of money. Take your time and read, write down questions that you think need answers and ask them during your face-to-face sessions. We can see you smiling. That's a good start.

### Objectives

By the end of this session, you should be able to:

(a) explain internal rate of return and its advantages and disadvantages
(b) calculate the internal rate of return for a single project and for a multiple project
(c) explain modified internal rate of return
(d) calculate the modified internal rate of return for a single project and for a multiple project

(e) explain the decision rules on IRR and MIRR

Now read on …

**6.1 Internal Rate of Return (IRR) Method**

Internal rate of return is defined as the discounted rate that is commonly used to determine how much of a return an investor can expect to realize from a particular project. The internal rate of return is the discount rate that occurs when a project break-even, or when the NPV equals 0 and the decision rule is to choose the project where the IRR is higher than the cost of financing. The greater the difference between the financing cost and the IRR, the more attractive the project becomes. Brealey (2006) states that internal rate of return rule is to accept an investment project if the opportunity cost of capital is less than the internal rate of return. The rationale behind this assertion is that an investment project yielding more than its opportunity cost of capital has a positive NPV, thus it is worthwhile investing.

It's possible that two mutually exclusive projects can have conflicting IRRs and NPVs, meaning that one project has lower IRR but higher NPV than another project. These issues can arise when initial investments between two projects are not equal. Just as it is the case with the NPV, one of the disadvantages of the IRR is the fact that the model gives a false sense of accuracy, since the computed present value is based on estimated and uncertain cash flows.

**6.2 Calculating the IRR of a project**

IRR is the rate which when applied to discount the cash flow makes the Net Present Value equal to zero. So, IRR of a project will be:

$$\text{IRR} = a + \frac{NPV_a}{NPV_a - NPV_b}(b - a)$$

Where

a = the discount rate given by the company

b = the discount rate that is mostly assumed.

$NPV_a$ = Net present value using discount rate 'a'

$NPV_b$ = Net present value using discount rate 'b'

This is what is called the interpolation formula since we use trial and error.

**Example 1**

Assume a project costs GHS2,000 today and pays GHS2,200 in one year. The return on this investment is 10%, for every dollar invested yields GHS1.10 in return. The 10% is the IRR of the investment.

From the above,

Since IRR is the rate at which NPV is zero, then will have

NPV = PV of inflow – PV of outflow

$$0 = \frac{2200}{(1+r)} - 2000$$

$$2000 = \frac{2200}{(1+r)}$$

$$1 + r = \frac{2200}{2000}$$

1 + r = 1.1
r = 0.1
r = 10%

**Example 2**
An investor has GHS10,000 to under a project. He has been presented with two possible projects and their expected cash inflows. Assuming that the rate of return is 10%, calculate the IRR for the projects and decide on which one to invest in.

| | Project A | Project B |
|---|---|---|
| Year 1 | GHS2,500 | GHS3,750 |
| Year 2 | 2000 | 3,750 |
| Year 3 | 1,875 | 2,500 |
| Year 4 | 1,250 | 2,250 |
| Year 5 | 3000 | 2,250 |

Solution
Present value of Project A and Project B using 10%

| YEAR | CASHFLOW (PROJECT A) | CASHFLOW (PROJECT B) | DISC. FACTORS (10%) | PRESENT VALUE (PROJECT A) | PRESENT VALUE (PROJECT B) |
|---|---|---|---|---|---|
| 0 | (10,000.00) | (10,000.00) | 1.0000 | (10,000.00) | (10,000.00) |
| 1 | 2,500.00 | 3,750.00 | 0.9091 | 2,272.73 | 3,409.09 |
| 2 | 2,000.00 | 3,750.00 | 0.8264 | 1,652.89 | 3,099.17 |
| 3 | 1,875.00 | 2,500.00 | 0.7513 | 1,408.72 | 1,878.29 |
| 4 | 1,250.00 | 2,250.00 | 0.6830 | 853.77 | 1,536.78 |
| 5 | 3,000.00 | 2,250.00 | 0.6209 | 1,862.76 | 1,397.07 |
| | | | | **(1,949.13)** | **1,320.40** |

Present value of Project A and Project B using 12%

| YEAR | CASHFLOW (PROJECT A) | CASHFLOW (PROJECT B) | DISC. FACTOR (12%) | PRESENT VALUE (PROJECT A) | PRESENT VALUE (PROJECT B) |
|---|---|---|---|---|---|
| 0 | (10,000.00) | (10,000.00) | 1.0000 | (10,000.00) | (10,000.00) |
| 1 | 2,500.00 | 3,750.00 | 0.8929 | 2,232.14 | 3,348.21 |
| 2 | 2,000.00 | 3,750.00 | 0.7972 | 1,594.39 | 2,989.48 |

| | | | | | |
|---|---|---|---|---|---|
| 3 | 1,875.00 | 2,500.00 | 0.7118 | 1,334.59 | 1,779.45 |
| 4 | 1,250.00 | 2,250.00 | 0.6355 | 794.40 | 1,429.92 |
| 5 | 3,000.00 | 2,250.00 | 0.5674 | 1,702.28 | 1,276.71 |
| | | | | **(2,342.20)** | **823.77** |

IRR = a + $\frac{NPV_a}{NPV_a - NPV_b}$ $(b - a)$

Internal Rate of Return for Project A ($IRR_A$)

$IRR_A$ = $10\% + \frac{-1,949.13}{-1,949.12 - (-2,342.20)}$ $(12\% - 10\%)$

$IRR_A$ = $10 + \frac{-1,949.13}{-1,949.12 + 2,342.20}$ $(12 - 10)$

$IRR_A$ = $10 + \frac{-1,949.13}{393.07}$ $(2)$

$IRR_A$ = $10 + (-4.9587\ x\ 2)$

$IRR_A$ = $10 + (-9.9174)$

$IRR_A$ = $10 - 9.9174$

$IRR_A$ = $0.083\%$

Internal Rate of Return for Project B ($IRR_B$)

$IRR_A$ = $10\% + \frac{1,320.40}{1,320.40 - 823.77}$ $(12\% - 10\%)$

$IRR_A$ = $10 + \frac{1,320.40}{496.63}$ $(12 - 10)$

$IRR_A$ = $10 + (2.6587\ x\ 2)$

$IRR_A$ = $10 + 5.3174$

$IRR_A$ = $15.3174\%$

$IRR_A$ = $15.32\%$

The investor should invest in project B because it has IRR greater than the required rate of return of the company.

**6.3 Decision rule**

If the IRR of a new project exceeds a company's required rate of return, that project is desirable so should be accepted. On the other hand, if IRR falls below the required rate of return, the project should be rejected. If the projects are mutually exclusive, accept the project with the highest IRR.

**6.4 Merits of IRR**

- It considers the time value of money principle.
- It takes into account the total cash inflow and outflow.
- It gives the approximate/nearest rate of return.

**6.5 Demerits of IRR**

- It involves complicated computational method.
- It is assumed that all intermediate cash flows are reinvested at the internal rate of return.
- It produces multiple rates which may be confusing for taking decisions.

**6.6 Modified Internal Rate of Return (MIRR) Method**

The modified internal rate of return (MIRR) assumes that positive cash flows are reinvested at the firm's cost of capital and that the initial outlays are financed at the firm's financing cost. By contrast, the traditional IRR assumes the cash flow from a project are reinvested at the IRR itself. The MIRR, therefore more accurately reflects the cost and profitability of a project.

The MIRR is used to rank investments or projects of unequal size. The first main problem with IRR is that multiple solutions can be found for the same project. The second problem is that the assumption that positive cash flows are reinvested at the IRR is considered impractical in practice. With the MIRR, only a single solution exists for a given project, and the reinvestment rate of positive cash flows is much more valid in practice.

**6.7 Illustration on MIRR**

Let's take this example to explain it more.

A project with an initial investment of GHS1,000 offers cash returns of GHS500, GHS600, and GHS400 at the end of years one, two and three respectively. The company's cost of capital is 10%. Calculate the MIRR.

**Solution**

Method 1

The first thing we need to do is to find the future value of the cash flow, thus compounding each cash flow forward to the end of the project using the firm's cost of capital. That is

$FV_1 = 500(1+0.1)^2$ $\qquad$ $FV_2 = 600(1+0.1)^1$

$= GHS605$ $\qquad$ $= GHS660$

The next thing is to add these values to get the modified cash flow:

Modified cash flow = GHS605 + GHS660 + GHS400

= GHS1,665

We also have to calculate the discount rate required to set this value when discounted equal to the outlay. To do this we need to use the formula below:

$$\text{Outlay} = \frac{modified\ cash\ flow}{(1+MIRR)^n}$$

From this formula, we can deduce that the MIRR is equal to the nth root of modified cash flow divided by the initial outlay.

$$\text{MIRR} = \sqrt[n]{\frac{modified\ cash\ flow}{outlay}} - 1$$

$$\text{MIRR} = \sqrt[3]{\frac{1665}{1000}} - 1$$

$= 1.1852 – 1$

$= 0.1852$

$= 18.52\%$

Method 2

This method is much more straightforward and employs a simple formula which is quick and easy to apply:

$$\text{MIRR} = \sqrt[n]{\left(\frac{PV}{Outlay}\right)} \times (1 + r) - 1$$

Take the present value (PV) of the project cash flows from the recovery phase divide by the outlay and take the 'n th' root of the result. Multiply the result by one plus the cost of capital deduct one and you have the answer.

Same question

$$PV = \frac{500}{(1+0.1)^1} + \frac{600}{(1+0.1)^2} + \frac{400}{(1+0.1)^3}$$

$= 1250.9391$

$$\text{MIRR} = \sqrt[3]{\left(\frac{1250.9391}{1000}\right)} \times (1 + 0.1) – 1$$

$= 1.1852 -1$

$= 0.1852$

$= 18.52\%$

**Summary**
In this session, we examined IRR and MIRR. We concluded that MIRR is useful for evaluating two projects with different time periods. The acceptance rule is that you accept the project with the highest IRR if they are mutually exclusive. We also looked at the merits and demerits of IRR. We encourage you to practice these issues discussed and learn how they are used in industry.

**Self-Assessment Questions**
**Exercise 5.6**

## UNIT 6: WORKING CAPITAL MANAGEMENT AND COST OF CAPITAL

**Unit Outline**
Session 1: Introduction and Working Capital Cycle
Session 2: Cash Management
Session 3: Receivables and Inventory Management

Session 4: Payables Management
Session 5: Cost of Individual Capital
Session 6: Weighted Average Cost of Capital

**Overview**

This is the final unit of Business Finance and you are welcome to this unit. We need to congratulate you for a good work done so far. In this unit, session one deals with the introduction to working capital management and working capital cycle. Session 2 looks at the management of cash and the various techniques used in managing cash in a company. In session 3, we examine receivables and inventory management and the models we use in management inventory. Since a business cannot do without credit, we take you through the management of payables in session 4; session 5 discusses the cost of individual capital such as preference shares, ordinary shares and bonds and finally, in session 6 we show you how to compute the weighted average cost of capital for a business or a project. This promises to be interesting and we encourage you to be resilient and never give up.

**Objectives**

By the end of this unit, you should be able to:

(a) define working capital and working capital management
(b) outline the objectives and importance of working capital management
(c) discuss the determinants of working capital management
(d) compute working capital cycles and working capital needs of a business
(a) state and explain the decisions in cash management
(b) discuss the reasons for holding cash and costs associated with holding cash
(c) determine the cash management techniques
(d) describe the models used for management cash and other marketable securities
(e) understand marketable securities and the different types available to a firm
(f) discuss how to manage inventories and receivables in an organization
(g) evaluate the management of accounts payables
(h) describe the cost of preference shares, ordinary shares and bonds
(i) determine WACC for a project or business organization

## SESSION 1: INTRODUCTION AND WORKING CAPITAL CYCLE

**Introduction**

We welcome you dearly to the first session of the last Unit in Business Finance. In the first session in Unit 1, we spoke and learned about decision areas in finance. One of such decision

areas is working capital management. In this session, we define working capital and show how we can calculate the working capital cycle of the business and the working capital needs of the company. Have a fruitful study time.

**Objectives**

By the end of this session, you should be able to:

(e) define working capital and working capital management
(f) outline the objectives of working capital management
(g) understand the importance of working capital management
(h) discuss the determinants of working capital management
(i) compute working capital cycles and working capital needs of a business

Now read on …

**1.1 Management of Capital**

This refers to the finance and investment of non-human resources that is, physical and monetary assets for the purpose of maximum benefit in terms of profitability (Hill, 1993).

**1.2 Definition of Working Capital**

A firm's total investment in current assets irrespective of financing sources. Working capital is the sum of a company's investment in current assets – primarily cash and cash equivalents, accounts receivable; and inventories. A firm's current assets (assets that can be converted into cash within a year) less its current liabilities (obligations due within a year) is the working capital of that firm.

Another related term is net working capital or circulating capital. This concept indicates the liquidity position of the firm and the extent to which working capital needs may be financed by permanent sources of funds. Pandey (1991) explains that working capital is defined as the sum of stocks, accounts receivable, cash and marketable securities. He focuses attention on two aspects of current asset management:

- Optimum investment in current assets;
- Financing of current assets.

**1.3 Working Capital Management**

Archer et al (1983) defines working capital management as involving managing the level and mix of current assets and current liabilities and long-term sources. Van Horne (1988) indicates that working capital management involves the administration of current assets and the financing (especially of current liabilities needed to support current assets.

Working capital management concerns itself with the short-term investment and finance decisions of a firm and has the goal of enabling a firm maintain adequate cash flow to meet its short-term operating costs and short-term obligations. It ensures efficiency by monitoring and using a firm's current assets and liabilities.

**1.4 Objectives of working capital management**

The main objectives of working capital management are:

- To minimize the cost of capital
- Maximize the return on current asset investment
- Maintain the working capital operating cycle

## 1.5 Importance of Working Capital Management

- An appropriate mix of working capital management is needed to maintain a firm's liquidity which is important so the firm can pay off its liabilities as they become due and avoid all the issues that come with insolvency.
- It is also important because the amount of working capital affects a firm's profitability because current assets must be financed and financing them costs money.
- Working capital management directly affects the firm's long-term growth and survival. This is because production and sales growth are supported by higher levels of working capital.
- Working capital forms, a large portion of the firm's total assets. The level of working capital depends on the type of industry. Firms in the manufacturing and retailing industry often keep more than half of their total assets in current assets.

## 1.6 Determinants of Working Capital Management

### 1.6.1 General nature and size of the business

How a business is conducted basically determines its working capital requirements. Thus, the general nature and size of a business influences the amount of working capital needed in the business. For instance, although manufacturing concerns invest in non-current assets as well as current assets, trading concerns have to maintain sufficient amount of cash, inventories and accounts receivables when compared with the former.

The credit policy of a firm or its terms of trade also influences its working capital needs. The more liberal and the more credit terms offered to customers, the longer the funds that are fixed up in accounts receivables.

### 1.6.2 Price level changes

Generally, rising price levels will require a firm to maintain higher amounts of working capital.
Under such situations, the same level of current assets will need increased investment when prices are increasing.

### 1.6.3 Growth and expansion activities

All other things being equal, industries that are static require less working capital than those that are growth industries. As businesses expand, its working capital requirements increase although not necessarily in proportion. However, the critical fact is that, the need for increase working capital funds does not follow the growth in business activities but precedes it. Advanced planning of working capital is therefore a continuing necessity for a growing concern. If this is not done, the company may have substantial earnings but little cash.

### 1.6.4 Availability of credit facilities from suppliers

The working capital requirements of a firm can also be influenced by the credit terms granted by its creditors. A firm will need less working capital if liberal credit terms are available to it. Similarly, the availability of credit from banks also influences the working capital needs of a firm. A firm which can get credit from a bank easily and on favourable conditions will operate with less working capital than a firm without such a facility.

### 1.6.5 Production Policy

If there are sharp variations in a firm's working capital because of seasonal variations, a firm may pursue a production policy which may reduce these sharp variations in working capital requirements. For example, a manufacturer of ceiling fans may maintain a steady production

activity during the peak business season. Such a production policy may dampen the fluctuations in working capital requirements.

### 1.6.6 Market conditions

The level of competition in the market also has a bearing on the working capital needs of the firm. When competition is keen, a larger inventory of finished goods is required to promptly serve customers who may not be inclined to wait because other manufacturers are ready to meet their needs.

Also, generous credit terms may have to be offered to attract customers in a highly competitive market. Thus, working capital needs tends to be high because of greater investment in finished goods inventory and accounts receivable.
If the market is strong and competition is weak, a firm can manage with a smaller inventory of finished goods because customers can be served with some delay. Also, under such circumstances, a firm can insist on cash payment and avoid locked up funds in accounts receivable.

### 1.6.7 Conditions of supply

The inventory of raw materials, spares, and stores depends on the conditions of supply. If the supply is prompt and adequate, the firm can manage with small inventory. On the other hand, if supply is unpredictable and small, then the firm will have to acquire inventories as and when they are available and carry larger inventory on an average.

## 1.7 Approaches to Working Capital Management

There are different approaches to working capital management. Financial ratio Analysis is used to monitor overall trends in working capital and to identify areas requiring closer management. Individual components of working capital can be effectively managed by using various techniques and strategies.

### 1.7.1 Financial Ratio Analysis

This is the process of calculating and comparing various ratios of amounts and balances taken from the financial statement. The main purposes of working capital ratio analysis are:

- To indicate working capital management performance
- To assist in identifying areas requiring closer management.

The following ratios are relevant the management of working capital;

a. Working Capital/Liquidity ratios (i.e. current and quick ratios)
b. Liquid interval measure
c. Stock turnover
d. Debtors ratio
e. Creditors ratio

**Summary Ratios**

Current Ratio = $\frac{Current\ Assets}{Current\ Liabilities}$

Acid Test Ratio = $\frac{Current\ Assets - Inventories}{Current\ Liabilities}$

Liquid Interval Measure = $\frac{Liquid\ Assets}{Average\ Operating\ Expenses}$

Stock turnover = $\frac{Cost\ of\ Sales}{Average\ Stock\ Level}$

Debtors collection period= $\frac{Average\ Trade\ Debtors\ (Accounts\ Receivables)}{Total\ Credit\ Sales}\ x\ 365\ days$

Creditors payment period= $\frac{Average\ Trade\ Creditors\ (Accounts\ Payables)}{Total\ Credit\ Purchases}\ x\ 365\ days$

**1.7.2 Working Capital Cycle**

Also known as Cash Operating Cycle, it refers to the length of time between the purchase of raw materials and the sale of finished goods. i.e. the length of time it takes to turn around stock, collect monies in respect of credit sales and finally settle all trade suppliers. The cycle starts when cash is converted to input like raw materials and labour and is completed when inventory and input is converted back to cash. Investment in working capital is influenced by 4 key elements/events in the production and sales cycle:

- Purchase of raw materials
- Payment of the purchase
- Sale of finished goods
- Collection of cash for the sales made

Cash conversion cycle is analysed as follows:

For a *manufacturing business*, it is shown as:

| | |
|---|---|
| Raw materials holding period | X |
| Payables payment period | (X) |
| Work-In-Progress holding period | X |
| Finished goods holding period | X |
| Receivables collection period | X |
| CASH CONVERSION CYCLE | X |

For a *retail or whole sale business*, it is shown as:

| | |
|---|---|
| Inventory holding period | X |
| Creditors payment period | (X) |
| Debtors collection period | X |
| CASH CONVERSION CYCLE | X |

**Illustration 1**

A Co. has the following estimated figures for the coming year;

| | |
|---|---|
| Sales | GH¢ 3.6 m |
| Average debtors | GH¢ 306,000 |
| Gross profit margin | 25% |
| Average creditors | GH¢130,000 |
| *Average stocks*: | |
| Finished goods | GH¢200,000 |
| WIP | GH¢350,000 |

Raw materials GH¢150,000

Stock levels are constant. What is the cash operating cycle?

Solution:

*W1*: Calculation of cost of sales:

Gross profit margin = 25% on sales

Gross profit = 25% * 3.6 m = GH¢900,000

Converting margin to mark-up,

Mark-up = $\frac{25}{100 - 25}$ = $\frac{25}{75} = \frac{1}{3}$ of cost of sales

⇨ 1/3 * CoS = GH¢900,000

CoS = GH¢900,000 * 3 = GH¢2.7 million

*W2*: Calculation of ratios:

Raw material holding period = $\frac{\text{average raw material}}{\text{Material usage or Cost of Sales}}$ x 365 days/52 weeks/12 months

=150,000/2,700,000 * 365 days = **20.28 days**

Creditors Payment period = $\frac{\text{Average Trade Creditors (Accounts payable)}}{\text{Credit Purchases or cost of sales}}$ × 365 days

= 130,000/2,700,000 *365days = **17.57 days**

WIP holding period $= \frac{Average\ WIP}{Production\ cost\ or\ Cost\ of\ Sales}\ x\ 365\ days$

WIP holding period $= \frac{350{,}000}{2{,}700{,}000}\ x\ 365\ days$

= **47.31 days**

Finished goods holding period = $\frac{Average\ Finished\ Goods}{Cost\ of\ Sales}\ x\ 365\ days$

Finished goods holding period = $\frac{200{,}000}{2{,}700{,}000}\ x\ 365\ days$

= **27.04 days**

Debtors Collection Period $= \frac{Average\ Trade\ Debtors}{Trade\ Credit\ Sales}\ x\ 365\ days$

Debtors Collection Period $= \frac{306{,}000}{3{,}600{,}000}\ x\ 365\ days$

= **31.03 days**

Finally, the cash conversion cycle is as follows:

| | |
|---|---|
| Raw materials holding period | 20.28 days |
| Payables payment period | (17.57 days) |
| Work-In-Progress holding period | 47.31 days |
| Finished goods holding period | 27.04 days |
| Receivables collection period | 31.03 days |
| CASH CONVERSION CYCLE | **108.90 days** |

Therefore, it takes the company 108.9 days to hold raw material, WIP, finished goods and turn around finished goods to generate cash to pay creditors.

**1.8 Working Capital Requirement/Investment Levels**

Sometimes, given various working capital component periods, one can generate the working capital required for operations. This is shown as follows:

| | |
|---|---|
| Current Assets (e.g. materials, WIP, debtors, etc.) | X |
| Current Liabilities (e.g. creditors, accruals, etc.) | (X) |
| WORKING CAPITAL REQUIRED | X |

Some *factors that affect working capital investment required* include:

- Nature of business
- Uncertainty in supplier deliveries
- Overall level of activity of business
- The company's credit policy
- The length of operating cycle

**Illustration 2**

XL Plc has the following expectations for the forthcoming period:

| | $'m |
|---|---|
| Sales | 10 |
| Materials | (6) |
| Other costs | (2) |
| Profit | 2 |

The following working capital ratios are expected to apply;

| | |
|---|---|
| Inventory days | 30 days |
| Receivables days | 60 days |
| Payables days | 40 days |

Required: Calculate the working capital required.

**Solution**

Inventory balance = $\frac{\text{Inventory days}}{365 \text{ days}}$ x Materials usage = 30 days/365 days x $6 m = **$ 0.49 m**

Receivables Balance = $\frac{\text{Receivables days}}{365 \text{ days}}$ x Sales = 60 days/365 days x $10 m = **$ 1.64 m**

Payables Balance = $\frac{\text{Payables days}}{365 \text{ days}}$ x Materials usage = 40 days/365 days x $6 m = **$ 0.66 m**

Working Capital requirement is therefore shown as:

| | |
|---|---|
| Inventory balance | $ 0.49 m |

| | |
|---|---|
| Receivables Balance | $ 1.64 m |
| Current Liabilities (e.g. creditors, accruals, etc.) | ($ 0.66 m) |
| WORKING CAPITAL REQUIRED | **$ 1.47 m** |

Therefore, $ 1,470,000 investment in working capital is required for operations.

**Summary**

In this session, we examined working capital management and the determinants of working capital of a business entity. We also looked at the objectives and importance of working capital management and calculated the working capital needs of the company. This should aid our process when we are sourcing for funding for our businesses. Remember the topic on capitalization? This will ensure that you got the right type of capital needed for the business.

**Self-Assessment Questions**

**Exercise 6.1**

## SESSION 2: CASH MANAGEMENT

### Introduction

Good day dear students. Guess you have some amount of cash in your pocket, bag or wallet. This cash you have needs proper management. This session explain what cash management is and the decisions we make concerning cash. What are the reasons for keeping the money in your pocket, bag or wallet? This session provides you with the answers. We explain the techniques and models used when managing cash and other marketable securities.

### Objectives

By the end of this session, you should be able to:

(j) explain the meaning of cash management
(k) state and explain the decisions in cash management
(l) discuss the reasons for holding cash
(m) examine costs associated with holding cash
(n) determine the cash management techniques
(o) describe the models used for management cash and other marketable securities
(p) understand marketable securities and the different types available to a firm

Now read on …

### 2.1 Meaning of Cash Management

Cash management is the process of managing a firm's liquid assets. Liquid assets consist of cash and assets easily converted into cash, such as marketable securities. Cash is the amount of currency the firm has on hand, checks, and bank balances. Cash is the firm's most liquid asset. Marketable securities are generally short – term investment instruments used by the firm to earn a return on temporarily idle funds. Marketable securities represent near cash because they can be readily converted into cash.

The goal of cash management is to determine the target cash balance required to maintain liquidity while minimizing the total costs related to the investment in cash. Cash management requires making risk – return trade-offs. Excessive cash balances involve unnecessary costs that reduce profitability. Inadequate cash balances increase the risk of being unable to meet required payments because of possible cash shortages. Thus, cash management involves having the optimum amount of liquid assets on hand at the appropriate time.

Cash management involves managing the monies of the firm in order to attain maximum cash availability and maximum interest income on any idle funds. At one end, the function starts when a customer writes a check to pay the firm on its accounts receivable. The function ends when a supplier, an employee, or the government realizes collected funds from the firm on an account payable or accrual. All activities between these two points fall within the realm of cash management. The firm's efforts to make customers pay their bills at a certain time fall within accounts receivable management. On the other hand, the firm's decisions about when to pay its bills involve accounts payable and accrual management.

### 2.1.1 Cash management involves three major decision areas.

All three major decision areas are incorporated into the firm's cash budget. These areas are as follows:

1. *Managing collections and disbursements*: This step involves minimizing idle cash balances by accelerating receipts and slowing disbursements. Management can control many factors that cause time lags in cash flows, but some factors are beyond its control.
2. *Determining appropriate cash balances*: This decision involves determining the minimum levels of cash needed to provide liquidity while minimizing the total costs of holding an investment in cash. Several approaches including mathematical models are available for determining the firm's optimal cash balances.
3. *Investing in idle cash*: this step concerns developing strategies for managing marketable securities and selecting appropriate short – term investment instruments.

Cash and bank balances should be kept to a minimum, as they (usually) earn nothing for the firm, but care must be taken to ensure that activities are not restricted through a shortage of ready cash to pay employees and creditors. Finance must obviously be set aside to meet taxation liabilities, pay dividends and invest in capital expenditure. However, until such payments become due, the cash may be profitably invested in short term investments.
Managing cash usually involves the following issues:

- The preparation and use of cash budgets.
- The management of short-term cash investments.
- The management of overdrafts and bank loans.
- The use of cash management models.
- Evaluating whether to use a centralized treasury department.

### 2.1.2 Reasons for Holding Cash Balances

There are four main reasons for holding cash balances. These are explained below.

#### 2.1.2.1 Transactions balances

Transactions balances are cash balances that are held for conducting normal business activities. In other words, they are cash balances needed to meet expected cash flows. The firm's need for transactions balances is reduced by the availability of commercial credit.

#### 2.1.2.2 Precautionary balances

These are cash balances held to meet unexpected cash flows. Precautionary balances provide a cushion against the uncertain cash flows due to seasonal or cyclical operations and unexpected problems. Theses balances are normally held in temporary interest-bearing investments.

#### 2.1.2.3 Compensating balances

These are cash balances needed to compensate banks for providing loans and services. Some banks require firms to maintain a certain amount of money in their checking accounts in lieu of explicit payments to the lending bank. The bank then lends this money to earn a return for the services provided to the firm.

#### 2.1.2.4 Speculative balances

This is the least important reason for holding liquid assets. These are cash balances held to take advantage of certain profit-making activities that require funds on short notice.

#### 2.1.2.5 Other reasons for holding cash balances

- Interest earnings

Cash balances can in the short term be put in bank deposits or invested in government securities to earn interest. Loans can be made to subsidiaries, associated companies and other companies for interest.

- Investment

Firms with cash surplus can give more credit to customers thereby generating additional sales. Cash balances can also be used by firms to provide loans to customers and make available lease finance to increase sales. Firms with surplus cash balances can undertake additional investment in fixed assets and working capital or even acquire other companies for cash.

### 2.1.3 Cost of Holding Cash

These include loss of interest and loss of purchasing power: inflation erodes the value of cash.

**Cost of holding too little cash**

- Opportunities can be missed. Examples are contracts, buying another business, etc.
- Loss of discounts from suppliers by not having cash to pay early.
- Inability to cope with emergencies such as fire outbreaks, strikes, competitors' action, bad weather, etc.
- Higher cost of borrowing because unexpected cash needs have to be met from temporary borrowings rather than drawing on cash balances.
- Annoyance of those to whom payment is due if payments are delayed, will lead to reluctance to supply and eventually lead to liquidation.
- Regular payments have to be made to top up the cash balances, e.g. Transaction cost of selling securities to release cash and arrangement fees for overdrafts.

## 2.2 Cash Management Techniques

This involves minimizing idle cash by accelerating receipts and slowing disbursements. Managing collections and disbursements require a thorough understanding of the term float.

### 2.2.1 Float

Float is an amount of money represented by checks outstanding and in process of collection. It is measured in dollar days. Float time is the time lag or delay between the moments of disbursement of funds on the part of the customer and the moment of receipt of funds on the part of the firm (seller).

This time lag, also called check book float, results in a difference between the firm's cash balance as shown on the bank's books and the amount shown on the firm's own books.

### 2.2.2 Accelerating collections

The objective of accelerating collections is to increase the speed of payment receipt without incurring excessive costs. The major purpose of speeding collections is to free cash in order to reduce the firm's total financing requirement.
There are several ways to accelerate collections

- Bill promptly and accurately. Reduce time involved in the process of preparing invoices.
- Send invoices from a location near the buyer. The firm's business should be geographically disbursed.
- Use concentration banking. This involves the use of decentralized collections centres and local banks to speed up collection of payments.
- Change the paying behaviour of customers. A firm may encourage its customers to pay sooner by using letters, telephone calls or personal visits and by supplying self-addressed envelopes in which to pay their bills. Economic incentives such as cash discounts may also be resorted to.
- Use wire transfers. A wire transfer is a means of electronically sending money between banks through the Federal Reserve System or a private bank wire system. This eliminates mailing and check- clearing time associated with other funds transfers methods and is often used to transfer funds from collection banks to the concentration bank.

### 2.2.3 Slowing disbursements

The objective of slowing disbursements is to delay paying suppliers and other creditors as long as possible without jeopardizing the firm's credit standing or incurring any additional finance charges.
There are several ways to delay disbursements

- Delay payments to suppliers. Pay on time, not before or after debts are due.
- Use drafts to pay bills. A draft is an order for the payment issued by one party to another party. Unlike checks, time drafts are not payable on demand.
- Use remote disbursements. Remote disbursements lengthen the time it takes for a check to clear in the banking system.

## 2.3 Marketable Securities

Marketable securities are held as a substitute for cash and as a temporary investment to meet seasonal needs or to cover unpredictable financing needs. Idle funds are typically invested in marketable securities, such as Treasury bills and Commercial paper. An alternative to holding excess cash in marketable securities is to use short – term borrowing as a source of need cash balances. This strategy is generally regarded as riskier because the firm may encounter difficulties in borrowing funds or repaying loans. The firm could also invest idle funds in securities offering higher expected returns such as stocks and bonds. However, these securities are not suitable for near cash reserves due to their risks. The appropriate level of marketable securities depends on several factors which include:

1. Predictability of cash flows: Other factors being equal, firms with less predictability cash flows should maintain a higher level of investment in marketable securities.
2. Transaction costs: The ability of marketable securities to buffer against cash shortages diminishes as the cost of buying and selling securities increases.

3. Interest rates: Higher interest rates on marketable securities make holding cash costlier. Consequently, investment in marketable securities generally increases as short – term interest rates rise.

**2.3.1 Factors to consider in selecting marketable securities.**

- Marketability: This is the ability to sell a security quickly and with the minimum possibility of a loss. Marketability is of prime importance for temporary investments.
- Default risk: This is the risk that a debt's interest or principal will not be paid.
- Maturity: This is the length of time remaining before a security is due to be repaid. Maturity should be short to reduce:
    1. Credit risk: the risk that the issuing firm's financial position will worsen.
    2. Interest rate risk: the risk that a loss will be incurred on the sale of a security sold before the maturity date due to an increase in interest rates.
    3. Purchasing power risk: the risk that inflation will reduce the purchasing power of the sum of money to be repaid.
- Rate of return: This is the yield of a security. A security's return is related to its risk. Generally, highly marketable securities with short maturities and low default risk have low yields.
- Minimum purchase size: This is the lowest size of a security which can be purchased.

**2.3.2 Types of marketable securities**

Money market instruments are the most suitable form of investment for idle funds. The money market is the market for short –term debt instrument.

A money market instrument is a high-grade security characterized by a high degree of safety of principal and maturities of one year or less. Money market instruments are divided into two major types:

- Discount paper – a money market instrument which sells for less than its par or face value. The buyer receives the par value at the security's maturity. The difference between the security's purchase price and par value represents the buyer's return.
- Interest bearing securities – these are debt instruments which pay interest based on the security's face value. They include;

**2.3.2.1 Treasury bills**

This is a short-term government security with a maturity of one year or less, issued at a discount from face value. A treasury bill is often called a risk-free security. Treasury bills are characterized by their lack of default risk, high degree of marketability, and exemption from state income taxes.

**2.3.2.2 Commercial paper**

This is a short – term, unsecured obligation with maturities ranging from 2 to 270 days, issued by banks, corporations, and other borrowers. Marketability for commercial paper is poor, but a firm that issue paper directly are generally prepared to repurchase the paper prior to maturity. The default risk varies but is normally low. Commercial paper is usually discounted but it can be interest bearing.

**2.3.2.3 Negotiable certificate of deposit (NCD)**

This is a debt instrument issued by a bank that usually pays interest at maturity. The default risk varies with the issuing institution. It has maturities ranging from a few weeks to several years.

**2.3.2.4 Money market mutual fund**
This is an open – ended mutual fund that invests in money market instruments. Money market mutual funds sell shares to investors and then pool the funds to acquire money market instruments. These funds permit small investors to participate directly in high – yielding securities that often are available only on relatively large denominations.

**3.4 Cash Management Models**
Cash Management models are aimed at minimizing the total costs associated with movements between:

- Current account (very liquid but not earning interest)
- Short-term investment (less liquid but earning interest)

The models are developed to answer the questions: at what point should funds be moved? How much should be moved in one go? The two popular models employed in cash management are as follows;

**3.4.1 Baumol Model** (Early 1950s)
Baumol noted that cash balances are very similar to inventory levels, and developed a model based on EOQ. Assumptions of the model are:

- cash use is steady and predictable.
- Cash inflows are known and regular.
- Daily cash needs are funded from current account.
- Buffer cash is held short term investments.
- The main carrying or holding cost of cash is the interest foregone from not investing cash.

The following formula calculates the amount of funds to inject into the current account or to transfer into short-term investments at one time.

$$Q = \text{Optimal cash} = \sqrt{\left(\frac{2 \cdot C_T \cdot D}{C_H}\right)}$$

where $C_T$ is transaction costs (e.g. Commission, etc.)
$C_H$ is cost of holding cash (e.g. Interest foregone)
D is demand for cash over the period

**Illustration 1**
Monthly excess cash generated is \$ 10,000 to be invested in short-term investments. Expected return on the investment is 5% p.a. Transaction cost (constant) is \$50.

a. Optimal amount of cash to be invested in each transaction = $Q = \sqrt{\left(\frac{2 \cdot C_T \cdot D}{C_H}\right)}$

Where $C_T$=\$50, D = \$10000*12= \$120,000, and $C_H$=5%=0.05

$$= \sqrt{\frac{2*50*120000}{0.05}} = \textbf{\$ 15,491.93}$$

b. Number of transactions arising each year = $\frac{D}{Q} = \frac{\$\ 120{,}000}{\$\ 15{,}491.93} = 7.75 \approx$ **8 transactions**

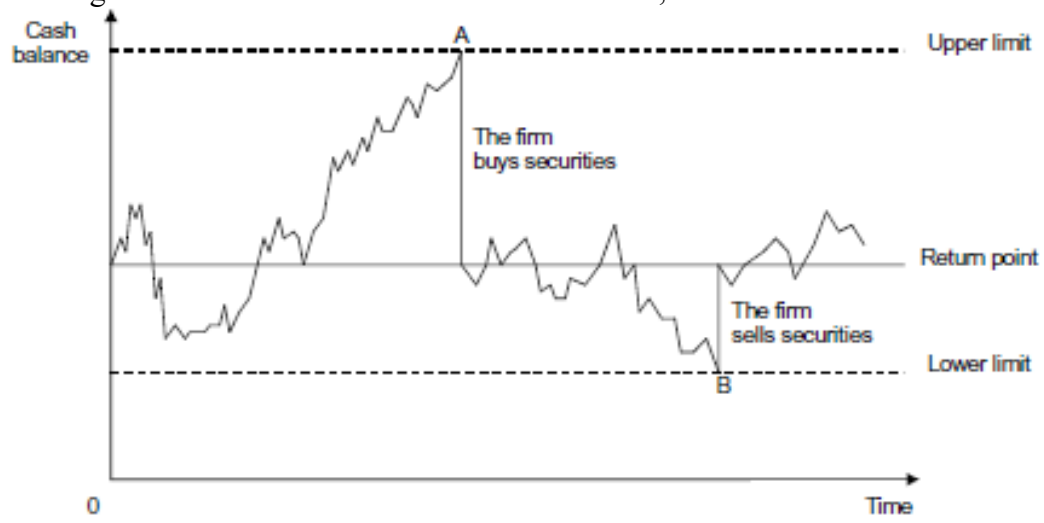
c. Cost of making 7.75 transactions p.a. = $C_T \cdot \frac{D}{Q}$ = \$ 50 (7.75) = **\$387.50**

d. Opportunity cost of holding cash = annual holding cost = $C_H.\frac{Q}{2}$ = 5% ($\frac{\$\ 15{,}491.93}{2}$) = **$ 387.30**

### 3.4.2 Miller-Orr Model

This model controls irregular movements of cash by setting upper and lower control limits on cash balances. The Miller-Orr model is used for setting the target cash balances. It has an advantage of incorporating uncertainty in the cash inflows and outflows.

The diagram shows how the model works over time;




The following is worth noting:

1. The model sets higher and lower control limits and a target cash balance (return point)
2. When cash balance hits the high limit, an amount equal to the difference between higher limit and return point is moved from cash to marketable securities (i.e. by buying marketable securities)
3. Conversely, when cash balance hits the lower limit, an amount equal to the difference between lower limit and return point is moved from marketable securities to cash (i.e. by selling marketable securities)

Formulas applicable are given as

$$Return\ point = Lower\ limit + \left(\frac{1}{3} \times spread\right)$$

The formula for the spread is:

$$Spread = 3\left(\frac{3}{4} \times \frac{transaction\ cost\ \times variance\ of\ cash\ flows}{interest\ rate}\right)^{\frac{1}{3}}$$

**NB:** The lower control limit is usually set by management based on how much risk of shortfall the firm is willing to accept. The upper limit is calculated by the model.

**Illustration 2**

The following data applies to a company.

1. The minimum cash balance is $8,000.

2. The variance of daily cash flows is 4,000,000, equivalent to a standard deviation of $2,000 per day.
3. The transaction cost for buying or selling securities is $50.
4. The interest rate is 0.025 per cent per day.
You are required to formulate a decision rule using the Miller-Orr model.

**Solution**
The spread between the upper and the lower cash balance limits is calculated as follows.

$$\text{Spread} = 3\left(\frac{3}{4} \times \frac{transaction\ cost \times variance\ of\ cashflows}{interest\ rate}\right)^{\frac{1}{3}}$$

$$= 3\left(\frac{3}{4} \times \frac{50 \times 4{,}000{,}000}{0.00025}\right)^{\frac{1}{3}}$$

$$= 3 \times (6 \times 10^{11})^{\frac{1}{3}}$$

$$= 3 \times 8{,}434.33$$

$$= \$\ 25{,}303\ \ say\ \$\ 25{,}300$$

The upper limit and return point are now calculated.

**Upper limit** = Lower limit + $25,300 = $8,000 + $25,300 = **$33,300**

**Return point** = lower limit + 1/3 x spread = $8,000 + 1/3 x $25,300 = $16,433, say **$16,400**

The decision rule is as follows.
If the cash balance reaches $33,300, buy $16,900 (= 33,300- 16,400) in marketable securities.
If the cash balance falls to $8,000, sell $8,400 of marketable securities for cash.

**Summary**
In this session, we discussed cash management, cash management techniques and the various reasons for holding cash. We also looked at marketable securities and the various marketable securities available to businesses. We looked at the Baumol and Miller-Orr Models of cash management. Sounds interesting right? Now we can management cash effectively when given the opportunity. That's our students. We are always proud of you.

**Self-Assessment Questions**
**Exercise 6.2**

## SESSION 3: RECEIVABLES AND INVENTORY MANAGEMENT

### Introduction

In accounting, inventories form part of the current assets of the business. The management of these inventories is a matter of concern for every finance manager. In this session, we explained what receivables management mean, what inventory management means and the reasons for holding inventory. At all times, a company need to hold some amount of inventory, hence in this session, we take you through how to calculate the various levels of inventory.

### Objectives

By the end of this session, you should be able to:

(a) explain receivables management
(b) outline the objectives of inventory management
(c) state the cost associated with holding inventory
(d) discuss the various models for inventory management
(e) compute the level of inventory in an organization

Now read on …

### 3.1 Receivables Management

Firms would, in general, rather sell for cash than on credit, but competitive pressures force most firms to offer credit. Firms offer credit in order to attract new customers and keep current customers.

Accounts receivable refers to money owed to a firm for goods or services sold on credit. This type of credit takes two basic forms:

- Trade commercial credit; this refers to credit that extends to other firms.
- Consumer or retail credit; credit which the firm extends to its final customers.

The level of accounts receivable depends on the volume of credit sales and the average collection period. Receivables are a component of a firm's working capital and represent investments. The goal of accounts receivable management is to ensure that firm's investment in accounts receivable is appropriate and contributes to shareholder wealth maximization. Managing accounts receivable consists of two major steps:

- Establishing credit policy
- Conducting credit analysis.

### 3.2 Inventory Management/Control

#### 3.2.1 Objectives of Inventory Control

- Reducing inventory to the lowest possible amount to minimize the level of capital employed to be funded (i.e. Liquidity)
- Ensuring that sufficient inventory is held so that it does not run out and disrupt business (i.e. Profitability)

#### 3.2.2 Reasons for holding inventory

The following are the reasons for holding inventory:

- To act a buffer in times of unusually high demand
- To avoid high costs of emergency orders
- To ensure continuity of production
- To take advantage of quantity discounts by ordering more at a time
- To reduce ordering costs by more items infrequently
- To hedge against seasonality of demand and supply

#### 3.2.3 Costs of holding inventory

1. Purchase costs/price
2. Holding Costs
    - cost of capital tied-up
    - Insurance
    - Risks of deterioration, obsolescence, and pilferage
    - Warehousing/storage costs
    - Stores administration costs
3. Systems Costs (i.e. people and computers)
4. Reorder/Set-up Costs:
    - Transport costs
    - Clerical and administrative expenses
    - Batch set-up costs for goods produced internally

5. Shortage Costs:
    - Production stoppages caused by lack of raw materials
    - Stock-out costs of finishes goods (i.e. any cost resulting from a delayed sale to a lost customer)
    - Emergency reorder costs

### 3.2.4 Models for Inventory Management

These are essentially models that establish stock Levels for Inventory Control. These stock levels include;

1. *Minimum Stock Level* ($L_{min}$):
   This is the level below which stock should not be allowed to fall. It is computed as;

   $L_{min}$ = RoL – (average usage x average lead time)

2. *Maximum stock Level* ($L_{max}$):
   This is a level above which stock must not be allowed to rise. It is computed as;

   $L_{max}$ = RoL + EOQ – (minimum usage x minimum lead time)

3. *Reorder Level* (RoL):
   This is the level of stock that prompts the reordering of a new batch. It is computed as;

   RoL = Maximum usage x Maximum lead time

4. *Economic Order Quantity* (EOQ):
   Also called *Reorder Quantity,* this is the quantity of stock at which the organization minimizes the total cost of holding and ordering inventory. i.e. it is the most economical quantity that equates total holding cost to total ordering cost. EOQ is computed as:

   $$EOQ = \sqrt{\left[\frac{2 \cdot C_o \cdot D}{C_H}\right]}$$

   where $C_o$ is cost per order
   $C_H$ is cost of holding one unit of stock per year
   D is annual demand for stock

   The *Total Cost (TC) of inventory* is measured as follows:

   TC = Purchase Cost + Annual Ordering Cost + Annual Holding Cost

   $$TC = P.D + \frac{C_o \cdot D}{Q} + \frac{C_H.Q}{2}$$

   Where P = purchase price per unit

   NB: For decision making purposes, the *Purchase Cost* component of TC may be ignored and only using the annual ordering and holding costs.

**Illustration 1**

The following data relates to an item of raw material BMX5:

| | |
|---|---|
| Cost of raw material | $10 |
| Usage per day | 100 units |
| Lead time | 20 to 30 days |

| | |
|---|---|
| Cost of ordering material | \$400 per order |
| Carrying costs | 10% per annum |

Additional Information:
Assume that each year consists of 48 working weeks of 5 days per week. You are required to calculate;

i. The re-order levels
ii. The re-order quantity
iii. The Maximum level
iv. The Minimum level
v. Number of Orders per year
vi. Total cost

**Solution**
$C_o$ =\$400
D = 100 units x 5 days x 48 weeks = 24,000 units
$C_H$= 10% x purchase price = 10% X \$10 = \$1 per unit p.a.

i. **EOQ** = $\sqrt{\left(\frac{2 \cdot C_o \cdot D}{C_H}\right)} = \sqrt{\left(\frac{2*\$400*24000}{\$1}\right)}$ = **4,382 units**

ii. **RoL** = Maximum usage x Maximum lead time = 100 units x 30 days
= **3,000 units**

iii. **$L_{max}$** = RoL + EOQ – (minimum usage x minimum lead time)
= 3000 + 4382 – (100 units x 20 days) = 7382 – 2000 = **5,382 units**

iv. **$L_{min}$** = RoL – (average usage x average lead time)
= 3000 – (1000 units x 25 days) = 3000 -2500 = **500 units**

v. **Number of Orders per year** = $\frac{D}{EOQ} = \frac{24{,}000 \text{ units}}{4{,}382 \text{ units}}$ = **5.5 times**

vi. **Total Cost** = TC = P.D + $C_o \cdot \frac{D}{Q} + C_H . \frac{Q}{2}$
= (\$10*24000) + (\$400*$\frac{24000}{4382}$) + (\$1*$\frac{4382}{2}$)
= 240,000 + 2190.78 +2191
= **\$244,381.78** per annum

**Summary**
You have been able to complete this session in good time. You now understood what receivables management mean and how it is different from inventory management. You can now identify the costs associated with holding inventory and also calculate inventory level to avoid a situation of running out of inventory and driving your customers away to your competitors.

**Self-Assessment Questions**
**Exercise 6.3**

## SESSION 4: PAYABLES MANAGEMENT

### Introduction

Under the short-term sources of finance, we mentioned trade credit as one. The management of this credit is crucial for the survival of the business. In this session, we introduce you to the proper management policies, the basic terminologies used in credit management, the credit period, the collection policies of the company and the mandatory 5Cs of credit.

**Objectives**
By the end of this session, you should be able to:
(a) explain the decisions variables in credit management
(b) describe the credit period of a business entity
(c) explain some basic credit terms
(d) discuss credit policies to be adopted by a business organization
(e) describe how the 5Cs of credit could be applied to credit management

Now read on …

**4.1 Payables or Credit Management**
A component of a firm's working capital arises from the necessity to extend credit to its customers. A firm's credit policy consists of two inter – related activities viz:

- The granting of credit
- The collection of receivables

Both of these aspects affect the composition of working capital. A more liberal credit policy implies higher levels of credit but will also result in higher levels of accounts receivable which must be financed. Credit management involves five basic decisions or policy variables.

1. Credit period; this refers to the length of time for which the firm is prepared to grant credit to its customers.
2. Credit instrument; this is the legal form the credit will take, e.g. IOU or receipt.
3. Credit standards; the decision as to which customers constitute "good" "bad" credit risks. It also requires the stipulation of a method for determining their credit – worthiness either on the basis of past performance, the analysis of their financial statements, or external reports of banks and / or credit agencies.
4. Collection policy; a policy for monitoring outstanding credits must be established, and a decision must be reached in case of slow payments.
5. Incentives; the firm must decide on the extent of the financial (or other) incentives which will be offered for prompt payment.

**4.2 Credit period**
A number of factors influence the length of the credit period. One of the most important is the buyer's inventory period and operating cycle. All other things being equal, the shorter these are, the shorter the credit period will normally be. Other factors that influence the credit period as outlined by Ross et al. (1995) are:

1. Extent to which goods are perishable: Perishable items have relatively rapid turnover and relatively low collateral value. Credit periods are thus shorter for such goods.
2. Consumer demand: Products that are well established generally have more rapid turnover. Newer or slow – moving products will often have longer credit periods associated with them to entice buyers.
3. Cost profitability and standardization: Relatively, inexpensive goods tend to have shorter credit periods. The same is true for relatively standardized goods and raw materials.
4. Credit risk: The greater the credit risk of the buyer, the shorter the credit period is likely to be (assuming that credit is granted at all).
5. The size of the account: If the account is small, the credit period may be shorter because small accounts are costlier to manage and the customers are less important.

6. Competition: When the seller is a highly competitive market, longer credit periods may be offered as a way of attracting customers.

**4.3 Credit terms**

After credit standards have been established and the credit worthiness of the customer has been assessed, the management of a firm must determine the terms and conditions on which trade credit will be made available, that is, the credit terms. This is influenced in industry by tradition, bargaining strength and product type. Credit terms have three components. These are:

- The credit periods
- Cash discount if any
- Cash discount period

The following are the basic credit terms:

- CIA (Cash in advance)
- CBD (cash before delivery); with this, payment must be made before the order is shipped.
- COD (Cash on delivery); the shipper collects the payment on behalf of the seller upon delivery.
- Cash payment; this is due when goods are delivered. Unlike COD, cash terms allow the customer up to about ten days to pay.
- Standard terms; "Net 30" or "Net 60". This means that payment is due in full 30 or 60 days from the date of the invoice.
- Discount terms ("2/10 net 30"); Discount terms include a discount percent, discount payment date, and net date. The buyer can take a 2% discount if the payment is made by the tenth day following the invoice date. Otherwise, the full amount is due 30 days following the invoice date.

**4.4 Collection policy**

Collection policy decisions normally become an issue only after the accounts are past due. When a customer fails to pay an invoice on the due date, it is often our natural reaction to think in terms of solicitors and legal action. It must be emphasized that this is a last resort. Since companies would not like to lose their customers, it is recommended that they establish a procedure for late collections along the following lines:

1. Send invoices promptly: Ensure that there is no delay between delivery of goods and dispatch of the invoice, so that the customer is made aware of the due date for payment as early as possible.
2. Letters: When an account is overdue by few days, a "friendly reminder" may be sent. If payment sis not received, one or two more letters might be sent with one of the letters becoming progressing.
3. Telephone calls: After the first couple of letters, the customer is phoned. If the customer is having financial problems, a compromise might be worked out. A partial payment is better than no payment.
4. A visit from the sales person responsible for that customer to request payment. A reminder to the sales person that commission is based on cash received not invoiced sales should be enforced.
5. Restriction of credit: Debtors can be denied additional credit / supplies if they exceed their credit limit. Insist on payment of previous order before dispatching goods if credit limit is breached.

6. Use of collection agency: The account can be turned over to a collection agency that specializes in collecting past due accounts. Collection agents usually charge a fee for more effort.
7. Legal action: If the bill is large enough, legal action may be used as a last resort, to obtain judgment against the debtor. In designing the collection procedures, we have to keep in mind that aggressive efforts to collect may results in loss of future sales. We also have to consider the customers' circumstances. For example, if the customer is in the midst of a labour stride, you may wish to avoid collection tactics detrimental to your relationship with that particular customer.

**4.5 The 5 C's of Credit**

Traditional financial management texts suggest that credit mangers would take note of the five C's of credit to evaluate the probability of default.

- Character: This attempt to measure the customers' willingness to pay. Levy and Sarnat (1994) submit that, it raises the fundamental question of whether or not he will try to honour his promise to pay. Most managers consider this issue crucial. If the answer to this question is strongly negative, it is unlikely that credit will be granted.
- Capital: This refers to the customer's ability to pay, i.e. his or her financial position. Here, an attempt is made to determine whether the customer has sufficient financial resources to meet the proposed obligation.
- Capacity: This is more subjective measure of the customer's ability to pay. Here, his previous "track record" with the firm is crucial.
- Collateral: This is simply the assets which the customer can offer to secure his debt.
- Condition: This refers to the customer's vulnerability to changes in business conditions, or other special events. That is, the effect of economic trends on the customer's position.

**Summary**

Congratulations dear students. You have done well so far. In this session, you learnt about credit management, credit periods and some basic credit terminologies that are used in credit management. We discussed the various decision variables in credit management and the credit policies that could be adopted by various business entities. We ended the session with the 5Cs of credit: character, capital, capacity, collateral and condition.

**Self-Assessment Questions**

**Exercise 6.4**

## SESSION 5: COST OF INDIVIDUAL CAPITAL

### Introduction

Dear lovely students, welcome to this session on the cost of individual capital. In this session, we shall tell you through the cost of ordinary shares, preference shares and cost of borrowing on long-term basis. The approaches used in the determination of these costs shall be thoroughly looked it. It's a bit technical but we trust you will make time to go through this successfully as we use more examples to make it simple for you.

### Objectives

By the end of this session, you should be able to:

(a) define cost of capital
(b) understand security market line and how it is used
(c) understand capital asset pricing model
(d) use the dividend approach in determining the cost of ordinary shares
(e) know and explain the various ways of determining the cost of different debts

Now read on …

### 5.1 Meaning of Cost of Capital

Cost of capital is the rate that a business must pay (explicitly or implicitly) to investors for its capital. Capital in this context refers to the funds a business uses to finance its assets and operations. The cost of capital can also be defined as the minimum rate of return required by suppliers of a business' capital. Cost of capital could also be seen as the minimum required rate of return that the business earns on new investments in order to maintain the market value of its ordinary shares.

Therefore, when we say that the required return on an investment is 10%, we usually mean that the investment will have a positive NPV only if its return exceeds 10%. Another way of interpreting the required return is to observe that the firm must earn 10% on the investment just to compensate its investors for the use of the capital needed to finance the project. This is why we could say that 10% is the cost of capital associated with the investment.

Assume we are evaluating a risk-free project. To determine the required return, we look at the capital markets and observe the current rate offered by risk-free investment and we use this to discount the project cash flow. Thus, the cost of capital for a risk-free investment is the risk-free rate. If this project were risky, then assuming all other information is unchanged, the required return is obviously higher. That is the cost of capital for this project (because it is risk) is greater than the risk-free rate, and the appropriate discount rate would exceed the risk-free rate.

Note that the terms 'required return', appropriate discount rate, and cost of capital are used interchangeably. The key fact is that cost of capital associated with an investment depends on the risk of that investment. Put differently, the cost of capital depends primarily on the *use of funds*, not the source. Given that a firm uses debt and equity capital, its overall cost of capital will be a mixture of the returns needed to compensate its creditors and its stockholders. Put

differently, a firms' cost of capital will reflect both its cost of debt capital and its cost of equity capital.

### 5.2 The Cost of Common Equity

There are two approaches to determining the cost of equity; the dividend growth model approach and the security market line (SML) approach/Capital Asset Pricing Model (CAPM)

#### 5.2.1 The Dividend Growth Model Approach

Assume a firm's dividend grows at a constant rate g, then, the price per share of the stock, $P_o$ can be written as:

$$Po = \frac{D_0\ (1+g)}{k_e - g} = \frac{D_1}{k_e - g}$$

Where:
$D_o$ = the dividend just paid
$D_1$ = the next period's projected dividend
$k_e$ = the required return on the stock.

We can re-arrange this to solve for $k_e$:

i.e. $k_e = \frac{D_1}{P_o} + g$

Since $k_e$ is the return that shareholders require on the stock, it can be interpreted as the firms cost of equity capital. To calculate $k_e$ using the dividend growth model approach we need to know $P_o$, $D_o$, and g. For publicly traded dividend paying firm, Po and Do can be observed directly. The expected growth rate in dividends must be estimated.

**Example 1**

K&K Services paid a dividend of $4 per share last year. The stock currently sells for $60 per share. You estimate that dividend will grow steadily at 6% year into the indefinite future. What is the cost of equity capital for K&K Services?

Solution

Using the dividend growth model, the expected dividend for the coming year, $D_1$ is

$D_1 = D_0(1+g)$
$= \$4(1.06)$
$= \$\ 4.24$

Substituting $D_1 = \$\ 4.24$ in $k_e = \frac{D_1}{P_o} + g$

$k_e = \frac{\$4.24}{\$60} + 6\%$

$k_e = 13.07\%$

The cost of equity is thus 13.07%

Did you notice that $k_e = \frac{D_0\ (1+g)}{P_o} + g$

$k_e = \frac{\$4\ (1+0.06)}{\$60} + 0.06 = 13.07$

The above formula serves as an estimate of the cost of equity capital, where new equity capital is obtained by retaining part of the firms' current period earnings. Of course, you know that common equity financing can come from one of two sources – retained earnings or sale of new shares. If the business retains earnings, it does not incur any floatation costs, so the investors required rate of return is the same as the formula above.

If the business issues new shares to raise equity capital, it incurs floatation costs; hence, we must adjust the formula for the floatation costs, F. We modify our original formula for ordinary shares to read:

$$k_e = \frac{D_1}{P_o - F} + g$$

**Example 2**
Assume: $D_1$ = \$2, $P_o$ = 40, F = \$4, g = 7%

$$\Rightarrow k_e = \frac{\$2}{\$40 - \$4} + 7\% = 12.6\%$$

**Example 3**
The Seyram Company's ordinary shareholders anticipate receiving a \$2.20 per share dividend next year based on the fact that they received \$2 last year and expect dividend to grow 10% next year. Furthermore, analysts predict that dividends will continue to grow at a rate of 10% into the foreseeable future. Given that Seyram's share is trading for \$50 per share, calculate the investors required rate of return.

Solution

$$k_e = \frac{D_1}{P_o} + g$$

$$k_e = \frac{\$2.20}{\$50} + 0.10 = 14.4\%$$

Now assuming Seyram decides to issue new ordinary shares at a cost of 15% of the current stock price, calculate the resulting cost of the new common equity capital.

Solution

$$k_e = \frac{D_1}{P_o - F} + g$$

$$k_e = \frac{\$2.20}{\$50 - \$7.50} + 0.1 = 15.18\%$$

Drawback of dividend growth model approach is that it's applicable to companies that pay dividends. It means that the approach is helpless in many situations. Even for companies that pay dividend, the model assumes that dividend growth is constant, which is never exactly the case.

**5.2.2 Estimating growth rate of dividend** (g)
In estimating the growth rate, we can use:

- Analysts' forecasts of future growth rates
- Historical growth rates (uses average changes in dividend overtime)

- Extrapolation based on past dividend patterns
- Gordon's earning retention model

Analyst's forecasts are available from different sources. These sources will have different estimates, so it might be helpful to obtain multiple estimates and then find the average. Using the ***Historical growth rates***, we might observe dividends for the previous, say 5 years, calculate the year-to-year growth rates and average them.

**Example 4**

Suppose you observe the following for a firm:

| Year | Dividend |
|---|---|
| 1995 | $ 1.15 |
| 1996 | $1.25 |
| 1997 | $1.50 |
| 1998 | $1.55 |
| 1999 | $1.70 |

We can calculate the % changes in the dividend for each year:

| Year | Dividend | $ Change | %Change |
|---|---|---|---|
| 1995 | $ 1.15 | - | - |
| 1996 | $1.25 | 0.10 | 8.70 |
| 1997 | $1.50 | 0.25 | 20 |
| 1998 | $1.55 | 0.05 | 3.33 |
| 1999 | $1.70 | 0.15 | 9.68 |

Averaging the four growth rates:

$\frac{8.70+20+3.33+9.68}{4} = \frac{41.71}{4} = 10.43\%$

**Extrapolation using past dividends**

Here, with information on earlier and current dividend paid, we can follow the trend using the simple formula:

$$g = \left(\frac{D_O}{D_{-n}}\right)^{\frac{1}{4}} - 1$$

Where

$D_0$ = current dividend per share

$D_{-n}$ = dividend ***n*** years ago

n = number of years of dividend growth

Using the same past dividend data in the previous example;

$$g = \left(\frac{D_O}{D_{-n}}\right)^{\frac{1}{n}} - 1 = \left(\frac{\$\,1.70}{\$1.15}\right)^{\frac{1}{4}} - 1 = 1.1027 - 1 = 0.1027 = \mathbf{10.27\%}$$

**5.2.3 Gordon's Earning Retention Model**

An assumption here is that dividend growth is dependent on the level of retained earnings in a business so that the higher the level of retentions in a business, the higher the potential dividend growth rate.

This model therefore uses the simple formula:

$\mathbf{g = b \cdot r_e}$

where b = earnings retention rate

$r_e$ = accounting rate of return (ARR) or return on equity (ROE)

**Example 5**

A company is about to pay an ordinary dividend of $0.16 a share. The share price is $2. The ARR on equity is 12.5% and 20% of earnings are paid out as dividends. Calculate the cost of equity for the company?

**Solution**

About to pay => cum-dividend share price = $P_0$ = $ 2

Ex-dividend share price = $P_0$ - $D_0$ = $2 - $0.16 = $ 1.84

$\mathbf{k_e} = \frac{D_1}{P_o - D_o} + g = \frac{D_0\,(1+g)}{P_o - D_o} + g$ but **b**=1-20% = 1- 0.20 = 0.80

Calculating ***g*** =>

$\mathbf{g = b \cdot r_e}$ = 0.80*12.5%= **10%**

Therefore, cost of equity =>

$$\mathbf{k_e} = \frac{D_0\,(1+g)}{P_o - D_o} + g$$

$$\mathbf{k_e} = \frac{D_0\,(1+g)}{P_o - D_o} + g$$

$$\mathbf{k_e} = \frac{\$0.16\,(1.10)}{\$2.00 - \$0.16} + 0.10$$

$$\mathbf{k_e} = \frac{\$0.16\,(1.10)}{\$1.84} + 0.10$$

$$\mathbf{k_e} = \frac{\$0.176}{\$1.84} + 0.10$$

$$\mathbf{k_e} = 0.0957 + 0.10$$

$$\mathbf{k_e} = 0.1957 = 19.57\%$$

Therefore, the dividend growth rate is 10% and the cost of equity is 19.57%

### 5.2.4 The SML Approach or Capital Asset Pricing Model (CAPM)

In risk-return analysis, the required or expected return on a risky investment depends on three things:

- the risk-free rate, $r_f$
- market risk premium, expected return on the market portfolio minus the risk-free rate, $r_m$ -$r_f$
- the systematic risk of the asset relative to average, which is called its beta coefficient, ß

Using the SML or CAPM approach the expected return on the company's equity, $k_e$ can be written as: $k_e = r_f + ß(r_m - r_f)$
Where: ß is the estimated beta for the equity. Therefore, the required return from the SML, $k_e$ is: $k_e = r_f + ß(r_m - r_f)$

**Example 6a**
Assume a market risk premium of 8.5%. Treasury bills are paying about 5.6%, and ß is 0.95. Estimate the cost of common equity.

Solution
$k_e = r_f + ß(r_m - r_f)$
$= 5.6\% + 0.95(8.5\%)$
$= 13.68\%$

**Example 6b**
Suppose shares in J Holdings have a beta of 1.2. The market risk premium is 8% and the risk–free rate is 6%. J Holdings' last dividend was $2 per share and the dividend is expected to grow at 8% indefinitely. The shares currently sell for $30. What is J Holdings' cost of equity capital?

Solution
$k_e = r_f + ß(r_m - r_f)$
$= 6\% + 1.2 \times 8\%$
$= 15.6\%$

**Using Dividend Growth Model**
Solution
The projected dividend $D_1$ is: $D_0(1 + g) = \$2 \times (1.08) = 2.16$
$\Rightarrow ke = \frac{D_1}{P_0} + g = \frac{2.16}{30} + 8\% = 15.2\%$

Our two estimates are reasonably close, giving us an average of 15.4% which is approximately the cost of equity capital for J Holdings.

**Example 7**
Jojo and Co Ltd just issued a dividend of $2.50 per share on its ordinary shares. The company is expected to maintain a constant 6% growth rate in its dividends indefinitely. If the share sells for $50 a share, what is the company's cost of equity?

Solution
$D_1 = Do(1xg) = 2.50(1+0.06) = \$2.65$

Substituting $D_1 = \$2.65$
$k_e = \frac{\$2.65}{\$50} + 6\% = 11.3\%$

**Example 8**

The T Company's ordinary share has a beta coefficient of 0.82. Furthermore, the risk-free rate is currently 7%, and the expected rate of return on the market portfolio of all risky assets is 16%. Estimate the Talbot's cost of capital.

Solution

$k_e = r_f + ß(r_m - r_f)$

$= 7\% + 0.82(16\% - 7\%) = 14.38\%$

**Example 9**

Big Joe PLC. Ordinary shares has a beta of 1.25. If the risk-free rate is 4% and the expected return on the market is 12%, what is Big Joe's cost of equity, capital?

Solution

$k_e = r_f + ß(r_m - r_f)$

$k_e = 4\% + 1.25(12\% - 4\%) = 14\%$

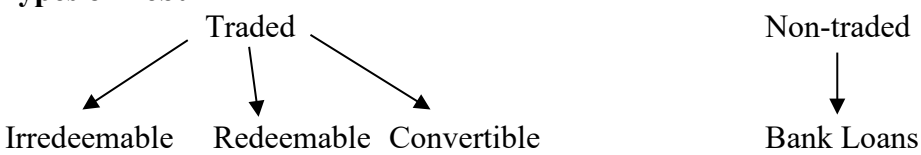
**5.3 The Cost of Debt**

This is the return that the firm's creditors demand on new borrowing. This can normally be observed either directly or indirectly, because the cost of debt is simply the interest rates in the financial markets. For example, if the firm already has bonds outstanding, then the yield to maturity on those bonds is the market-required rate on the firm's debt.

Alternatively, if we knew that the firm's bonds were rated, say AA, we can simply find out what the interest rate on newly issued AA rated bond is. The simplest case will be a $1,000 bond paying $100 annual coupon, thus providing a 10% yield.

To properly estimate the cost of debt, there is the need to identify the specific types of debt and their associated costs.

**Types of Debt**




**Note the following:**

- *Traded debts* are those debts that can be easily bought & sold. *Non-traded debts* are not.
- The terms loan notes, bonds, loan stock, and marketable debt, are used interchangeably. Gilts are debts issued by government.
- *Irredeemable debt* – no repayment of principal, interest in perpetuity
- *Redeemable debt* – interest paid until redemption of principal
- *Convertible debt* – may be later converted to equity
- Debt is usually quoted in nominal value blocks of
  - $1000 where market price is near to $1000.eg. a bond market price of $850 is more likely to have a par value of $1000 than say $100 par value.
  - $100 where market price is near to $100.eg. a bond market price of $110 is more likely to have a par value of $100 than say $1000 par value.
- The terms *ex-interest* and *cum-interest* are used in much the same way as *ex-dividend* and *cum-dividend*.

### 5.3.1 Irredeemable Debt

Since the company does not intend to repay the principal but to pay interest forever, the market value (current price) of an irredeemable debt can be analysed as the PV of perpetuity.

$$MV = P_0 = \frac{Interest}{r}$$

Where r = debt holders' required return
MV = market value, and
$P_0$ = current price of bond

⇨ Debt holders' required return = r = $\frac{Interest}{MV}$

But the *cost of irredeemable* debt to the firm will be the *tax-adjusted debtholders' required return.*

⇨ $K_d = r\,(1\text{-}T) = \frac{Interest\,(1-T)}{MV}$

where T = corporate tax rate

**Example 10**

A company has in issue 10% irredeemable debt quoted at $800 cum-interest. If corporate tax rate is 30%, what will be required return by debt providers and the cost of debt to the company?

**Solution**

Par value = $1000 since the market value $800 is reasonably closer to $1000 than $100.
Coupon = Interest = 10% * 1000 =$100
Cum-Interest price of debt= $800
Ex-Interest price of debt = Cum-Interest price of debt – immediate coupon
= $800 -$100 = $ 700 = MV

*Debt holders' required return* = r = $\frac{Interest}{MV} = \frac{\$100}{\$700} = 0.1429 = 14.29\%$

*Company's cost of debt* = $K_d = \frac{Interest\,(1-T)}{MV} = \frac{\$100(1-0.30)}{\$700} = 0.10 = 10\%$

### 5.3.2 Redeemable Debt

The company will pay interest for a number of years and then repay the principal (sometimes at premium or discount to the original loan amount). The *market value* (MV) of a redeemable debt is simply the sum of the PVs of the interest and the redemption payment.

Market Value/Price = $MV_0 = \frac{Int_1}{1+r} + \frac{Int_2}{(1+r)^2} + \cdots + \frac{Int_t + P_t}{(1+r)^t}$ ==> Equation 1

Since the interest on the debt is generally fixed, i.e. $Int_1=Int_2=Int_t=Int$, we can apply the annuity formula as:

Market Value/Price = $MV_0 = Int.\left(\frac{1}{r} - \frac{1}{r(1+r)^t}\right) + \frac{P_t}{(1+r)^t}$ ==> Equation 2

Where:

| | | |
|---|---|---|
| Int. | = | periodic coupon |
| r | = | debt holders' required return |
| $P_t$ | = | redemption amount (at par, premium or discount) |

**NB**: Either *Equation 1* or *2* can be used. However, where ***t*** is relatively large, say *100 years*, only *Equation 2* can be possibly used because it will be time-consuming and cumbersome to use Equation 1.

However, from previous discussions, we established that the company's cost of debt is the tax-adjusted debt holders' required return. Hence, the cost of redeemable debt is obtained by adjusting the coupon interest rate with tax effect. The following equations, particularly *Equation 4* are therefore used by altering the equations above;

Market Value/Price = $MV_0 = \frac{Int_1\,(1-T)}{1+k_d} + \frac{Int_2\,(1-T)}{(1+k_d)^2} + \cdots + \frac{Int_t\,(1-T) + P_t}{(1+k_d)^t}$ => Equation 3

Market Alternatively, using PV of annuity approach:

Value/Price = $MV_0$ = Int. (1-T) $\left(\frac{1}{k_d} - \frac{1}{k_d(1+k_d)^t}\right) + \frac{P_t}{(1+k_d)^t}$ ==> Equation 4

Where:

| | | |
|---|---|---|
| Int. | = | periodic coupon |
| $k_d$ | = | company's cost of debt, i.e. after-tax cost of debt |
| $P_t$ | = | redemption amount (at par, premium or discount) |

Just like the choice between *Equations 1 and 2*, it is preferable to use *Equation 4*.

**Calculating the cost of redeemable debt**

There are *two ways* of estimating the cost of redeemable debt:

- *IRR technique*
- *Trial and Error Technique (i.e. the traditional YTM method)*

**IRR Technique**

The Cost of Redeemable Debt is simply the value of $k_d$ for which NPV (net investment) is zero, i.e. $k_d$ is more or less the IRR.

Hence:

Net investment = NPV = 0 = $-MV_0$ + Int. (1-T) $\left(\frac{1}{k_d} - \frac{1}{k_d(1+k_d)^t}\right) + \frac{P_t}{(1+k_d)^t}$ => Equation 5

This equation is therefore used to estimate the **cost of redeemable debt** by relying on the **computational interpolation formula** below, given *two assumed discount rates* and their *corresponding NPVs* as we have done under the topic Capital Budgeting.

$$K_d = IRR = a + \frac{NPV_a}{NPV_a - NPV_b}(b-a) = \frac{NPV_a}{NPV_a - NPV_b}(b-a) + a$$

**Trial and Error Technique** (i.e. the traditional YTM method)

Since yield to maturity is the return investors expect on a bond, it is the cost of debt ($k_d$) to the issuing company. The YTM formula unlike the IRR technique helps to establish *not the*

*actual* but *an approximate value* close to the actual $k_d$. The general formula for estimating the yield to maturity $r_d$ is:

$$k_d = \frac{\text{Annual Interest payment} + \dfrac{\text{Principal payment} - \text{price of the bond}}{\text{Number of years to maturity}}}{0.6\,(\text{price of the bond}) + 0.4\,(\text{principal payment})}$$

**Example 11**

Suppose the Global Tool Company issued a 30 year, 7% bond 8 years ago. The bond is currently selling for 96% of its face value. What is Global Tool cost of debt?

Solution

Let $k_d$ = cost of debt. Then:

$$k_d = \frac{\text{Annual Interest payment} + \dfrac{\text{Principal payment} - \text{price of the bond}}{\text{Number of years to maturity}}}{0.6\,(\text{price of the bond}) + 0.4\,(\text{principal payment})}$$

$$k_d = \frac{70 + \frac{1000-960}{22}}{0.6(960) + 0.4(1000)} = \frac{70 + 40/22}{570+400} = \frac{70 + 1.18}{976} = 7.29\%$$

**Example 12**

Assume a debt issue pays \$101.5 per year in interest, has a 20 year life, and its currently selling for \$940. Find the cost of this debt.

Solution:

$$k_d = \frac{101.5 + \frac{1000-940}{20}}{0.6(940) + 0.4(1000)} = \frac{101.50+3}{564+400} = 10.84\%$$

What happens if the firm incurs flotation costs? The formula needs to be adjusted to take care of the flotation costs such as brokerage commissions and legal and accounting fees incurred in issuing the debt. The flotation costs are deducted from the price of the bond. In the example above, suppose we were told that the firm paid \$58 per bond as flotation costs, then

$$k_d = \frac{101.5 + \frac{1000-(940-58)}{20}}{0.6(940-58) + 0.4(1000)} = \frac{101.50 + 5.9}{529.2 + 400} = 11.56\%$$

Cost of debt (yield to maturity) as shown above indicates how much a corporation has to pay on a before tax basis. Hence, we must adjust it for tax considerations. The after-tax cost of debt is actually the yield to maturity times one minus the tax rate i.e. $k_d$ (after tax) = $k_d(1-T_c)$. In the example above, assuming the firm is paying a 35% tax. The after-tax cost of debt would be:

$k_d$ (after tax) = 11.56% (1-0.35) = 11.56% (0.65) = 7.51%

**Example 13**

African Pot Restaurant located in Elmina plans a bond issue for the near future and wants to estimate its current cost of debt capital. After talking with the firm's investment banker, the

firm's chief financial officer has determined that a 20-year maturity bond with a $1,000 par value and 8% coupon can be sold to investors for$908.32 at floatation costs of $58.32 per bond when it issues the securities. Estimate the cost of this debt assuming the firm is in the 34% corporate tax bracket.

Solution:

**YTM method**:

$k_d$ (before tax) = $$\frac{80 + \frac{1000-(908.32-58.32)}{20}}{0.6(908.32-58.32) + 0.4(1000)} = \frac{80 + 7.5}{510 + 400} = 9.62\%$$

After tax cost of debt = $k_d(1-T_c)$ = 9.62(1- 0.34) = **6.35%**

Alternatively, we can use the *risk-adjusted coupon* straight way to obtain the *after-tax cost of debt* as follows:

$k_d$ (after tax) = $$\frac{80(1-0.34) + \frac{1000-(908.32-58.32)}{20}}{0.6(908.32-58.32) + 0.4(1000)} = \frac{52.8 + 7.5}{510 + 400} = \mathbf{6.63\%}$$

**IRR Technique**:

$$NPV = 0 = -MV_0 + Int.\ (1-T)\left(\frac{1}{k_d} - \frac{1}{k_d(1+k_d)^t}\right) + \frac{P_t}{(1+k_d)^t}$$

If r = 5%; then:

$$NPV = k_d = -(908.32 - 58.32) + 80\ (1-0.34)\left(\frac{1}{0.05} - \frac{1}{0.05(1+0.05)^{20}}\right) + \frac{1{,}000}{(1+0.05)^{20}}$$

$$NPV = k_d = -850 + 52.8\left(\frac{1}{0.05} - \frac{1}{0.05(1.05)^{20}}\right) + \frac{1{,}000}{(1.05)^{20}} = 184.89$$

If r = 10%; then:

$$NPV = k_d = -(908.32 - 58.32) + 80\ (1-0.34)\left(\frac{1}{0.10} - \frac{1}{0.10(1+0.10)^{20}}\right) + \frac{1{,}000}{(1+0.10)^{20}}$$

$$NPV = k_d = -850 + 52.8\left(\frac{1}{0.10} - \frac{1}{0.10(1.5)^{20}}\right) + \frac{1{,}000}{(1.5)^{20}} = -251.84$$

Using *interpolation*,

$$K_d = IRR = a + \frac{NPV_a}{NPV_a - NPV_b}(b-a) = \frac{NPV_a}{NPV_a - NPV_b}(b-a) + a$$

$$K_d = \frac{184.89}{184.89-(-251.84)}(10\% - 5\%) + 5\%$$

$$K_d = \frac{184.89}{184.89+251.84}(5\%) + 5\%$$

$K_d = \frac{184.89}{436.73}(5\%) + 5\%$

$K_d = 0.4234\ (5\%) + 5\% = 7.1\%$

However, from *advanced mathematical calculations* or a *programmed calculator*, the *actual value of* $k_d$ using IRR is **6.66%**. Difference in result is due to *choose of discount rates* and *rounding-off.*
**NB:** The *closer* the *chosen discount rates* to the *actual value of* $k_d$, the *more accurate* the *interpolated value.*

With regard to the YTM approach, using the *risk-adjusted coupon* straight away to obtain the *after-tax cost of debt* gives a more-closer-to-actual value than finding the *pre-tax cost of debt* separately before adjusting it for tax relief, i.e. 6.63% is closer to the actual value of 6.66% than does 6.35%.

**5.3.3 Convertible Debt**
This is a form of loan note that allows the investor to choose between taking the redemption proceeds or converting the loan note into a pre-set number of shares.To calculate the cost of convertible debt, you should:

- Calculate the value of the conversion option using available data
- Compare the conversion option with the cash option assuming that all investors will choose the option with the higher value
- Calculate the IRR of the flows as for redeemable debt

The cost of capital of convertible to debt is harder to determine. The calculation will depend on whether or not conversion is likely to happen.

(a) If conversion is **not** expected, the conversion value is ignored and the bond is treated as **redeemable debt**, using the IRR method described above.
(b) If conversion **i**s expected, the IRR method for calculating the cost of redeemable debt is used, but the **number of years to redemption** is replaced by the **number of years to conversion** and the **redemption value** is replaced by the **conversion value** i.e. the market value of the shares into which the debt is to be converted.

**Conversion value** = $P_0\ (1 + g)^n R$

| | | |
|---|---|---|
| where | $P_0$ | is the current ex-dividend ordinary share price |
| | g | is the expected annual growth of the ordinary share price |
| | n | is the number of years to conversion |
| | R | is the number of shares received on conversion |

**Example 14**
A company has issued 8% convertible bonds which are due to be redeemed in five years' time at 5%. They are currently quoted at $85 per $100 nominal. The bonds can be converted into 20 shares in five years' time. The share price is currently $4 and is expected to grow at a rate of 7% pa. Assume a 30% rate of corporate tax. Calculate the cost of the convertible debt.

**Solution**
Cash redemption value =$100*105%

=$105

Conversion value = $P_0 (1 + g)^n R$
= $4 (1+ 0.07)^5 *20$
= $112.20

As the cash redemption value is $105, investors would **choose to convert** the bonds so the **conversion value** is used in the IRR calculation.

Hence, using IRR

$$NPV = -MV_0 + Int.\ (1-T)\left(\frac{1}{k_d} - \frac{1}{k_d(1+k_d)^t}\right) + \frac{P_t}{(1+k_d)^t}$$

$$NPV = -85 + 8(1-0.30)\left(\frac{1}{k_d} - \frac{1}{k_d(1+k_d)^5}\right) + \frac{112.2}{(1+k_d)^5}$$

$$NPV = -85 + 5.6\left(\frac{1}{k_d} - \frac{1}{k_d(1+k_d)^5}\right) + \frac{112.2}{(1+k_d)^5}$$

If r = 10%; then:

$$NPV_1 = -85 + 5.6\left(\frac{1}{0.10} - \frac{1}{0.10(1+0.10)^5}\right) + \frac{112.2}{(1+0.10)^5}$$

$$NPV_1 = -85 + 5.6\left(\frac{1}{0.10} - \frac{1}{0.10(1.10)^5}\right) + \frac{112.2}{(1.10)^5} = 5.90$$

If r = 12%; then:

$$NPV_2 = -85 + 5.6\left(\frac{1}{0.12} - \frac{1}{0.12(1+0.12)^5}\right) + \frac{112.2}{(1+0.12)^5}$$

$$NPV_2 = -85 + 5.6\left(\frac{1}{0.12} - \frac{1}{0.12(1.12)^5}\right) + \frac{112.2}{(1.12)^5} = -1.15$$

Using *interpolatio*n,

$$K_d = IRR = a + \frac{NPV_a}{NPV_a - NPV_b}(b-a) = \frac{NPV_a}{NPV_a - NPV_b}(b-a) + a$$

$$K_d = \frac{5.90}{5.90-(-1.15)}(12\% - 10\%) + 10\%$$

$$K_d = \frac{5.90}{5.90+1.15}(2\%) + 10\%$$

$$K_d = \frac{5.90}{7.05}(2\%) + 10\% = 10.02\%$$

Therefore, the cost of convertible debt is 10.02%

**5.3.4 Non-Tradable Debt**

The cost of short-term funds such as bank loans and overdrafts are the current interest being charged on such funds. Alternatively, the cost of debt of ordinary or straight bonds could be

used. Bank loans and other non-tradable fixed interest loans simply need to be adjusted for tax relief. Hence, cost of non-tradable debt = interest rate (1- corporate tax rate) = r (1 –T)

**Example 15**
A firm has a fixed rate of bank loan of $1 million. It is charged 11% pa. The corporate tax rate is 30%. What is the cost of the loan?

**Solution;**
Cost of Loan = r (1 –T) = 11%*(1-0.30) = 11%*0.70 = **7.7%**

**5.4 Cost of Preferred Stock**
Preferred stock has a fixed dividend paid every period forever, so a share of preferred stock is essentially perpetuity. The cost of preferred stock, $k_P$ is: $k_P = D/P_o$
Where: D = fixed dividend
$P_o$ = current price per share of the preferred share

**Example 16**
On January 1, 2004, S&T had several issues of preference shares that traded on the GSE. One issue paid $7.80 annually per share and sold for $96.5 per share. The other paid $2.13 per share annually and sold for $25.875 per share. What is S&T's cost of preference share?

**Solution**
$k_P = D/P_o$
= $ 7.8/$96.5
= 8.1%

Using the 2nd issue, the cost is:
$k_P = D/P_o$
= 2.13/25.875 = 8.2%

Therefore S&T's cost of preference share appears to be in the 8.1% - 8.2% range. Once again, where flotation costs are incurred when new preferred stocks are sold, we must adjust the formula to reflect the flotation costs. Therefore: $k_p = D/P_o - F$

**Example 17**
On January 20, 2000, Goldfields had an issue of preference share that traded on the GSE. The issue paid an annual dividend of $3.75 per share and the preferred stock price closed at $60.375. Assume that if the firm were to sell an issue of preference share with the same characteristics as its outstanding issue, it would incur flotation costs $2.375 per share and the shares would sell for their January 20, 2000, closing price. What is Goldfields cost of preference shares?

Solution:

$$k_p = \frac{\$3.75}{\$60.375-\$2.375}$$

= 6.47%

**Summary**
Interesting journey for you. But finally, you have made it. You were able to go through all the 17 examples in this session that took you through the various approaches to determining

the cost of equity and debt. You have noticed that the concept of time value of money had been applied in most of the instances and this points to the fact that when sourcing for funding, we need to know the cost involved since no business person is Father Christmas.

**Self-Assessment Questions**
**Exercise 6.5**

## SESSION 6: WEIGHTED AVERAGE COST OF CAPITAL

### Introduction

Finally, finally, finally!!! We are here today. We are in this session going to discuss WACC and what it means. What the market value of equity is and still borrow from our previous lessons what we learned.

### Objectives

By the end of this session, you should be able to:

(a) explain what WACC is
(b) discuss factors to be considered when calculating WACC
(c) understanding the computation of WACC

Now read on …

### 6.1 What is Weighted Average Cost of Capital?

A firm's average cost of capital (ACC) is the average cost of the funds, normally represented by the WACC. The computed WACC represents the cost of capital currently employed. Alternatively, the cost of raising the next increment of capital can be determined- this is what is termed the *Marginal Cost of Capital* (MCC). A firm's MCC is the additional cost he firm will pay to raise an additional dollar of capital.

**6.2 Calculating WACC**

To find an average cost, the various sources of finance must be weighted according to the amount of each held by the company. The weights for the sources of finance could be *book values* (i.e. historic cost of finance) or *market values* (i.e. current opportunity cost of finance). However, as much as possible, the use of *market values* is preferred to the use of *book values*.

WACC is calculated as follows:
Let E stand for the market value of the firm's common equity. E is calculated by multiplying the number of outstanding shares by the price per share.
Let D represent the combined market value of the debt.
Let P represent the value of the firm's preferred equity. P is calculated by multiplying the number of preferred shares by the price per share.
Thus: **V** = **E**+**D**+**P**

We can calculate the *weights* or *percentages* of the total capital represented by the debt and equity as: **100%** = 1 = $^{E}/_{V} + {}^{D}/_{V} + {}^{P}/_{V}$

If the total market value of a company's ordinary shares were calculated as $200m, the total market value of the company's preference share was $150m and the total market value of the company's debt was $50m, then:

$$^{E}/_{V} = {}^{\$200m}/_{\$400m} = 50\%$$
$$^{P}/_{V} = {}^{150m}/_{400m} = 37.5\%$$
$$^{D}/_{V} = {}^{50m}/_{400m} = 12.5\%.$$

This implies that 50%, 37.5 and 12.5% of the firm's financing is common equity, preference equity and debt respectively. The firm's overall cost of capital, also called Weighted Average Cost of Capital (WACC), can be calculated as follows:

$$\text{WACC} = \mathbf{(^{E}/_{V})\ k_e + (^{P}/_{V})\ k_p + (^{D}/_{V})\ k_d\ (1\text{-}T_c)}$$

where

$k_d$ = pre-tax cost of debt
$k_d(1\text{-}T_c)$ = post-tax cost of debt

This is the overall return the firm must earn on its existing assets to maintain the value of its stock.

**Example 1**

BQ Ltd has 1.4million shares outstanding. The stock currently sells for $20 per share. BQ's debt is publicly traded and was recently quoted at 93% of face value. It has a total face value of $5m, and it's currently priced to yield 11%. The risk-free rate is 8% and the market risk premium is 7%. You have estimated BQ's beta as 0.74. If the corporate tax rate is 34%, what is the WACC of BQ?

**Solution**

Using SML, $k_e = r_f + ß(r_m - r_f)$

$= 8\% + 0.74(7\%) = 13.18\%$

Total value of equity is 1.4 m ($20) = $ 28m. The pretext cost of debt is the current yield to maturity on the outstanding debt, 11%. The debt sells for 93% of its face value, so the current market value is 0.93 x $5m - $ 4.65m. The total market value of the equity and debt together is $28m+$4.65m = $32.65m

WACC = $(^E/_V) \times k_e + (^D/_V) R_D \times (1-T_c)$ But $^E/_V = {}^{28}/_{32.65} = 85.76\%$, $D/_V = {}^{100-85.76=14.24\%}$

$= .08576 (13.18\%) + 0.1424(11\%) (1-034)$

$= 12.34\%$

**Example 2**

(a) Kwick System Co. Ltd sold bonds to the public five years ago to finance its operations. The bond was recently quoted at 89% of face value. The bond has a total face value of $8.5 million. And it's currently priced to yield 9%. Kwick System's shares currently sell for $14 per share. Fifty-three percent (53%) of Kwick System's 3.2 million shares of stock are outstanding. To raise additional capital, Kwick System issued 500 preference stock that paid $6.5 annually per share and sold for $95 per share. Treasury bills pay 8% and the expected return on the market is 17%. You have estimated that Kwick System has a beta of 0.92. If the corporate tax rate is 36%, Calculate Kwick System's Average Weighted Cost of Capital.

(b) Assume that no information is provided on risk free rate, beta and expected market returns. Instead, you are told that Kwick System paid a dividend of $4 per share last year and will grow steadily at 6% per year into the future. The stock currently sells for $60 a share. Calculate Kwick System's Average Weighted Cost of Capital.

**Solution**

(a) Using SML, $k_e = r_f + ß(r_m - r_f)$

$= 8\% + 0.92(17\%-8\%)$

$= 16.28\%$

Kwick System's share outstanding = 1,696,000

Total value of common equity outstanding = 1,696,000($14) = $23,744,000

Cost of debt is 9%

Bond (debt) sells for 89% of par value of $8.5m.

Therefore, current market value of debt = 0.89($8.5m) = $7.565m

Cost of preferred stock = 6.5/95 = 6.8%

Total value of preferred stock =500($95) = $0.0475m

Total market value of common, preferred equity and debt

= $(7.565m+23.744m+ 0.0475m)

$^E/_V$ = 23.744m/31.3565m = 0.7572

$^P/_V$ = 0.0475m/31.3565m = 0.0015

$^D/_V$ = 7.565m/31.3565m = 0.2413

WACC = $(^E/_V)k_e + (^P/_V)k_p + (^D/_V)k_d(1-T_c)$

$= 0.7572(16.28\%) + 0.0015(6.8\%) + 0.2413(9\%) (1 - 0.36)$

$= 12.32 + 0.01 + 1.39$

$= 13.72\%$

(b) Using the dividend growth model:

$k_e = \frac{D1}{Po} + g$ but $= D_1 (1+g) = \$4(1.06) = \$4.24$

$\Rightarrow k_e = \frac{\$4.24}{\$60} + 6\% = 13.07\%$

Substitute $k_e = 13.07\%$ in the WACC equation:

$= 0.7572(13.07\%) + 0.0015(6.8\%) + 0.2413(9\%)(1 - 0.36)$

$= 9.90 + 0.01 + 1.39$

$= 11.30\%$

**Summary**

Congratulation for coming to the end of the unit and the course. In this session, we took you through the calculation of WACC and how it works in practice. The cost of capital and concept of time value of money were still relevant. We wish you all.

**Self-Assessment Questions**

**Exercise 6.6**